Class 8

Physics

(Solutions)

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1. Units and Dimensions

Solutions

LEVEL: I

1. Correct option: (C) 1000m.

$1\text{fermi}=10^{-15}\text{m}$

$1\text{Nautical mile}=1852\text{m}$

$1\text{century}=100\text{ years}$

The International System of Units (SI) uses the kilometre as the unit of distance.
A metre can measure both length and distance. It is a SI unit with the symbol "m." The unit of measurement for length and distance is the metre.

$1\text{m}=\frac{1}{1000}\text{km}$

The conversion factor is 1000 to convert kilometres to meters.

2. Correct option: (B) Ampere.

The unit of electric current is the Ampere (A).

3. Correct option: (A) Radian.

The radian, denoted by the symbol "rad," is the SI-derived unit of plane angle. The plane angle that an arc with length equal to the circle's radius subtends at the circle's centre is known as a radian. As a result, we discovered that a plane angle is measured in radians.

4. Correct option: (B) 10^{-6}m**.**

Since micro is the submultiple, the value of 1 micron is equivalent to 10^{-6}m. However, the micron is the unit used to measure microscopic distance.

5. Correct option: (A) Light year.

The term "astronomical unit" (AU) refers to the average distance between the Sun and the Earth. Light Year: A light year is the distance that light travels in a vacuum in one year. The light year is used to calculate the separation between celestial bodies.

6. Correct option: (A) Light year.

The distance travelled by light in a vacuum in one year is called a light year.

7. Correct option: (B) $1.5\times10^{11}\text{m}$**.**

$1\text{ A.U}=1.5\times10^{11}\text{m}$

$1.5\times10^{11}\text{m}$ metres are equivalent to one astronomical unit. The distance between the Earth and the Sun is measured in terms of length. From the greatest distance (aphelion) to the smallest (perihelion), and once a year back again, that distance changes as the Earth orbits the Sun.

8. Correct option: (A) $9.5\times10^{15}\text{m}$**.**

A light-year is a unit of measurement used to express astronomical distances; it is around 9.46 trillion kilometres $9.5\times10^{12}\text{km}$ long or $9.5\times10^{15}\text{m}$.

9. Correct option: (D) 1609m.

$1\text{mile}=1609\text{ m}$

The appropriate conversion factor is 1609. It displays how many metres there are in a mile. Consequently, 1609 metres are equal to 1 mile.

10. Correct option: (A) 10^{-15}m**.**

The SI unit of length is sometimes also referred to as the femtometre. A Fermi is a remarkably short length. It is equivalent to 10^{-15}m.

11. Correct option: (A) 10^{-10}m**.**

We are aware that an angstrom is a very small unit of length. The diameter of elementary particles as well as the radius of an atom are typically expressed using the Angstrom unit.

12. Correct option: (C) $10^{-8}\,\text{sec}$**.**

$1\text{shake} = 10\,\text{Nanoseconds}$

$1\text{shake} = 10^{-8}\,\text{sec}$

An informal metric unit of time known as a shake is used to measure the timing of various nuclear reactions in nuclear physics. It is roughly equal to the length of a nuclear chain reaction involving quick neurons. It can be used in circuits as well.

13. Correct option: (C) $1.67\times10^{-27}\,\text{kg}$**.**

$1\text{A.M.U} = 1.67\times10^{-27}\,\text{kg}$

Its mass is one-twelfth that of an atom of carbon-12.

14. Correct option: (B) 1000 kg.

$1\text{ mtertic tonne} = 1000\text{kg}$

Although the S.I. standards refer to the metric tonne as the "tonne," the United States government prefers to use the term "metric tonne."

15. Correct option: (B) 14.59kg.

$1\text{slug} = 14.59\text{kg}$

Based on normal gravity, the international foot, and the avoirdupois pound, one slug weighs 32.1740 lb (14.59390 kg). An object with a mass of 1 slug would weigh around 32.2 lbf, or 143 N, at the Earth's surface.

16. Correct option: (C) 454 grams.

$1\text{pound} = 454\text{grams}$.

A pound is a unit of weight that is roughly equivalent to 454 grammes. In the FPS system, it serves as the unit of weight measurement.

17. Correct option: (B) $10^5\,\frac{\text{N}}{\text{m}^2}$**.**

$1\text{atomsphere} = 10^5\,\frac{\text{N}}{\text{m}^2}$

The SI unit of pressure, known as a pascal (Pa), is known as newtons per square metre $\frac{\text{N}}{\text{m}^2}$.

The pressure produced by a one newton force applied over a one square metre area is known as a pascal.

As a result, when we state that 1 atmosphere is equal to $10^5\,\frac{\text{N}}{\text{m}^2}$, we imply that $10^5\,\frac{\text{N}}{\text{m}^2}$ pascals or newtons per square metre is the equivalent to 1 atmosphere's pressure.

18. Correct option: (B) .

$1\text{bar} = 10^5\text{pa}$

In Europe and other regions that use the metric system, the unit of measurement known as BAR is 100000 pascals. It is roughly equivalent to the atmospheric pressure at sea level on Earth.

19. Correct option: (C) 756W.

$1\text{horsepower} = 746\text{W}$

The foot-pound-second (fps) or English system's horsepower (hp) unit is occasionally used to measure how quickly mechanical energy is used up.One horsepower is about equal to 746 watts (W) or 0.746 kilowatts (kW).

20. Correct option: (D) $1.6\times10^{-19}\text{J}$**.**

$1\text{e.v}=1.6\times10^{-19}\text{J}$

A nuclear energy unit called an electron volt (eV) is equal to $1.6\times10^{-19}\text{J}$, which is the amount of energy that an electron with a charge of $1.6\times10^{-19}\text{C}$ can absorb when it is accelerated across a potential difference of one volt.

21. Correct option: (B) 10^{8}gauss

$1\text{weber}=10^{8}\text{gauss}$

Weber is the magnetic flux unit in the SI.

22. Correct option: (A) 10^{4}gauss**.**

$1\text{Tesla}=10^{4}\text{gauss}$

Gauss or Tesla units are used to express magnetic induction. The Tesla of the International System of Units (SI) is equivalent to the gauss from the Centimetre Gramme- Second (CGS) system.

23. Correct option: (B) $\frac{10^{3}}{4\pi}\frac{\text{A}}{\text{m}}$**.**

$1\text{oversted}=\frac{10^{3}}{4\pi}\frac{\text{A}}{\text{m}}$

The coherent derived unit of the auxiliary magnetic field H in the centimetre-gram-second (CGS) system of units is the oersted (symbol Oe).It is the same as 1 dyne for every maxwell.

24. Correct option: (C) 10^{-28}m^2**.**

$1\text{barn}=10^{-28}\text{m}^2$

A barn is a measure of area (symbol b). It was initially used in nuclear physics to express the cross sections of nuclei and nuclear reactions. Nowadays, it is used in all branches of high energy physics to express the cross sections of any scattering process and is best understood as a representation of the likelihood that two small particles will interact. The area of a barn is 10 28 m2 (100fm 2), which is roughly equivalent to the cross-sectional area of a nucleus.

25. Correct option: (D) poise.

In memory of French biologist Jean Léonard Marie Poiseuille, the cgs unit of viscosity is poise (P). Additionally, it is frequently used as centipoise (cP), especially in ASTM standards.

26. Correct option: (C) 365 days.

The solar year, also known as the tropical year or the year of the seasons, is the interval between two consecutive vernal equinoxes (the period when the Sun appears to cross the celestial equator travelling north). It is 365 days, 5 hours, 48 minutes, and 46 seconds long.

27. Correct option: (C) 4.

A leap year is a year that is divisible by 4. As far as we are aware, a leap year is one that is divisible by 4. Additionally, a number is 4 dividable if the sum of its digits in the units and tens places is also 4.

Only 2004 is therefore divisible by 4.

28. Correct option: (B) Kelvin.

According to the International System of Units, Kelvin, denoted by the letter K, is the S.I. unit of temperature. In the fields of science and engineering, the Kelvin scale is commonly acknowledged or utilised.

29. Correct option: (B) 30.48 cm.

1 foot is equal to approximately 30.48 centimetres.

30. Correct option: (B) lunar month.

In roughly 27 days, the moon makes one revolution around the planet. The sidereal month encompasses this time frame. In 29.5 days, the moon goes through all of its phases. The synodic month is the length of time needed to complete all of the cycles.

LEVEL: II

1. Correct option: (A) Shake.

The shake is not the unit of length. An informal metric unit of time known as a shake is used to measure the timing of various nuclear reactions in nuclear physics. It is roughly equal to the length of a nuclear chain reaction involving quick neurons.

2. Correct option: (B) 4min.

A solar day is the length of time it takes for the earth to revolve around the sun on its axis. A sidereal day is the length of time it takes for the earth to revolve around a distant star on its axis. The time difference between solar and sidereal days is 4 minutes. Due to the earth's own motion (revolution) around the sun, the solar day is longer because it requires a bigger angle of rotation to maintain the same location with respect to the sun. Its location in relation to the distant star remains essentially constant.

3. Correct option: (D) Fermi.

In Fermi, nuclear radii are measured. $1\text{fermi}=10^{-15}\text{m}$

Nuclear radii are measured using the Fermi unit, although this unit is no longer in use. Femtometre is the preferred unit of measurement instead of Fermi. The letter fm serves as its emblem. A quadrillionth of one Fermi is equivalent to 10^{-15}m, or 1 Fermi.

4. Correct option: (A) S.I.

The SI unit system is based on more than just mass, length, and time.

The seven fundamental units of this unit system are the mole, the mass, the length, the time, the temperature, the luminous intensity, and the current.

5. Correct option: (C)1852m.

$1\text{Nautical mile} = 1852\text{m}$

A nautical mile is a unit of length used for defining territorial seas and for usage in air, sea, and space navigation.

6. Correct option: (B) bar.

The pressure is measured in bar. Therefore, bar cannot be used to measure length.

A suitable unit of measurement for the pressures produced by low frequency vapour cloud explosions that are frequently taken into account as part of accidental loading risk analyses is the bar. Outside of the United States, bars frequently discuss turbocharger boost in the automotive industry. Tyre pressure is frequently stated in bars.

7. Correct option: (B) kilogram.

One of the International System of Units' (SI) seven base units is the kilogramme (kg). Because it is used to measure mass, a fundamental physical quantity, it is regarded as a fundamental unit. The SI (International System of Units) Unit is the kilogram. It is a Metric System unit, and the 10-base system is all that is used in the calculation. The fundamental

units are those that the International System of Units defines. These units are referred to as "fundamental units" since they are independent of all other units.

8. Correct option: (B) 100 years.

$1\text{century} = 100 \text{ years}$

In historical and chronological contexts, a century is defined as a span of 100 years and is frequently used as a unit of measurement. It denotes a length of time that is commonly understood and employed as a unit in several disciplines, such as history, demography, and cultural studies.

9. Correct options: (A), (B) and (C).

Explanation: (A) A solar day is the length of time it takes for the Earth to complete one rotation of its axis with respect to the Sun. Between two successive noons is when it occurs. Since it symbolises time, it has a unit of time.

(B) The interval between two subsequent new or full moons is known as a lunar month. A lunar month is typically 29.5 days long on average. Its actual duration fluctuates over the course of the lunar year. It has a unit of time since it signifies time.

(C) A leap year is one in which there is an extra day added to the year so that it is in synchrony with the astronomical or seasonal year. Instead of the typical 365 days, a leap year has 366 days, with February 29 serving as an extra day. Once every four years, it happens. It has a unit of time since it signifies time.

(D) A light-year is a measurement of length. It is the maximum distance light can cover in a year. The speed of light is around 300,000 kilometres per second. It may therefore cover roughly 10 trillion kilometres in a single year. It is not a time unit.

So, the mean solar day , Lunar month and leap year are unit of time.

Correct options: (A),(B) and (C).

10. Correct options: (A),(B) and (C).

Explanation:

(A) The International System of Units (SI) uses the steradian (symbol: sr) or square radian as the unit of solid angle. It is similar to the radian, which measures planar angles, and is used in three-dimensional geometry.

(B) One mole (symbol mol) comprises precisely 6.02214×10^{23} elementary entities. The mole is a unit of measurement for substance amount (symbol n). The International System of Units (SI) uses it as a foundation unit.

(C) In the International System of Units, the Candela serves as the fundamental unit of luminous intensity. Candela is a unit of measure for the luminance emitted by a point light source per unit solid angle in a specific direction. One of the seven SI units, it.

(D) 'Metre', 'Kilogramme', and 'Second' are the SI units for length, mass, and time, respectively. Although length, mass, and time can all be described in terms of other units, these are the ones that are universally accepted.

So, Correct options: (A),(B) and (C).

11. Correct options: (A) ,(B),(C) and (D).

Explanation:

(A) The pound is a weight measurement unit that is frequently used in the US. Weight, on the other hand, is a measurement of the force of gravity acting on an object, and it varies depending on the location. On the other hand, mass is a fundamental feature of matter and is constant wherever it is. As a result, the pound is not regarded as a common unit of mass.

(B) Slug is a term for a unit of mass that is frequently used in engineering and physics in the United States. It is the mass that accelerates by a square of one foot per second when a

force of one pound is applied. The slug, however, is not an accepted mass unit under the International System of Units (SI).
(C) Atomic mass units, or AMUs, are a type of mass unit frequently used in atomic and nuclear physics. It is specified as being one-twelfth of a carbon-12 atom's mass, or around $1.66\times10^{-27}\text{kg}$. Atomic and subatomic particle masses are expressed using the atomic mass unit.
(D) Quintals are a type of mass measurement that are frequently employed in several nations, especially in agriculture. It is equivalent to 220.46 pounds or 100 kilogrammes. Although it is a legitimate mass unit, it is not extensively used internationally and is not regarded as a standard SI unit.
So, Correct options: (A) ,(B),(C) and (D).

12. Correct option: (A),(B),(D)

$1\text{quintal}=100\text{kg}$

$1\text{quintal}=10^{5}\text{grams}$

$1\text{quintal}=10^{8}\text{milligrams}$

So, Correct option: (A),(B),(D).

13. Correct option: (A) 10^{-1}m **.**

$1\text{deci meter}=10^{-1}\text{m}$

One metre (1 dm) is equal to one tenth of a metre because the word "Deci" means "one tenth." One tenth of a metre, or 100 cm, is equal to 10 cm. Ten cm are equal to one decimeter.

14. Correct option: (C) 10^{-3} **.**

$1\text{milliseconds}=10^{-3}\text{seconds}$

A millisecond is a measure of time in the International System of Units that is equal to 1000 microseconds and one thousandth of a second (0.001 or 10^{-3} or $\frac{1}{1000}$) of a second.

15. Correct option: (B) 10^{-12} **.**

Pico is a unit of measurement that is equivalent to 10^{-12} .
It represents a trillionth. It is a metric unit prefix and is represented by the letter p.
Pico metres correspond to 10^{-12} units.

16. Correct option: (C) Area.

Area is not a fundamental quantity. It is a derived number that is computed by multiplying two lengths together, such as when calculating the area of a rectangle by multiplying its length and width.Basic quantities that are independent of other physical quantities are known as fundamental quantities. Example Temperature, mass, time, and length.
Derived quantities are those that can be derived from basic quantities. For instance, volume, speed, and area.

17. Correct option: (D) Electric current.

Electric current is a fundamental quantity due to its independence, SI unit, direct measurement, conservation of quantity, and universality. It is a fundamental electrical quantity that is used to define other electrical quantities and is crucial to comprehending how electrical circuits and power systems behave.

18. Correct option: (C) 1 kelvin.

Water's 'triple point' temperature is 273 K. The temperature and pressure at which a material's three phases—gas, liquid, and solid—coexist in thermodynamic equilibrium are

known as the triple point of that substance. Water has a triple point at 273.16 K and 611.73 Pa. This is the starting point of the Kelvin scale and it equals $0.01^{\circ}\mathrm{C}$, which is the temperature at which water freezes.

19. Correct Matching is: a-R; b-S; c-P; d-Q

a) 1 angstrom = 10^{-10}m

b) 1 mile = 1.6km

(c) 1 light year = 9.5×10^{12}km

(d) 1 barn = $10^{-28}\mathrm{m}^2$

So, the correct matching is (a)-(R), (b)-(S),(c)-(P), (d)-(Q)

20. Correct Matching is: a-R; b-P; c-S; d-Q

1 kilometer = 1000 m

b) 1 day = 86,400 second

c) 1 metric ton = 10 quintal

d) 1 year = 365 days

So, the correct matching is (a)-(R), (b)-(P), (c)-(S), (d)-(Q).

21. Correct Answer is: $x=4$

$1\mu\mathrm{m}=10^{x}\mathrm{A}^{\circ}$

The following conversion factor can be used to change from micrometres (μm) to angstroms (A°): $1\mu\mathrm{m}=10000\mu\mathrm{m}$

We know that, $1\mu\mathrm{m}=10^{4}\mathrm{A}^{\circ}$

So, the value of $x=4$

22. Correct Answer is: the value of, $x=9$

$1\mathrm{nm}=10^{-x}\mathrm{m}$

We know that, $1\mathrm{nm}=10^{-9}\mathrm{m}$

Therefore, $x=9$.

Dimensions
LEVEl-1

1. Correct option: (B) Pascal.

One newton per square metre is equivalent to one pascal $\left(\frac{\mathrm{N}}{\mathrm{m}^2}\right)$ (abbreviated as Pa), the SI unit of pressure.

2. Correct option: (C) Thrust.

The dimensional formula of force = MLT^2

The dimensional formula of thrust = MLT^2

So, the dimensional formula of force is equal to dimensional formula of thrust.

3. Correct option: (B) Work and Energy.

We know that, Work = Force

$\mathrm{Work}=\mathrm{Force}\times\mathrm{displacement}$

$W=F\times D$

$W=\left[MLT^{-2}\right]\left[L\right]$

$W=\left[ML^{2}T^{-2}\right]$

Or,

Energy is given by as,

$$K = \frac{1}{2}mv^2 = [M]\left[LT^{-1}\right]^2 = \left[ML^2T^{-2}\right]$$

The work and energy pairs of physical quantities have the same dimensional formula.
So, Correct option: (B) Work and Energy.

4. Correct option: (A) $\left[\mathrm{ML^2T^{-2}}\right]$.

Calorie is unit of energy, which is given by as,

Energy=F.r

So, the dimensions can be calculated as,

$$\text{Calorie}=\left[MLT^{-2}\right][L]=\left[ML^2T^{-2}\right]$$

So, the dimensions of calorie is $\left[\mathrm{ML^2T^{-2}}\right]$.

5. Correct option: (C) Gravitational constant.

The dimensional formula of gravitational constant is $\left[\mathrm{M^{-1}L^3T^{-2}}\right]$.

6. Correct option: (C) time.

Impulse is given by as,

$$\text{Impulse} = F\times t = \left[M^1L^1T^{-2}\times T^1\right]=\left[M^1L^1T^{-1}\right]$$

$$\text{Angular veolcity} = \omega = \left[M^\circ L^\circ T^{-1}\right]$$

So, in both quantities, time has the same dimension.

7. Correct option: (B) $\left[M^1L^0T^{-2}\right]$.

$$\left[\mathrm{Jm^{-2}}\right]=\frac{\left[M^1L^2T^{-2}\right]}{\left[L^{-2}\right]}=\left[M^1L^0T^{-2}\right]$$

So, the dimensional formula of $\left[\mathrm{Jm^{-2}}\right]$ is $\left[M^1L^0T^{-2}\right]$.

8. Correct option: (A) radians.

Radians are used to measure a body's angular displacement (θ). Rad is the symbol for it.

9. Correct option: (C) $\left[T^{-1}\right]$.

The angular velocity of body is given by as,

Pressure×Volume=Work $\omega = \dfrac{V}{r}$

Now dimension of angular velocity is given by as,

$$\omega = \frac{\left[LT^{-1}\right]}{L}$$

$$\omega = \left[T^{-1}\right]$$

So, the dimensions formula of angular velocity is $\left[T^{-1}\right]$.

10. Correct option: (A) Work.

Dimensions of work $=\left[ML^2T^{-2}\right]$

Now, Pressure×volume$=\left[ML^{-1}T^{-2}\times L^3\right]=\left[ML^2T^{-2}\right]$

So, Pressure×Volume=Work

11. Correct option: (C) Mass and Time.

Linear momentum is given by as,

$P = mv = \left[M^1 L^1 T^{-1} \right]$

Angular momentum is given by as,

$L = mvr = \left[M^1 L^2 T^{-1} \right]$ $L = mvr = \left[M^1 L^2 T^{-1} \right]$

So, In terms of mass and time, linear and angular momenta have same dimensions.

12. Correct option: (D) Radian.

The radian, a dimensionless unit used to measure angles, is not a physical quantity. It is a ratio of an arc's length to a circle's radius. Radians are pure numbers since they don't have any physical dimensions.

13. Correct option: (B) Energy.

One thousand watt-hours make up a kilowatt-hour, or kwh, which serves as a measure of energy. The total energy in kilowatt-hours is the product of the power in kilowatts and the time in hours if the energy is being transmitted or used at a steady rate (power) over a specific period of time.

14. Correct option: (D) Volt.

A unit of work is not a volt. It is a voltage unit.

When a force acts on a body, it is said to be doing work in physics if the point of application is moved in the force's direction.

15. Correct option: (D) Joule.

A newton metre or newton-meter is a derived unit that is defined as the work done, or energy required, to apply a force of one newton for a distance of one metre. It is represented by the symbols N.m.

16. Correct option: (B) Ampere.

The S.I unit of electric current is Ampere.

17. Correct option: (D) Pascal metre.

The dimension of energy is given by as,

$E = \left[M^1 L^2 T^{-2} \right]$

The dimension of pascal metre is given by as,

$\text{Pasacal metre} = \left[M^1 L^{-1} T^{-2} \times L^1 \right] = \left[M^1 L^0 T^{-2} \right]$

Watt second, Kilowatt hr, newton mt., have the unit of energy.

Which is $\left[M^1 L^2 T^{-2} \right]$.

Hence, Pascal metre does not have the unit of energy.

18. Correct option: (A) kelvin.

According to the International System of Units, Kelvin, which is denoted by the letter K, is the SI unit of temperature.

19. Correct option: (D) coefficient of friction.

Because it is the ratio of one force to another, the coefficient of friction has no units. There is no dimension and no unit for the coefficient of friction. Since it is a scalar, the physical quantity is unaffected by the force's direction. The things creating friction will determine the coefficient of friction.

20. Correct option: (C) Relative density.

Because it is a ratio of two-dimensionally identical quantities, the relative density has no defined unit. Notably, when we divide a quantity in two dimensions, they cancel out and

the resulting ratio has no dimensions. The density of a fluid as compared to a reference fluid (water) is known as its specific density.

21. Correct option: (B)

The dimensions of force is, $F = \left[\mathrm{M^1L^1T^{-2}} \right]$

The dimensions of acceleration is , $a = \left[\mathrm{M^0L^1T^{-2}} \right]$

Divide the unit of force by the unit of acceleration, we get

$$\frac{F}{A} = \frac{\mathrm{M^1L^1T^{-2}}}{\mathrm{M^0L^1T^{-2}}} = \left[\mathrm{M^1L^0T^0} \right] = \mathrm{Mass}$$

If the unit of force is divided by the unit of acceleration, then derived unit is mass.

22. Correct option: (B) Power.

Dimension formula of joule (energy) $\left[ML^1T^{-2} \right]$ $J = \left[M^1L^2T^{-2} \right]$

Dimension formula of second, $s = \left[T \right]$

Now, dimension formula of, $\frac{\mathrm{J}}{\mathrm{s}} = \frac{\left[M^1L^2T^{-2} \right]}{\left[T \right]} = \left[M^1L^2T^{-3} \right]$

Dimension formula of power is given by as,

$$\text{Power} = f \times v = \left[MLT^{-2} \right]\left[LT^{-1} \right] = \left[ML^2T^{-3} \right]$$

So, $\frac{\mathrm{J}}{\mathrm{s}}$ is the unit of power.

23. Correct option: (A) power.

Dimension formula of power = $\left[ML^2T^{-3} \right]$

Dimension formula of acceleration = $\left[LT^{-2} \right]$

Dimension formula of force constant= $\left[ML^0T^{-2} \right]$

Dimension formula of stress= $\left[ML^1T^{-2} \right]$

Hence. Power has dimensions of 2 in length.

24. Correct option: (D) Energy.

The right response is energy. The unit of energy most frequently used in atomic and nuclear physics is the electron volt. An electron volt is the amount of energy that a single electron gains when it is charged and transferred over an electric potential difference of one volt.

25. Correct option: (B) Energy per unit volume.

The dimension formula of pressure is given by as,

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Pressure} = \left[ML^{-1}T^{-2} \right]$$

Now dimension formula of energy per unit volume,

$$\text{Pressure} = \frac{\text{Energy}}{\text{Voulme}} = \left[\frac{ML^2T^{-2}}{L^3} \right] = \left[ML^{-1}T^{-2} \right]$$

So, the energy per unit volume is dimensionally correct.

26. Correct option: (B) Momentum.

Dyne is unit of force.

So, the dimension formula of Dyne -s is given by as,

$Dyne-s=\left[MLT^{-2}\right]\left[T\right]=\left[MLT^{-1}\right]$

Dimension formula of force = $\left[MLT^{-2}\right]$

Dimension formula of momentum = $\left[MLT^{-1}\right]$

Dimension formula of energy= $\left[ML^2T^{-2}\right]$

Dimension formula of power = $\left[ML^2T^{-3}\right]$

So, the Dyne-seconds is unit of momentum.

27. Correct option: (A) momentum and impulse

Dimension formula of momentum = $\left[MLT^{-1}\right]$

Dimension formula of impulse= $f\times t=\left[MLT^{-1}\right]$

Dimension formula of energy= $\left[ML^2T^{-2}\right]$

Dimension formula of pressure is,

$\text{Pressure}=\dfrac{\text{Force}}{\text{Area}}$

$\text{Pressure}=\left[ML^{-1}T^{-2}\right]$

Dimension formula of power = $\left[ML^2T^{-3}\right]$

The dimension formulas of momentum and impulse are the same.

28. Correct option: (A) $\left[ML^2T^{-2}\right]$ **.**

Dimension formula of energy is given by as

Energy= $\left[ML^2T^{-2}\right]$

29. Correct option: (C) 8times.

Given:

$\text{m}=2m$

$\text{kg}=\dfrac{\text{kg}}{2}$

$\text{s}=\dfrac{\text{s}}{2}$

We know that energy in term of force and displacement is given by as,

Unit of energy = $\dfrac{\text{kgm}^2}{\text{s}^2}$

Now,

$E=\dfrac{\text{kgm}^2}{\text{s}^2}$

$E=\dfrac{\text{kg}}{2}\times\dfrac{\left(2\text{m}^2\right)}{\left(\dfrac{\text{s}}{2}\right)^2}$

$E=8\left(\dfrac{\text{kgm}^2}{\text{s}^2}\right)$

$E=8E$

So, the energy is 8 times its original value.

30. Correct option: (A) $M^2L^0T^{-2}, MLT^{-2}$.

Given:

$$F = \frac{X}{\text{Linear density}} + Y$$

Dimension formula of force= MLT^{-2}

Dimension formula od linear density= ML^{-1}

Therefore,

$$MLT^{-2} = \frac{X}{ML^{-1}}$$

$$X = M^2L^0T^{-2}$$

$$Y = f = MLT^{-2}$$

So, the dimension formula of X is $M^2L^0T^2$ and the dimension formula of Y is MLT^{-2}.

LEVEL -II

1. Correct option: (C) Force and displacement.

Given: product of P and Q is, $P \times Q = \left[ML^2T^{-2}\right]$

Now, let $P = M^AL^BT^C, Q = M^1L^bT^c$

$P \times Q = M^{A+a}L^{B+b}T^{C+c} = M^1L^2T^{-2}$

A+a=1.B=b=2,C+c=2

$\Rightarrow \frac{P}{Q} = M^{A+a}L^{B+b}T^{C+c} = MT^{-2}$

A-a=1,B-b=0,C-c=0

ÞA=1,a=0,B=1,b=1,C=-2,c=0

So,

$P = MLT^{-2}$=Force

$Q = M^0L^1T^0$=displacment

So, P and Q respectively are force and displacement.

2. Correct option: (B) 16.

Given:

$F = 4F$

$L = 2L$

$T = \frac{1}{2}T$

Power is given by as,

$$P = F.v = \frac{F \times L}{T}$$

Now,

$$P_1 = \frac{4F \times 2L}{\frac{T}{2}}$$

$$P_1 = 16P$$

So, the unit of power is changed by factor 16.

3. Correct option: (B) 2.

Given:

$$(\text{force})^x = \frac{(\text{mass})^2 (\text{radius})^2}{(\text{time period})^4}$$

$$J = [ML\,T]$$

We have to find value of $\frac{J}{2E}$.

$$Mr^2 = \text{M}^1\text{L}^2$$

Dimension formula of force = $[MLT^{-2}]$

Therefore,

$$\left(\text{MLT}^{-2}\right)^x = \frac{(\text{mass})^2 (\text{radius})^2}{\text{T}^4}$$

$$\left(\text{MLT}^{-2}\right)^x = \frac{(\text{m})^2 (\text{L})^2}{\text{T}^4}$$

$$\left(\text{MLT}^{-2}\right)^x = \text{M}^2\text{L}^2\text{T}^{-4}$$

$x = 2$

So, the value of x is 2.

4. Correct option: (B) Energy.

Dimension of P =Dimensions of $\frac{a}{V^2}$ .....(1)

Dimension of V =Dimensions of b .....(2)

Using equation (1) and (2),

Dimensions of a =Dimensions $P \times$ Dimensions of V^2 .

Dimensions of a = $\left[\text{ML}^{-1}\text{T}^{-2}\right] \times \left[\text{L}^3\right]^2$

Dimensions of a = $\left[\text{ML}^5\text{T}^{-2}\right]$.

Unit of a =Unit of $P \times$ Unit of V^2

Unit of a = $= \frac{\text{N}}{\text{m}^2} \times \text{m}^6 = \text{Nm}^4$

Using equation (2),

$[b] = [V] = \left[\text{M}^0\text{L}^3\text{T}^0\right]$

unit of b =unit of $V = m^3$ =volume.

So the dimensions of $\frac{a}{b}$ is given by as,

$$\frac{a}{b} = \frac{\left[\text{ML}^5\text{T}^{-2}\right]}{\left[\text{M}^0\text{L}^3\text{T}^0\right]}$$

$$\frac{a}{b} = \left[\text{ML}^2\text{T}^{-2}\right]$$

So, the dimensions of $\frac{a}{b}$ are same as energy.

5. Correct option: (A) moment of inertia.

Dimension formula of moment inertia= $Mr^2 = \mathrm{M^1L^2}$

Dimension formula of, $J = \left[\mathrm{ML^2T^{-1}}\right]$

Dimension formula of, $E = \left[\mathrm{ML^2T^{-2}}\right]$

Now dimension formula of $\frac{J}{2E}$ is given by as,

$$\frac{J^2}{2E} = \frac{\left(\mathrm{M^1L^2T^{-1}}\right)^2}{2\left(\mathrm{M^1L^2T^{-2}}\right)}$$

$$\frac{J}{2E} = \frac{\mathrm{M^2L^4T^{-2}}}{\mathrm{ML^2T^{-2}}}$$

$$\frac{J}{2E} = \mathrm{ML^2T^0}$$

So, the $\frac{J}{2E}$ has the dimensions of moment of inertia.

6. Correct option: (B) momentum.

The dimensions of Planck constant, $h = \left[\mathrm{ML^2T^{-1}}\right]$

The dimensions of wavelength, $\lambda = \left[\mathrm{L}\right]$

Therefore, the dimensions of $\frac{h}{\lambda}$ is given by as,

$$\frac{h}{\lambda} = \frac{\left[\mathrm{ML^2T^{-2}}\right]}{\left[\mathrm{L}\right]} = \left[\mathrm{MLT^{-1}}\right] = \text{momentum}$$

So, $\frac{h}{\lambda}$ has the dimensions of momentum.

7. Correct option: (A) $\left[\mathrm{F^0aT^2}\right]$.

The length is expressed as follows:

$\mathrm{L=KF^{\alpha}A^{\beta}T^{\gamma}}$

The dimension formula of length is $\left[\mathrm{M^0L^1T^0}\right]$

The dimension formula of force is $\mathrm{M^1L^1T^{-2}}$.

The dimension of acceleration is , $a = \left[\mathrm{M^0L^1T^{-2}}\right]$

The dimension of time is $\left[\mathrm{M^0L^0T^1}\right]$.

Substitute all the value we get,

$$\left[\mathrm{M^0L^1T^0}\right] = \left[\mathrm{M^1L^1T^{-2}}\right]^{\alpha}\left[\mathrm{M^0L^1T^{-2}}\right]^{\beta}\left[\mathrm{M^0L^0T^1}\right]^{\gamma}$$

$$\left[\mathrm{M^0L^1T^0}\right] = \left[\mathrm{M^{\alpha}L^{\alpha+\beta}T^{-2\alpha-2\beta+\gamma}}\right]$$

$\Rightarrow \alpha=0, \alpha+\beta=1.$

$\Rightarrow \beta=1$

$\Rightarrow -2\alpha-2\beta+\gamma=0$

$\Rightarrow \gamma=2$

So, $L=\left[F^0\, a\, T^2\right]$

Therefore, the dimensional formula for length is $\left[F^0 a T^2\right]$.

8. Correct option: (C) LT^{-2}, L, T **.**

In accordance with the dimensional homogeneity principle

Dimension of v =dimension of at =dimension of $\frac{b}{t+c}$

Dimension of a = $\frac{[v]}{[t]}=\frac{LT^{-2}}{T}=\left[LT^{-2}\right]$

Dimension of b = $[v][t]=\left[LT^{-1}T\right]=[L]$

Dimensions of c = $[t]=[T]$

So, the dimensions of a , b and c are respectively LT^{-2}, L, T .

9. Correct options: (A), (B) and (C)

The dimension formula of distance, displacement, wavelength is same. Which is $\left[M^0L^1T^0\right]$.

The dimension formula of speed is $\left[M^0L^1T^{-1}\right]$.

10. Correct option: (B and (C).

Given:

$\text{Force}=100\text{dyne}=\left[MLT^{-2}\right]$

$\text{velocity}=10\frac{\text{cm}}{\text{s}}$

$\text{Energy}=500\text{erg}$

$\frac{\text{Energy}}{\text{force}}=\frac{\left[ML^2T^{-2}\right]}{\left[MLT^{-2}\right]}=[L]=\frac{500}{100}=\text{cm}$

$\frac{\text{Energy}}{(\text{velcoity})^2}=M^1=\frac{500}{(10)^2}=5\text{gm}$

So, the unit of length is 5cm and the unit of mass is 5gm.

11. Correct option: (A) and (B).

Let $M=(\text{some number})(V)^a(F)^b(T)^c$ .....(1)

Dimensions of mass=$\left[M^1L^0T^0\right]$ .....(2)

By equating the dimensions of both sides, we obtain using equation (1) and (2),

$M^1L^0T^0=(1)\left[L^1T^{-1}\right]^a\left[M^1L^1T^{-2}\right]^b\left[T^1\right]^c$

$= M^bL^{a+b}T^{-a-2b+c}$

Get, a=-1,b=1,c=-1

Therefore,

$M=(\text{some number})\left(V^{-1}F^1T^1\right)$

$[M]=\left[V^{-1}F^1T^1\right]$

Similarly for Energy,

$[E]=[\text{Some number}]=[\text{V}]^{a}[\text{F}]^{b}[\text{T}^{c}]$ .....(3)

Dimensions of energy, $E=[\text{ML}^{2}\text{T}^{-2}]$ .....(4)

By equating the dimensions of both sides, we obtain using equation (3) and (4),

$[\text{ML}^{2}\text{T}^{-2}]=[\text{M}^{0}\text{L}^{0}\text{T}^{0}][\text{LT}^{-1}]^{a}[\text{MLT}^{-2}]^{b}[\text{T}]$

$[\text{ML}^{2}\text{T}^{-2}]=[\text{M}^{b}\text{L}^{a+b}\text{T}^{-a-2b+c}]$

1=b;2=a+b;-2=-a-2b+c

Get, a=1;b=1,c=1

Therefore,

$E=(\text{some number})=\text{V}^{1}\text{F}^{1}\text{T}^{1}$

$[E]=[\text{V}^{1}][\text{F}^{1}][\text{T}^{1}]$

So, the dimensions formula of mass is $[\text{V}^{-1}\text{FT}]$.

The dimensions formula of energy is $[\text{VFT}]$.

12. Correct option: (A), (B) and (C).

The dimensions formula of distance and displacement is $[\text{M}^{0}\text{L}^{1}\text{T}^{0}]$.

The speed and velocity units are the same $\frac{\text{m}}{\text{s}}$.

The dimensions formula of speed and velocity is $[\text{ML}^{1}\text{T}^{-1}]$

The dimensions formula of work and energy is $[\text{M}^{1}\text{L}^{2}\text{T}^{-2}]$.

The dimensions formula of Planck constant is $[\text{ML}^{2}\text{T}^{-1}]$.

The dimensions formula of torque is $[\text{M}^{1}\text{L}^{2}\text{T}^{-2}]$.

So, the dimensions formula of Planck constant and torque are different.

13. Correct option: (B) $[\text{M}^{1}\text{L}^{2}\text{T}^{-3}]$.

Dimension formula of power is given by as,

$\text{Power}=\text{f}\times\text{v}=[\text{MLT}^{-2}][\text{LT}^{-1}]=[\text{ML}^{2}\text{T}^{-3}]$

14. Correct option: (D) $[\text{MLT}^{-1}]$

Dimension formula of impulse= $\text{f}\times\text{t}=[\text{MLT}^{-1}]$

15. Correct option: (DI I.

[I] stands for the dimensional formula of electric current.

Electric current is a fundamental quantity that is unrelated to any other physical quantities, according to the dimensional formula [I]. It has characteristics with other basic concepts like length ([L]), time ([T]), and mass ([M]).

16. Correct option: (D) $[\text{W}^{1}\text{F}^{-1}\text{v}^{-1}]$.

The work done is given by as,

$\text{Work done}=\text{Force}\times\text{distance}$

$W=F\times S$

$\text{and distnace}(s)=\text{velocity}(v)\times\text{time}(t)$

Now, work done is given by as,

$W = F.v.t$

$t = \frac{W}{F.v}$

$[T] = \left[W^{1}F^{-1}v^{-1} \right]$

So, the dimensional formula of time is $\left[W^{1}F^{-1}v^{-1} \right]$.

17. Correct option: (B) $\left[F^{1}M^{-1}T^{2} \right]$**.**

Let Length=$F^{a}M^{b}T^{c}$

Dimensions of length is given $\left[M^{0}L^{1}T^{0} \right]$.

Therefore,

$\left[M^{0}L^{1}T^{0} \right]=F^{a}M^{b}T^{c}$

$\left[M^{0}L^{1}T^{0} \right]=\left(MLT^{-2} \right)^{a} M^{b}T^{c}$

$\left[M^{0}L^{1}T^{0} \right]=M^{a+b}L^{a}T^{-2a+c}$

a+b=-,a=1

b=-1

-2a+c=0

c=2

$\left[M^{0}L^{1}T^{0} \right]=F^{1}M^{-1}T^{2}$

So, the dimension formula of length is $\left[F^{1}M^{-1}T^{2} \right]$.

18. Correct option: (D) $F^{1}L^{-1}T^{2}$**.**

The force is given by as,

$[F]=[ma]=MLT^{-2}$

Thus, we have obtained,

$M=F^{1}L^{-1}T^{2}$

So, the dimensions formula of mass is $F^{1}L^{-1}T^{2}$.

19. Correct Matching is: a-R; b-S; c-Q; d-P

Spring constant, $K = \frac{F}{X} = \frac{M^{1}L^{1}T^{-2}}{L^{1}} = M^{1}L^{0}T^{-2}$

Pascal is unit of pressure, $P = \frac{F}{A} = \frac{M^{1}L^{1}T^{-2}}{L^{2}} = M^{1}L^{-1}T^{-2}$

Hertz is unit of frequency $= \frac{1}{t} = M^{0}L^{0}T^{-1}$

Joule is unit of energy so dimension of joule is $ML^{2}T^{-2}$.

20. Correct Matching is: a-Q; b-P; c-S; d-R

The dimensions formula of work and energy is $\left[M^{1}L^{2}T^{-2} \right]$

So, the dimension of length in work is 2 and the dimension of time in energy is -2.

Dimension formula of density= $M^{1}L^{-3}T^{0}$

So, dimension of length in density is -3.

Dimension formula of impulse= $f \times t=\left[MLT^{-1} \right]$

So, dimension of length in impulse is -1.

21. Correct Answer is: Dimension of time in work is -2.

The dimensions formula of work is $\left[M^1L^2T^{-2}\right]$

So, the dimension of time in work is -2.

22. Correct Answer is: Dimension of mass in force is 1.

The dimension formula of force is $M^1L^1T^{-2}$.

So, the dimension of mass in force is 1.

2. Motion in One Dimension

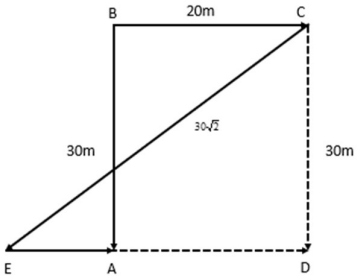
Solutions

LEVEL-I

One Dimension – Motion Along Straight Line

1. Correct option: (C) 10m long west.4

Let AB represent the person's displacement towards the north, BC represent the person's displacement towards the south, and CE represent the person's displacement towards the south-west. Finding the individual's departure from the source is necessary, EA



Let's consider the right angle ECD, using Pythagoras theorem,

$ED^2+CD^2=EC^2\cdots\cdots(1)$

From the diagram, $CD=AB=30m$

$EC=30\sqrt{2}m$

$ED=EA+AD$

$AD=BC=20m$

$\therefore ED=EA+20$

Put the value of ED in equation(1),

$(EA+20)^2+(30)^2=(30\sqrt{2})^2$

$(EA+20)^2+900=1800$

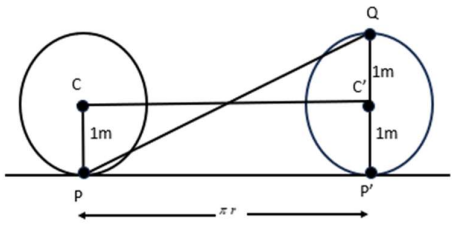
$EA+20=30$

$EA=10m$

So, the displacement of the person from the origin will be 10m along the west.

2. Correct option: (C) $\sqrt{\pi^2+4}$.

Given: $r=1m$



Wheel's linear displacement during a half-revolution= πr

Point P : reaches at P' vertically 2m above the ground after a half rotation.

The displacement of the point which was initially in contact with ground=PQ

$$\text{PQ} = \sqrt{(\pi r)^2 + 2r^2}$$

$$\text{PQ} = r\sqrt{\pi^2 + 4}$$

$$\therefore r = 1\text{m}$$

$$\text{PQ} = \sqrt{\pi^2 + 4}$$

So, the displacement of the point of the wheel initially in contact with the ground is $\sqrt{\pi^2+4}$.

3. Correct option: (B) $\frac{40}{13}$ **.**

For Forward journey:

$$d = 6\text{km}, v_1 = 2.5\frac{\text{km}}{\text{hr}}$$

$$t_1 = \frac{d}{v_1} = \frac{6}{2.5} = 2.4\text{hr}$$

For backward journey:

$$d = 6\text{km}, v_2 = 4\frac{\text{km}}{\text{hr}}$$

$$t_2 = \frac{d}{v_2} = \frac{6}{4} = 1.5\text{hr}$$

The average speed is given by as,

$$V_{avg.} = \frac{\text{Total distnace}}{\text{Total time}}$$

$$V_{avg.} = \frac{v_1 + v_2}{t_1 + t_2}$$

$$V_{avg.} = \frac{6+6}{2.4+1.4}$$

$$V_{avg.} = \frac{40}{13}\frac{\text{km}}{\text{hr}}$$

So the average speed is $\frac{40}{13}\frac{\text{km}}{\text{hr}}$.

4. Correct option: (D) $36\frac{\text{km}}{\text{h}}$ **.**

Given: Velocity for one third distance, $v_1 = 20\frac{\text{km}}{\text{h}}$

Velocity for two third of distance, $v_2 = 60\frac{\text{km}}{\text{h}}$

Now,

Time required to travel a one third of the distance, $t_1 = \frac{\frac{d}{3}}{v_1} = \frac{\frac{d}{3}}{20}$

Time required to travel a two third of the distance, $t_2 = \frac{\frac{2d}{3}}{v_2} = \frac{\frac{2d}{3}}{60}$

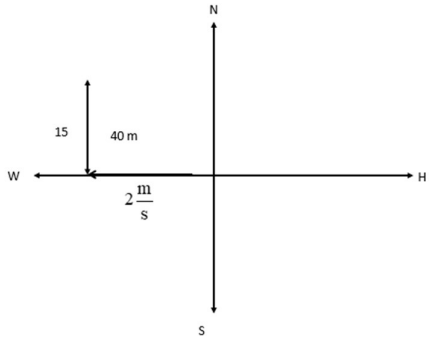
The average velocity is given by as,

$$V_{avg.} = \frac{\text{Total distnace}}{\text{Total time}}$$

$$V_{avg.} = \frac{d}{t_1 + t_2} = \frac{d}{\frac{d}{3} + \frac{2d}{3}} = 36\frac{\text{km}}{\text{h}}$$

So, the average speed is $36\frac{\text{km}}{\text{h}}$.

5. Correct option: (B) $\frac{42}{25}, \frac{6}{5}$**.**



Total time is given by as,

$$\text{Total time} = \frac{30}{2} + \frac{40}{\frac{3}{2}}$$

$$\text{Total time} = 15 + \frac{80}{3} = \frac{125}{3}$$

Distance= 70m

Now, displacement is give by as,

$$\text{displacent} = \sqrt{30^2 + 40^2} = 50\text{m}$$

Now average speed is given by as,

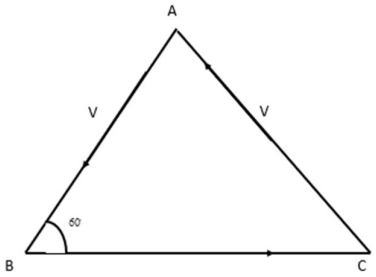
$$\left\langle V_{speed} \right\rangle = \frac{70}{\frac{125}{3}} = \frac{42}{25}\frac{\text{m}}{\text{s}}$$

Now average displacement is given by as,

$$\left\langle V_{Velocity} \right\rangle = \frac{50}{\frac{125}{3}} = \frac{6}{5}\frac{\text{m}}{\text{s}}$$

average speed and average velocity is $\frac{42}{25}\frac{\text{m}}{\text{s}}, \frac{6}{5}\frac{\text{m}}{\text{s}}$.

6. Correct option: (C) $\frac{2a}{3v}$.



Now, Velocity of A is d along AB the velocity of B is along BC.
The Component of velocity along BA is given by as,
Thus, the separation AB decreases at the rate,

$$V-\left(-\frac{V}{2}\right)=\frac{3V}{2}$$

The time required to reduce the separation AB from d to zero is determined by this rate constant.

$$t=\frac{a}{\frac{3V}{2}}$$

$$t=\frac{2a}{3V}$$

So, the person meets after time $\frac{2a}{3v}$.

7. Correct option: (D) 10 sec.

Given:

$v_1=10\text{ms}^{-1}$

$v_2=5\,\text{ms}^{-1}$

The relative velocity of parrot with respect to train is given by as,

$$\Rightarrow\left[v_1-(-v_2)\right]=\left[10-(-5)\right]=15\text{ms}^{-1}$$

Therefore, the time taken by the parrot to cross the train is given by as,

$$\Rightarrow\frac{150}{15}=10\text{s}$$

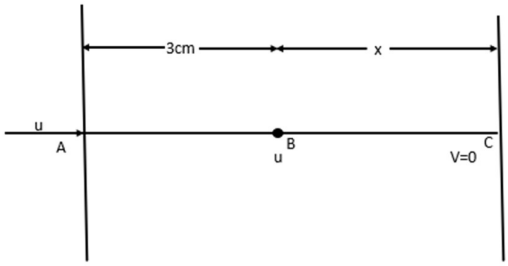
So, the time taken by the parrot to cross the train is 10s.

8. Correct option: (B) 1 cm.

Let the initial velocity of the bullet is u.

After the penetrating 3 cm its velocity becomes $\frac{u}{2}$.

Now using equation of motion,



$v^2 = u^2 - 2as$

$\left(\frac{u}{2}\right)^2 = u^2 - 2a(3)$

$\left(\frac{u}{2}\right)^2 = u^2 - 6a$

$a = \frac{u^2}{8}$

As it advances, it will cross the distance x and come to a stop at point C.

For distance BC, $v = 0, u = \frac{u}{2}, s = x, a = \frac{u^2}{8}$

Now from the equation of motion,

$v^2 = u^2 - 2as$

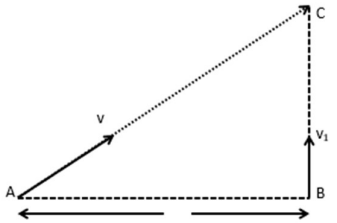
$\left(\frac{u}{2}\right)^2 = u^2 - 2a(3)$

$0 = \left(\frac{u}{2}\right) - 2\left(\frac{u^2}{8}\right).x$

$x = 1\text{cm}$

So, the bullet will penetrate 1cm more.

9. Correct option: (B) $\sqrt{\frac{a^2}{\left(v^2 - v_1^2\right)}}$**.**



After time 't' from the beginning, have the two boys meet at point C. Then

$\text{PR} = vt, \text{QR} = v_1 t$

Consequently, in a right triangle PQR,

$(\text{PR})^2 = (\text{PQ})^2 + (\text{QR})^2$

$\left(v^2 t^2\right) = a^2 + v_1^2 t^2$

$v^2 t^2 - v_1^2 t^2 = a^2$

$t^2\left(v^2 - v_1^2\right) = a^2$

$t = \sqrt{\frac{a^2}{v^2 - v_1^2}}$

$(PR)^2 = (PQ)^2 + (QR)^2$

$(v^2t^2) = a^2 + v_1^2t^2$

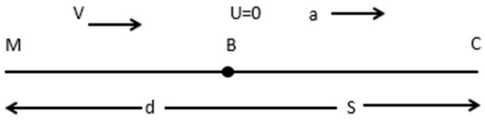
$v^2t^2 - v_1^2t^2 = a^2$

$t^2(v^2 - v_1^2) = a^2$

$t = \sqrt{\frac{a^2}{v^2 - v_1^2}}$

So, the required value of time is $\sqrt{\frac{a^2}{(v^2 - v_1^2)}}$.

10. Correct option: (B) $\sqrt{2ad}$ **.**



Let us consider a passenger going at constant speed v should be allowed to board the bus at location C in time t.

For bus:

Initial speed of bus is, $u = 0$

Now using the equation of motion,

$s = ut + \frac{1}{2}at^2$

$s = 0 + \frac{1}{2}at^2$

Now for passenger, $d + s = vt$

$\therefore d + \frac{at^2}{2} = vt$

$\Rightarrow at^2 - 2vt + 2d = 0$

The roots of above equitation is real and which given by as,

$(-2v)^2 - 4a(2d) \geq 0$

$\Rightarrow v^2 \geq 2ad$

$\Rightarrow v \geq \sqrt{2ad}$

So, the minimum velocity with which the passenger should run towards the bus so as to catch it is $\sqrt{2ad}$.

11. Correct option: (B) $22\,\text{sec}$ **.**

Given: Velocity of car A moving with uniform velocity, $v_A = 44\text{ms}^{-1}$

Velocity of car B is, $v_B = 0\text{ms}$

Acceleration, $a = 2\frac{\text{m}}{\text{s}^2}$

Acceleration is given by as,

$a = \frac{\text{Change in velocity}}{\text{time}}$

$a = \frac{v_A - v_B}{\text{time}}$

$t = \frac{44 - 0}{2}$

$t = 22\,\text{sec}$

$t = \frac{44-0}{2}$

$t = 22\,\text{sec}$

So, the required value of time is $22\,\text{sec}$.

12. Correct option: (D) $18\frac{\text{m}}{\text{s}}$**.**

Given:

Velocity of body, $u = 10\frac{\text{m}}{\text{s}}$.

Acceleration of body is, $a = 2\frac{\text{m}}{\text{s}^2}$

Time, $t = 4\,\text{sec}$

Acceleration is given by as,

$a = \frac{\text{Change in velocity}}{\text{time}}$

$a = \frac{v-u}{t}$

$2 = \frac{v-10}{4}$

$v = 10+8$

$v = 18\frac{\text{m}}{\text{s}}$

So, the velocity after an interval of 4 sec is $18\frac{\text{m}}{\text{s}}$.

13. Correct option: (A) 36m.

Given: Initial velocity of body is, $u = 0\frac{\text{m}}{\text{s}}$

Acceleration of body, $a = 8\frac{\text{m}}{\text{s}^2}$

$n = 5$

As we know, the formula for the distance travelled in nth second is,

$S_n = u + \frac{a}{2}(2n-1)$

So, the distance travelled in 5th second is,

$S_{5^{th}} = 0 + \frac{8}{2}(2\times 5-1) = 36\text{m}$

So, the distance travelled in 5th second is 36m.

14. Correct option: (D) 180m.

First case:

The initial velocity of the car, $u_1 = 10\frac{\text{m}}{\text{s}}$

The final velocity of the car, $v_1 = 0$

Distance, $s = 20\text{m}$

Second case;

The initial velocity of the car, $u_2 = 30\dfrac{\text{m}}{\text{s}}$

The final velocity of car, $v_2 = 0$

Acceleration for first case,
Using equation of motion,

$v_1^2 = u_1^2 + 2as$

$0 = (10)^2 + 2 \times a \times 20$

$a = -\dfrac{10 \times 10}{2 \times 20}$

$a = -2.5\dfrac{\text{m}}{\text{s}^2}$

Now finding the stooping distance foe second case,
Again using the equation of motion,

$v_2^2 = u_2^2 + 2as$

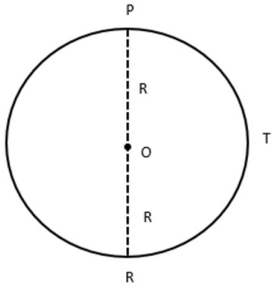
$0 = (30)^2 + 2 \times (-2.5) \times s$

$s = \dfrac{30 \times 30}{2 \times 2.5}$

$s = 180\text{m}$

So, the stooping distance is 180m.

15. Correct option: (D) $\dfrac{3\pi}{2}$.



Displacement = $\text{POR} = 2R$

Distance $= 2\pi\text{R} + \dfrac{2\pi R}{2} = 2\pi R$

The ratio of distance travelled by the body to its displacement,

$\dfrac{\text{Distance}}{\text{Displacement}} = \dfrac{3\pi R}{2R} = \dfrac{3\pi}{2}$

So, the ratio of the distance travelled by the body to its displacement is $\dfrac{3\pi}{2}$

16. Correct option: (A) $\dfrac{7}{5}$.

Given:

$u = 0\text{ms}^{-1}$

The distance travelled by in n^{th} second is given by as,

$$S_n = u + \frac{a}{2}(2n-1)$$

Therefore, ratio of distance travelled in 4th and 3rd sec is,

$$\frac{S_{4^{th}}}{S_{3^{rd}}} = \frac{0 + \frac{a}{2}(2\times 4 - 1)}{0 + \frac{a}{2}(2\times 3 - 1)}$$

$$\frac{S_{4^{th}}}{S_{3^{rd}}} = \frac{2\times 4 - 1}{2\times 3 - 1} = \frac{7}{5}$$

So, ratio of distance travelled in 4th and 3rd sec is $\frac{7}{5}$.

17. Correct option: (B) Car will cover less distance before rest.

When a truck and a car are driving at the same velocity,

$a = \frac{v^2 - u^2}{2s}$ and $m \propto \frac{1}{a}$

Therefore,

$m \propto s$

This suggests that the distance a car travels before stopping will be less.

18. Correct option: (C) $s_1 = \frac{1}{3}s_2 = \frac{1}{5}s_3$

The initial velocity of body is, $u = 0 \text{ms}^{-1}$

Now using equation of motion,

$$s_n = ut + \frac{1}{2}at^2$$

$$s_1 = \frac{1}{2}a(5)^2 = \frac{25a}{2}$$

$$\Rightarrow s_1 + s_2 = \frac{1}{2}a(10)^2 = \frac{100a}{2}$$

$$\Rightarrow s_2 = \frac{100a}{2} - s_1 = 75\frac{a}{2}$$

Also,

$$s_1 + s_2 + s_3 = \frac{1}{2}a(15)^2 = \frac{225a}{2}$$

$$\Rightarrow s_3 = \frac{225a}{2} - s_2 - s_1 = \frac{125a}{2}$$

Now,

$$s_1 : s_2 : s_3 = \frac{25a}{2} : \frac{75a}{2} : \frac{125a}{2}$$

$$s_1 = \frac{1}{3}s_2 = \frac{1}{5}s_3$$

So, the relation between s_1, s_2 and s_3 is $s_1 = \frac{1}{3}s_2 = \frac{1}{5}s_3$.

19. The correct answer is (5:9) option is not available.

Given: Initial speed is , $u = 0\text{ms}^{-1}$.

Distance travelled after starting at rest in n^{th} seconds is given by as,

$$s = \frac{1}{2}(2n-1)$$

For $A : n = 5$

So, $s_A = \frac{a_1}{2}(2\times 5-1) = \frac{9}{2}a_1$

For B: $n = 3$

So, $s_B = \frac{a_2}{2}(2\times 3-1) = \frac{5}{2}a_2$

$$\therefore x + 50 = ut$$

But we $\frac{t^2}{2} + 50 = ut$ know that,

$$\Rightarrow t^2 - 2ut + 100 = 0$$

$$s_A = s_B$$

$$\frac{9}{2}a_1 = \frac{5}{2}a_2$$

$$a_1 : a_2 = 5 : 9$$

So, the ratio of $a_1 : a_a$ is 5:9.

Note: The option is not available in question.

20. Correct option: (C) 10ms^{-1}**.**

Given: the acceleration of bus is, $a = 1\text{ms}^{-2}$.

Assume that the student boarded the bus at time t and travelled a distance of x.

Velocity of the student, $u = 0\text{ms}^{-1}$.

Initial velocity of bus is, $u_1 = 0$

Now using the equation of motion,

$$x = ut + \frac{1}{2}at^2$$

$$x = \frac{1}{2}at^2$$

$$x = \frac{t^2}{2}$$

Now distance travelled by student $= x + 50$

$$\therefore x + 50 = ut$$

$$\frac{t^2}{2} + 50 = ut$$

$$\Rightarrow t^2 - 2ut + 100 = 0$$

The above equation have real roots i.e. its discriminant ≥ 0.

$$(2u)^2 - 4\times 100 \geq 0$$

$$u^2 - 100 \geq 0$$

$$\Rightarrow u > 10\text{ms}^{-1}$$

So, the minimum value of u is 10ms^{-1}.

21. Correct option: (C) $5.5\frac{\text{m}}{\text{s}}$**.**

The second equation of motion is given by as,

$$s_n = ut + \frac{1}{a}at^2$$

Now distance travelled by in 5 second is,

$$s_5 = u(5) + \frac{a(5)^2}{2} = 40 \cdots\cdots (1)$$

And the distance travelled by in 10 second is,

$$s_{10} = u(10) + \frac{a(10)^2}{2} = 40 + 65 = 105 \cdots\cdots (2)$$

Now solving equation (1) and (2), we get,

$$u = 5.5\frac{\text{m}}{\text{s}}$$

So, the initial velocity is $5.5\frac{\text{m}}{\text{s}}$.

22. Correct option: (C) $S = \frac{1}{72}\text{ft}^2$**.**

If t_1 be the period of time during which the vehicle accelerates from rest at the rate $\vec{V}$, and the distance travelled in time t_1 is,

Using equation of motion,

$$S = ut + \frac{1}{2}at^2$$

$$S = S_1 = 0(t_1) + \frac{1}{2}ft_1^2 = \frac{1}{2}ft_1^2 \cdots\cdots (1)$$

And the velocity at time t_1 is $v_1 = ft_1$.

Consequently, the distance covered in time t_1 will be,

$$S_2 = v_1 t = ft_1 t \cdots\cdots (2)$$

If the car decelerates at a rate of $\frac{f}{2}$ to come to rest in time t_2, the distance travelled in that period can be calculated using,

Using equation of motion,

$$v^2 - u^2 = 2as$$

$$0 - v_1 = 2\left(-\frac{f}{2}\right)S_3$$

$$\text{or } S_3 = \frac{(ft_1)^2}{f} = ft_1^2 \cdots\cdots (3)$$

Now using,

$S_3 = 2S_1 = 2S$

Given $S_1 + S_2 + S_3 = 15S$

or $s + ft_1 t + 2s = 15S$

or $ft_1 t = 12S$

or $12S = ft_1 t \cdots\cdots (4)$

Now using equation(1) and (2), we get

$$\frac{12S}{S} = \frac{ftt_1}{\frac{1}{2}ft_1^2}$$

$$t_1 = \frac{t}{6}$$

Now, from equation (1), we get

$$S = \frac{1}{2}f\left(\frac{t}{6}\right)^2 = \frac{ft^2}{72}$$

So, if the total distance traversed in $15s$, then $s = \frac{ft^2}{72}$.

23. Correct option: (C) $\sqrt{\frac{u^2+v^2}{2}}$.

Let's consider length of the train is L.

So, using equation of motion we get,

$v^2 - u^2 = 2aL$

$$\Rightarrow L = \frac{v^2 - u^2}{2a}$$

Additionally, let the train's intermediate point's speed be v_m.

Therefore,

$$v_m^2 - u^2 = 2a \times \frac{L}{2} = aL$$

$$\Rightarrow v_m^2 - u^2 = a \times \frac{v^2 - u^2}{2a} \cdots\cdots (2)$$

Now using equation (1) and (2), we get

$$v_m = \sqrt{\frac{v^2 + u^2}{2}}$$

So, the required value of velocity is $\sqrt{\frac{v^2+u^2}{2}}$.

24. Correct option: (A) $\frac{2v(v_1+v_2)}{(2v+v_1+v_2)}$.

Let the total distance travelled is d and time taken to travel the first half distance be t_1 and second half t_2.

Distance of fist half = $\frac{d}{2}$

Time of first half = $t_1 = \frac{\frac{d}{2}}{v_0} = \frac{d}{2v_0}$

Let the total time taken to travel the remaining distance $t_2, \frac{d}{2} = \frac{t_2 v_1}{2} + \frac{t_2 v_2}{2}$

$$\frac{d}{2} = \frac{t_2 v_1 + t_2 v_2}{2}$$
$$d = t_2 \left[v_1 + v_2 \right]$$
$$t_2 = \left[\frac{d}{v_1 + v_2} \right]$$

Therefore total time $= t_1 + t_2 = \frac{d}{2v_0} + \frac{d}{v_1 + v_2}$

The average speed is given by as,

$$V_{avg.} = \frac{\text{Total distnace}}{\text{total time}}$$

$$V_{avg.} = \frac{d}{\frac{d}{2v_0} + \frac{d}{v_1 + v_2}} = \frac{d}{\frac{d(v_1 + v_2 + d(2v_0)}{2v_0 (v_1 + v_2)}}$$

$$V_{avg.} = \frac{d}{\frac{d[v_1 + v_2 + 2v_0]}{2v_0 (v_1 + v_2)}} = \frac{2v_0 (v_1 + v_2)}{2v_0 + v_1 + v_2}$$

The average speed is $\frac{2v_0 (v_1 + v_2)}{2v_0 + v_1 + v_2}$.

25. Correct option: (B) 0.5ms^{-1}**.**

The velocity of particle, $v_x = 0.4 \frac{\text{m}}{\text{s}}$

$$v_y = 0 \frac{\text{m}}{\text{s}}$$

$$a_y = 0.15 \frac{\text{m}}{\text{s}^2}$$

$$t = 2 \text{sec}.$$

Now using equation of motion,

$$\left(v_y \right)_{\text{final}} = v_y \hat{i} + a_y t$$

$$\left(v_y \right)_{\text{final}} = 0 + (0.15) \times 2 = 0.3 \frac{\text{m}}{\text{s}}$$

So, $v = v_x \hat{i} + v_y \hat{j} = 0.4 \hat{i} + 0.3 \hat{j}$

The magnitude of resultant velocity is given by as,

$$v_{net} = |v| = \sqrt{(0.4)^2 + (0.3)^2} = 0.5 \frac{\text{m}}{\text{s}}$$

So, the magnitude of resultant velocity is $0.5 \frac{\text{m}}{\text{s}}$.

26. Correct option: (B) 10 sec.

Given:

$v = 30\text{ms}^{-1}$

$u = 20\text{ms}^{-1}$

$t = 5\,\text{sec}$

Now using equation of motion,

$v = U + at$

$30 = 20 + 5a$

$a = 2\text{ms}^{-2}$

Now, $U = 0$

Now using equation of motion.

$20 = 2t'$

$t' = 10\,\text{sec}$

So, the required time is 10sec.

27. Correct option: (D)

28. Correct option: (B)

29. Correct option: (C)

30. Correct option: (A) $\left(\frac{\alpha\beta}{\alpha+\beta}\right)\times t$ **.**

Let t_1 be the time acceleration and $t - t_1$ be the time for deceleration.

Now using equation of motion,

$v = U + at$

$v = 0 + \alpha t_1 \cdots\cdots(1)$

or

$v = U + at$

$v = 0 + \beta(t - t_1) \cdots\cdots(2)$

Now using equation (1) and (2),

$\alpha t_1 = \beta\left(t - t_1\right)$

$\Rightarrow t_1 = \frac{\beta t}{\alpha+\beta}$

$\Rightarrow v = \left(\frac{\alpha\beta}{\alpha+\beta}\right)\times t$

So, the maximum velocity is $\left(\frac{\alpha\beta}{\alpha+\beta}\right)\times t$.

LEVEL-II

1. Correct option: (D) $14\frac{\text{m}}{\text{sec}}$ **.**

Given: the acceleration, $a = 2t + 5$

Time, $t = 2\,\text{sec}$

Acceleration is given by as,

$a = \frac{dv}{dt}$

Therefore, the velocity is given by as,

$v = \int a dt$

$v = \int (2t + 5) dt$

Distance covered in 2esc is,

$v = \int_0^2 (2t + 5) dt$

$v = \int_0^2 (2t) dt + = \int_0^2 (5) dt$

$v = \left[\frac{2t^2}{2} \right]_0^2 + 5\left[t \right]_0^2$

$v = \left[(2)^2 - (0)^2 \right] + 5(2 - 0_$

$v = 4 + 10$

$v = 14 \frac{\text{m}}{\text{s}}$

So, the velocity after $2\,\text{sec}$ is $14 \frac{\text{m}}{\text{sec}}$.

2. the correct answer is 1250m.

Given:

$u_A = u_B = 72\text{kmh}^{-1} = 72 \times \frac{5}{18} \text{ms}^{-1}$

Now using the newton's law of motion,

$s = ut + \frac{1}{2} at^2$

$s_B = u_B t + \frac{1}{2} at^2$

$s_B = 20 \times 50 + \frac{1}{2} \times 1 \times (50)^2$

$s_b = 1000 + 1250$

$s_B = 2250\text{m}$

Distance covered by the train A is represent by s_A,

$s_A = u_A \times t$

$s_A = 20 \times 50 = 1000\text{m}$

Now , original distance between the two trains,

$s = s_B - s_A$

$s = 2250 - 1000$

$s = 1250\text{m}$

The original distance is 1250m.

Note: The option is not available in question.

3. Correct option: (C) $\frac{4}{7}$.

From the graph:

The maximum velocity is represented by v_m.

Area of $v-t$ graph = distance travelled

Now, the average velocity is given by as,

$$V_{avg.}=\frac{8s}{t_1+t_2+t_3}$$

$$\frac{1}{2}v_m\times t_1=s$$

$$t_1=\frac{2s}{v_m}$$

$$t_2\times v_m=2s$$

$$t_2=\frac{2s}{v_m}$$

$$\frac{1}{2}\times t_3\times v_m=5s$$

$$t_3=\frac{10s}{v_m}$$

So, the average velocity is given by as,

$$V_{avg.}=\frac{8s}{\frac{2s}{v_m}+\frac{2s}{v_m}+\frac{10s}{v_m}}$$

$$\frac{V_{avg}}{v_m}=\frac{4}{7}$$

The ratio of the average velocity to maximum velocity is $\frac{4}{7}$.

4. Correct option: (B) 2 s.

Given:

$$U=0\frac{\text{m}}{\text{s}}$$

$$t=4\,\text{sec}$$

Acceleration due to gravity, $a=9\frac{\text{m}}{\text{s}^2}$

Now using equation of motion,

$$s=ut+\frac{1}{2}at^2$$

$$s=0\times4+\frac{1}{2}\times9\times4$$

$$s=72\text{m}$$

$$\frac{1}{4}\text{ of }72=18m$$

Now we have, $s=18\text{m},u=0\frac{\text{m}}{\text{s}},a=9\frac{\text{m}}{\text{s}^2}$

Again using equation of motion,

$$s = ut + \frac{1}{2}at^2$$

$$18 = 0 \times t + \frac{1}{2} \times 9 \times t^2$$

$$18 = \frac{1}{2} \times 9t^2$$

$$t^2 = 2 \times 2$$

$$t^2 = 4$$

$$t = 2$$

Time taken to cover one fourth the distance starting from the top is 2 s.

5. Correct options: (A) and (C).

Given:

$a_1 = 2\text{ms}^{-2}$

$a_2 = -4\text{ms}^{-2}$

Initial velocity , $u = 0\,\text{ms}^{-1}$

Now using equation of motion,

$v = u + at$		$v = u + at$
$v_0 = a_1 t_1$	or	$v_0 = a_2 t_2$
$v_0 = 2t_1$		$v_0 = 4t_2$

So, the total time is,

$$t_1 + t_2 = 6$$

$$\frac{v_0}{2} + \frac{v_0}{4} = 6$$

$$v_0 = 8\text{ms}^{-1}$$

It is possible to determine distance in constant accelerated motion by using $V_{avg.}t$ so,

$$AC = V_{avg.}t = \frac{v_0 + 0}{2}t_1$$

$$CB = V_{avg.}t = \frac{v_0 + 0}{2}t_2$$

Total distance travelled is given by as,

$$S = AC + CB$$

$$S = \frac{1}{2}v_o t_2 + \frac{1}{2}v_0 t_2$$

$$S = \frac{1}{2}v_0\left[t_1 + t_2\right]$$

$$S = \frac{1}{2} \times 8 \times 6$$

$$S = 24\text{m}$$

So, Correct option: (A) and (C).

6. Correct options: (A) and (D).

Using equation of motion,

$s = ut + \frac{1}{2}at^2$

When, $u = 0, s = \frac{1}{2}at^2$

So, the displacement of the body is directly proportional to the square of the time.

Therefore, $s \propto t^2$ Since the particle is moving in a straight line since acceleration is constant and the particle is starting from rest, there is no change in the direction of velocity.

7. Correct options: (A), (B) and (C).

Let us consider speed at midpoint is represent by v and acceleration is represent by a.

Let's consider distance from the midpoint to each is s.

Now, using the equation of motion is,

$v^2 = u^2 + 2as$

$v^2 = (2)^2 + 2as$

$v^2 = 4 + 2as \cdots\cdots (1)$

$(14)^2 = v^2 + 2as$

$196 = v^2 + 2as \cdots\cdots (2)$

From using equation (1) and (2), we get

$v^2 - 4 = 196 - v^2$

$v = 10\frac{\text{m}}{\text{s}}$

So, the speed at the mid-point of AB is $10\frac{\text{m}}{\text{s}}$.

Considering a point P in light of the stated question, $\text{AP:PB} = 1:5$

Let, AP=x and PB=5x,

Using equation of motion,

$v^2 = 2^2 + 2ax$

$v^2 = 4 + 2ax \cdots\cdots (3)$

$14^2 = v^2 + 2a \times 5x$

$196 = v^2 + 10ax \cdots\cdots (4)$

Using equation (3) and (4), we get,

$v^2 - 4 = \frac{196 - v^2}{5}$

$v = 6\frac{\text{m}}{\text{s}}$

So, the speed at a point P such that AP:PB=1:5 is $6\frac{\text{m}}{\text{s}}$.

Let's say it takes t_1 seconds to travel from point A to the halfway point.

and the time t_2 it took to get from A to B.

Now,

$$t_1 = \frac{v_{mid} - 1}{a} = \frac{10 - 2}{a} = \frac{8}{a}$$

$$t_2 = \frac{14 - v_{mid}}{a} = \frac{14 - 10}{a} = \frac{4}{a}$$

So,

$$\frac{t_1}{t_2} = \frac{1}{2}$$

$$2t_1 = t_2$$

The time to go from A to the mid-point of AB is double of that to go from mid-point to B.
So, Correct option: (A), (B) and (C).

8. Correct option: (B) 2c,b+2ct **.**

The particle's velocity should be determined as,

$$v = \frac{dx}{dt}$$

Given: $x = a + bt + ct^2$

So, the magnitude of the velocity is given by as,

$$v = \frac{dx}{dt}$$

$$v = \frac{d\left(a + bt + ct^2\right)}{dt}$$

$$v = b + 2tc$$

Now, we have to find the magnitude of the acceleration is given by as,

$$a = \frac{dv}{dt}$$

$$a = \frac{d\left(b + 2tc\right)}{dt}$$

$$a = 2c$$

So, Correct option: (B) 2c,b+2ct .

9. Correct options: (B), (C) and (D).

(A) Distance covered: The circumference of a circle is the distance a particle travelling along a circular route travels. $C = 2\pi r$, where $'r'$ is the radius, is the formula for calculating a circle's circumference. The particle has therefore travelled a distance of $2\pi r$

(B) Displacement: The straight-line distance and direction between an object's beginning and final positions are measured by the vector variable known as displacement. The displacement and the travelled distance are not identical for a particle travelling in a circle. It depends on the circle's beginning and ending locations.

Since the particle is moving on a circular path in this instance, its displacement is equal to the circle's radius, or $2r$

(C) Acceleration: Centripetal acceleration (a_c) is the term used to describe the acceleration of an item travelling in a circular direction. It is always pointed in the direction of the circle's centre, and the formula for calculating its size is as follows:

$$a_c = \frac{V^2}{r}$$

(D) The rate of change of angular displacement with respect to time is measured by angular velocity (ω). The following formula can be used to determine the angular velocity of a particle travelling in a circle:

$$\omega = \frac{V}{r}$$

So, Correct option: (B), (C) and (D).

10. Correct option: (C) 5 Sec.

Given:

$$v_A = 10\frac{\text{m}}{\text{s}}$$

$$a_B = 2\frac{\text{m}}{\text{s}^2}$$

Now, minimum distance between then would be when, $v_B = v_B$

Therefore,

$$v = u + at$$

$$v_B = u_B + a_B t$$

$$10 = 0 + 2t$$

$$t = 5\,\text{sec}$$

So, the require value of time is $5\,\text{sec}$.

11. Correct option: (B) 15m.

$$v_A = 10\frac{\text{m}}{\text{s}}$$

$$a_B = 2\frac{\text{m}}{\text{s}^2}$$

Now, minimum distance between then would be when, $v_B = v_B$

Therefore,

$$v = u + at$$

$$v_B = u_B + a_B t$$

$$10 = 0 + 2t$$

$$t = 5\,\text{sec}$$

Now, they will now have a minimum distance after 5 seconds,

In 5 seconds,

$$s_A = 50m$$

$$s_B = \frac{1}{2} \times 2 \times (5)^2 = 25\text{m}$$

$$\text{x} = \left((40+25)-50\right)$$

$$x = 15\,\text{m}$$

12. Correct option: (C)

13. Correct option: (B) $\left(\frac{\alpha\beta}{\alpha+\beta}\right) \times t$ **.**

Let t_1 be the time acceleration and $t - t_1$ be the time for deceleration.

Now using equation of motion,

$v = U + at$ or $v = U + at$

$v = 0 + \alpha t_1 \cdots\cdots (1)$ $v = 0 + \beta(t - t_1) \cdots\cdots (2)$

Now using equation (1) and (2),

$\alpha t_1 = \beta\left(t - t_1\right)$

$\Rightarrow t_1 = \dfrac{\beta t}{\alpha + \beta}$

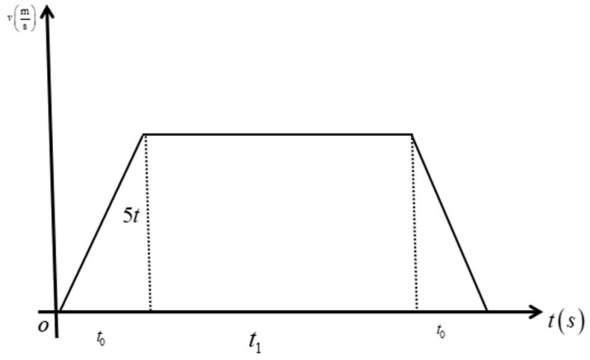
$\Rightarrow v = \left(\dfrac{\alpha\beta}{\alpha + \beta}\right) \times t$

So, the maximum velocity is $\left(\dfrac{\alpha\beta}{\alpha + \beta}\right) \times t$.

14. Correct option: (B)

15. Correct option: (A)

16. Correct option: (C) 500m.



Accelerating case:

Initial velocity, $u = 0$

Acceleration, $a = \omega$

Consider time interval is t_0 .

So, the final velocity is given by as,

$v = u + at$

$v = 0 - \omega t_1$

or $t_0 = \dfrac{v}{\omega}$

Decelerating case:

$v_{\max} = 5t$

Initial velocity, $v_{\max} = 5 \times 5$

$v_{\max} = 25 \text{ms}^{-1}$

Acceleration, $a = -\omega$

So, the final velocity is given by as,

$v = u + at$

$0 = v - \omega t_1$

or $t_1 = \dfrac{v}{\omega}$

or $t_1 = t_0$

Now, if the car travels with uniform velocity for time t_1, then we have,

$\tau = 2t_0 + t_1 = 25\text{s}$

We know that average velocity is given by as,

$$\text{Average velocity}, (v_{av}) = \frac{\text{Total displacement}}{\text{total time}}$$

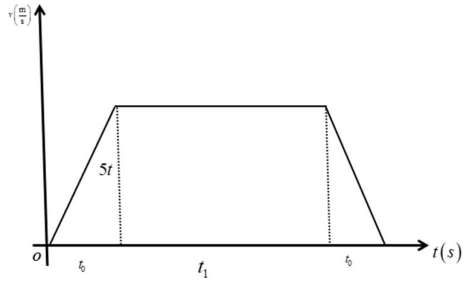
$$(v_{av}) = \frac{S}{\tau}$$

$$\frac{72 \times 5}{18} = \frac{S}{25}$$

$S = 500\text{m}$

So, the total distance travelled by the car is 500m.

17. Correct option: (B) 25ms^{-1}**.**



Accelerating case:

Initial velocity, $u = 0$

Acceleration, $a = \omega$

Consider time interval is t_0.

So, the final velocity is given by as,

$v = u + at$

$v = 0 - \omega t_1$

or $t_0 = \frac{v}{\omega}$

Decelerating case:

$$\text{Average velocity}, (v_{av}) = \frac{\text{Total displacement}}{\text{total time}}$$

Initial velocity, $(v_{av}) = \frac{S}{\tau}$

$$\frac{72 \times 5}{18} = \frac{S}{25}$$

$S = 500\text{m}$

Acceleration, $a = -\omega$

So, the final velocity is given by as,

$v = u + at$

$0 = v - \omega t_1$

or $t_1 = \frac{v}{\omega}$

or $t_1 = t_0$

Now, if the car travels with uniform velocity for time t_1, then we have,

$\tau = 2t_0 + t_1 = 25\text{s}$

We know that average velocity is given by as,

$$\text{Average velocity}, (v_{av}) = \frac{\text{Total displacement}}{\text{total time}}$$

$$(v_{av}) = \frac{S}{\tau}$$

$$\frac{72 \times 5}{18} = \frac{S}{25}$$

$S = 500\text{m}$

Now, $S = \text{Area under the graph}$

$$500 = \frac{1}{2}vt_0 + vt_1 + \frac{1}{2}vt_2$$

$$v\left(t_0 + t_1\right) = 500$$

$$v\left(\tau - t_0\right) = 500$$

$$\omega t_0\left(25 - t_0\right) = 500$$

$$t_0^2 - 25t_0 + 100 = 0$$

Solving it , we get, $t_0 = 5\text{s as} t_0 \neq 20\text{s}\left(t_0 < \frac{25}{2}\text{s}\right)$

From the graph maximum velocity is given by as,

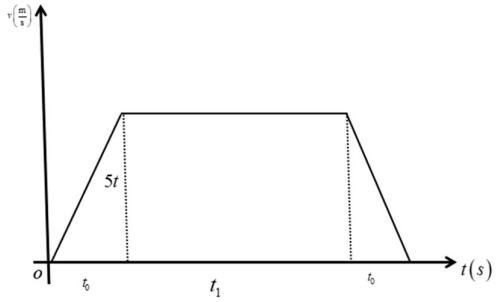
$v_{\max} = 5t$

$v_{\max} = 5 \times 5$

$v_{\max} = 25\text{ms}^{-1}$

So, the maximum speed of the car is 25ms^{-1}.

18. Correct option: (D) 15s.



Accelerating case:

Initial velocity, $u = 0$

Acceleration, $a = \omega$

Consider time interval is t_0.

So, the final velocity is given by as,

$v = u + at$

$v = 0 - \omega t_1$

or $t_0 = \frac{v}{\omega}$

Decelerating case:

$$\text{Average velocity}, (v_{av}) = \frac{\text{Total displacement}}{\text{total time}}$$

Initial velocity, $(v_{av}) = \frac{S}{\tau}$

$$\frac{72 \times 5}{18} = \frac{S}{25}$$

$$S = 500\text{m}$$

Acceleration, $a = -\omega$

So, the final velocity is given by as,

$v = u + at$

$0 = v - \omega t_1$

or $t_1 = \frac{v}{\omega}$

or $t_1 = t_0$

Now, if the car travels with uniform velocity for time t_1, then we have,

$\tau = 2t_0 + t_1 = 25\text{s}$

We know that average velocity is given by as,

$$\text{Average velocity}, (v_{av}) = \frac{\text{Total displacement}}{\text{total time}}$$

$(v_{av}) = \frac{S}{\tau}$

$\frac{72 \times 5}{18} = \frac{S}{25}$

$S = 500\text{m}$

Now, $S = \text{Area under the graph}$

$500 = \frac{1}{2} v t_0 + v t_1 + \frac{1}{2} v t_2$

$v(t_0 + t_1) = 500$

$v(\tau - t_0) = 500$

$\omega t_0 (25 - t_0) = 500$

$t_0^2 - 25t_0 + 100 = 0$

Solving it , we get, $t_0 = 5\text{s}$ as $t_0 \neq 20\text{s} \left(t_0 < \frac{25}{2}\text{s} \right)$

Therefore, $t_1 = 25 - 2 \times 5 = 15\text{s}$

So, the required time is 15s.

19. Correct Matching is: a-PQ; b-R; c-R; d-S

Explanation: (a) The net displacement is of the body is zero therefore the average velocity also is zero.

(b) the average speed is given by as,

$$u_{av} = \frac{\text{Total distnace}}{\text{Time of flight}}$$

$$u_{av} = \frac{2H_{\max}}{2ug}$$

$$u_{av} = \frac{\frac{2u^2}{2g}}{\frac{2u}{g}}$$

$$u_{av} = \frac{u}{2}$$

If velocity of projectile is v_0.

Therefore,

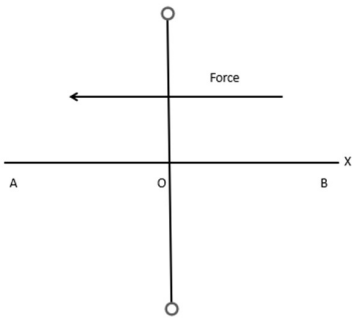
$$v_{av} = \frac{v_0}{2}$$

(c) $T_{ascent} = \frac{v_0}{2}$

(d) $T_{descent} = \frac{v_0}{g}$

So, the correct matching is (a)-(P) (Q), (b)-(R), (c)-(R), (d)-(S).

20. Correct Matching is: a-QR; b-R; c-QR; d-QR



a) Initial velocity > 0

Since the direction of the force is in the (-Ve) x axis, a particle travelling in the (+ve) x axis will experience a decrease in speed.

Or, the particle will move with increasing speed in the (-Ve) x axis direction if it is moving in that direction.

b) Initial velocity < 0

In this instance, the particle is, let's say, halfway between $\overline{OA}$ and because force has been applied in the (-Ve) x axis direction, it can proceed in that direction with increasing speed.

c) If the particle is in the region between $\overline{OB}$ and $\overrightarrow{BO}$, its speed will increase if it is travelling in the BO direction and decrease if it is moving in the $\overline{OB}$ direction.

d) If the particle is between $\overline{OA}$, then due to the imparted force, it will also move with decreasing speed if the initial velocity is in the $\overline{OB}$ direction, else it will move with rising speed.

Motion Under Gravity
LEVEL-I

1. Correct option: (C) $\sqrt{a}:\sqrt{b}$.

Given: initial speed, $u=0$

The specific gravity is 9.8ms^{-1}..

Now using equation of motion,

$$s=ut+\frac{1}{2}at^2$$

$$s=0+\frac{1}{2}gt^2$$

$$t=\sqrt{\frac{2s}{g}}$$

$$t\propto\sqrt{s}$$

Therefore, the ratio of the time taken by the two bodies to cover these distances is given by as,

$$\frac{t_a}{t_b}=\sqrt{\frac{a}{b}}$$

2. Correct option: (A) Acceleration due to gravity in vacuum depends upon the mass of the body.

When thrown from the same height in a vacuum, the iron and wooden balls will fall to the earth exactly at the same moment under the influence of gravity because there is no air resistance.

In the absence of air resistance, all bodies fall with a same acceleration, roughly equal to 9.8ms^{-1}.

3. Correct option: (A) 80m.

Given: time, $t=4\sec$

Initial velocity, $u=0$

Using equation of motion,

$$s=ut+\frac{1}{2}gt^2$$

$$s=0\times4+\frac{1}{2}\times10\times4^2$$

$$s=80\text{m}$$

So, the height of the tower is 80m.

4. Correct option: (A) $\sqrt{u^2+2gh}$.

Using the first equation of motion,

$$v^2=u^2+2as$$

$$a=g, s=h$$

$$v=\sqrt{u^2+2gh}$$

So, the velocity is $\sqrt{u^2+2gh}$.

5. Correct option: (A) 1:3:5.

Using the second equation of motion,

$$s = ut + \frac{1}{2}at^2$$

Now distance travelled in n^{th} second is given as,

$$s_{n^{th}} = u + \frac{1}{2}g(2n-1)$$

$$s_1 = \frac{g}{2} \times (2 \times (2) - 2) = \frac{g}{2}$$

$$s_1 = \frac{g}{2} \times (2 \times (2) - 1) = \frac{3g}{2}$$

$$s_3 = \frac{g}{2} \times (2 \times (3) - 1) = \frac{5g}{2}$$

$$s_1 : s_2 : s_3 = \frac{g}{2} : \frac{3g}{2} : \frac{5g}{2} = 1:2:3$$

As a result, the ratio of the distance covered in the first, second, and third seconds of a body falling freely from rest is 1:3:5.

6. Correct option: (C) 610m.

Given:

$$g = 9.8\frac{\text{m}}{\text{s}}$$

$$u = 12\frac{\text{m}}{\text{s}}$$

$$t = 10\,\text{sec}$$

Now using equation of motion, we get,

$$s = 12 \times 10 + \frac{1}{2} \times 9.8 \times (10)^2$$

$$s = 120 + 0.5 \times 980$$

$$s = 610\text{m}$$

So, the displacement of stone is 610m.

7. Correct option: (B) $\frac{4u^2}{g}$.

Given: $v = 3u$

Using third equation of motion,

$$v^2 = u^2 + 2as$$

$$(3u)^2 = u^2 + 2gs$$

$$s = \frac{9u^2 - u^2}{2g}$$

$$s = \frac{4u^2}{g}$$

So, the height of tower is $\frac{4u^2}{g}$.

8. Correct option: (B) 3h.

Given: initial velocity of body, $u = 0$

Acceleration, $a = -g$

Displacement, $g =$

Now using third equation of motion,

$v^2 = u^2 + 2as$

$v^2 = 0 + 2(-g)(-g)$

$v^2 = 2gh \cdots\cdots (1)$

After travelling distance h_1, velocity become double i.e $2v$.

Again using third equation of motion,

$(2v)^2 = v^2 + 2gh_1$

$4v^2 = v^2 + 2gh_1 \cdots\cdots (2)$

Now using equation (1) an (2)

$4(2gh) = 2gh + 2gh_1$

$2gh_1 = 6gh$

$h_1 = 3h$

So, it has to travel 3h distance more to gain double the velocity.

9. Correct option: (B) 2s.

Case:1

Ball dropped from a tower, $H = 100\text{m}$

$u = 0$

$a = +g = 10\frac{m}{s^2}$

Case:2 A second ball is simultaneously sent vertically along with initial velocity, $u = 50\frac{\text{m}}{\text{s}}$.

$a = -g = -10\frac{\text{m}}{\text{s}^2}$

Allow these two balls to collide at second t.

The height that $1st$ ball descended is then revealed. H the height reached by the second ball equals the tower's height.

$\Rightarrow \frac{1}{2}gt^2 + ut - \frac{1}{2}gt^2 = H$

$50t = 100$

$t = 2\text{sec}.$

So, two balls cross each other after 2 sec.

10. Correct option: (B) $2g$.

Given:

$u = u$

$a = g$

If the velocity acquired is numerically equal to the

$v^2 = u^2 + 2as$

Displacement, $v^2 = 0 + 2gv \quad |v| = |s|$

$v = 2g$

Therefore, using the equation of motion we get,

$v^2 = u^2 + 2as$

$v^2 = 0 + 2gv$

$v = 2g$

So, the velocity acquired is $2g$.

11. Correct option: (A) 12.25ms^{-1}**.**

Given:

$\text{u}=0$

$\text{h}=44.1\text{m}$

$\text{g}=9.8\text{ms}^{-2}$

Using equation of motion,

$$h = ut + \frac{1}{2}gt^2$$

$$44.1 = 0 + \frac{1}{2}(9.8)t^2$$

$$t = 3\,\text{sec}$$

Now initial speed is $2u$.

Stone 2 is thrown one second after stone 1, hence it travels the same distance in one second less time than stone 1. Both stones hit the water at the same time.

Therefore, time taken by stone 2,

$t_1 = t - 1 = 3 - 1 = 2\,\text{sec}$

Since is acceleration is constant.

Now using equation of motion,

$$h = ut + \frac{1}{2}gt_1^2$$

$$44.1 = u \times 2 + \frac{1}{2} \times (9.8) \times (2)^2$$

$$u = 12.25\frac{\text{m}}{\text{s}}$$

So, the initial speed of the second stone is 12.25ms^{-1}.

12. Correct option: (C) $\frac{3h}{4}$ **.**

Given:

$u = 0$

$$v = 10\frac{\text{m}}{\text{s}}$$

$\text{height} = h$

$\text{time} = t$

Since. We know that,

$v = u - gt$

$\because u = 0$

$10 = 10t$

$t = 1\text{sec}$

At,

$t = \frac{1}{2}s, u^2 = 2 \times gh$

$\frac{100}{2 \times 10} = h$

$h = 5\text{m}$

Now using equation motion is,

$h = ut + \frac{1}{2}gt^2$

$h = 10 \times \frac{1}{2} - \frac{1}{2} \times 10 \times \frac{1}{4} = \frac{15}{4} = \frac{3h}{4}$

So, the height covered is $\frac{3h}{4}$.

13. Correct option: (B) $25.1\frac{\text{m}}{\text{s}}$**.**

Given: Distance travelled, $s = 30\text{m}$

Time, $t = 1\text{sec}$

Acceleration due to gravity, $g = 9.8\frac{\text{m}}{\text{s}^2}$

Now using second equation of motion,

$s = ut + \frac{1}{2}at^2$

$30 = u(1) + \frac{1}{2} \times 9.8 \times (1)^2$

$30 + u + \frac{1}{2} \times 9.8 \times 1$

$30 = u + 4.9$

$u = 25.1\frac{\text{m}}{\text{s}}$

So, the velocity of the projection is $25.1\frac{\text{m}}{\text{s}}$.

14. Correct option: (D) $\frac{u}{\sqrt{2}}$**.**

A body is accelerated up in a vertical projection.

$H_{\max} = \frac{u^2}{2g}$

Where, $H_{\max}$ = maximum height attained by the projectile

u = initial velocity

g = acceleration due to gravity

Now using third equation motion,

$20\frac{\text{m}}{\text{s}}$

It is known as acceleration due to gravity when it occurs when a body is driven higher. Gravity's acceleration has a negative sign since it acts in the downward direction, assuming the upward direction is positive.

As a result, the equation above becomes,

$v^2 = u^2 - 2gh$

Where, $h =$ maximum height attained

Substitute instead of "because we need to determine the velocity at half the maximum height: $h = \frac{H_{\max}}{2}$

Now,

$H_{\max} = \frac{u^2}{2g}$

$h = \frac{u^2}{2g}$

Substituting the value of $h = \frac{u^2}{2g}$ in equation(1),

$H_{\max} = \frac{u^2}{2g}$

$v^2 = u^2 - 2g\frac{u^2}{4g}$

$v^2 = u^2 - \frac{u^2}{2}$

$v^2 = \frac{u^2}{2}$

$v = \frac{u}{\sqrt{2}}$

So, the maximum height is $\frac{u}{\sqrt{2}}$.

15. Correct option: (B) 55m.

From the diagram,

Distance up to 4 sec:

Distance (0 to 1 sec) $= \frac{1}{2}\times 1\times 2 == 10\text{m}$

For ,(1-2sec) $= 20\times(2-1) = 20\text{m}$

For, (2-3sec)

$= (3-2)\times 10 + \frac{1}{2}(3-2)\times(20-10)$

$= 1\times 10 + \frac{1}{2}\times 1\times 10 = 15$

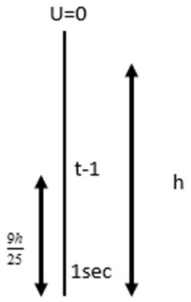
For (3-4sec) $=(4\text{m}-3)\times 10=10$

Total distance $=10+20+15+10=55\text{m}$

So, the distance covered by the particle in 4 seconds is 55m

16. Correct option: (C) 122.5m.

Initial velocity, $u=0$



Now using equation of motion,

$$h=\frac{1}{2}gt^2\cdots\cdots(1)$$

$$\frac{16h}{25}=\frac{1}{2}g(t-1)^2\cdots\cdots(2)$$

Now using equation (1) and (2),

$$\frac{16}{25}\left(\frac{1}{2}gt^2\right)=\frac{1}{2}g\left(t-1\right)^2$$

$$16t^2=25t^2-50t+25$$

$$0=9t^2-50t+25$$

$$0=9t^2-45t-5t+25$$

$$0=9t\left(t-5\right)-5\left(t-5\right)$$

$$0=\left(9t-5\right)\left(t-5\right)$$

$$t=5\text{sec}$$

So, the particle total in 5 sec,

$$H=\frac{1}{2}gt^2$$

$$H=\frac{1}{2}\times 10\times 25$$

$$H=125\text{m}$$

So, the total distance of tower is 125m.

17. Correct option: (C) 49m.

Given:

$$u=14.7\frac{\text{m}}{\text{s}}$$

$$t=5\text{sec}$$

$$a=-9.8\frac{\text{m}}{\text{s}^2}$$

Now using the equation of motion,

$$s = ut + \frac{1}{2}at^2$$

$$s = 14.7 \times 5 - \frac{1}{2} \times 9.8 \times (5)^2$$

$$s = -49\text{m}$$

So, the height of tower is 49m.

18. Correct option: (C) 1 sec, 4.9m.

Given:

$$g = 9.8\text{ms}^{-2}$$

$$a = 0.2\text{ms}^{-2}$$

$$a_{rel} = (9.8 + 0.2) = 10\text{ms}^{-2}$$

The displacement is given by as,

$$s = \frac{1}{2}a_{rel}t^2$$

$$t = \sqrt{\frac{2 \times s}{a_{rel}}}$$

$$t = \sqrt{\frac{2 \times 5}{10}}$$

$$t = 1\text{sec}$$

Now distance covered of elevator is $= \frac{1}{2} \times 0.2 \times 1 = 0.1\text{m}$

Displacement of bolt $= 5 - 0.1 = 4.9\text{m}$

Time taken by bolt to strike the floor is 1 sec.

Distance travelled is 4.9m.

19. Correct option: (C) 6s.

Distance travelled in 2s is given by as,

$$h = \frac{1}{2}gt^2$$

$$h = \frac{1}{2} \times 10 \times 2^2 = 20\text{m}$$

Velocity after 2s is given by as,

$$v = gt$$

$$v = 10 \times 2 = 20\frac{\text{m}}{\text{s}}$$

Since there is no acceleration, the body might cover the final 80 metres $(100 - 20 = 80\text{m})$ at a constant $20\frac{\text{m}}{\text{s}}$ speed.

So, the time taken for the remaining journey is, $2 + 4 = 6\text{s}$

$$t = \frac{80}{20} = 4s$$

There for total time= $2 + 4 = 6\text{s}$.

20. Correct option: (D) 6g.

Given:

$u = 0$

$t = 2\,\text{sec}$

Height of second ball is given by as,

$$h_2 = ut + \frac{1}{2}gt^2$$

$$h_2 = \frac{1}{2}g \times (2)^2$$

$$h_2 = 2g$$

Height of first ball is given by as,

$$h_1 = \frac{1}{2}g \times (4)^2 = 8g$$

Separation after 2 sec is ,

$$\text{Separation} = 8g - 2g = 6g$$

So, the separation after 2 sec is 6g.

21. the correct option (A) 45m.

Given: Initial velocity, $u = 0$

Let us consider the total height of tower is h .

Using the equation of motion we get,

$$H = ut + \frac{1}{2}gt^2 = \frac{1}{2}gt^2 \cdots\cdots (1)$$

Height before last second is given by as,

$$H - 25 = \frac{1}{2}g(t-1)^2$$

$$\frac{1}{2}gt^2 - 25 = \frac{1}{2}gt^2 + \frac{1}{2}g - gt$$

$$5t^2 - 25 = 5t^2 + 5 - 10t$$

$$30 = 10t$$

$$t = 3\text{s}$$

Substituting in equation(1),

$$H = \frac{1}{2} \times 10 \times (3)^2 = 45\text{m}$$

So, height of tower is 45m.

22. Correct option: (C) $y - x = g$ **.**

For first case:

Using the formula:

$$x = \frac{g}{2}(2n-1)$$

$$\left(\frac{2x}{g}\right) + 1 = 2n \cdots\cdots (1)$$

For second case:

$$y = \frac{g}{2}(2n+1)$$

$$\left(\frac{2y}{g}\right) - 1 = 2n \cdots\cdots (2)$$

Using equation (1) and (2) we get,

$$\left(\frac{2x}{g}\right) + 1 = \left(\frac{2y}{g}\right) - 1$$

$$2g = 2y - 2x$$

$$g = y - x$$

So, the answer is $y - x = g$.

23. correct option (B) $\sqrt{2}v$.

Given: $u = 0$

Using third equation of motion,

$$v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2gh$$

$$v = \sqrt{2gh} \cdots\cdots (1)$$

For falling through the next h ,

$$v_2^2 = u^2 + 2gh_2$$

$$v_2^2 = 4gh$$

Now using equation (1),

$$v_2^2 = 4gh = 2v^2$$

$$v_2 = \sqrt{2}v$$

So, the acquired velocity is $\sqrt{2}v$.

24. Correct option: (B) 9.8m.

Given: The average velocity of freely falling body, $V_{av} = 7\frac{\text{m}}{\text{sec}}$.

$$V_{avg} = \frac{\text{displacement}}{\text{time}}$$

Let time is 't',

$$= H = \frac{1}{2}gt^2$$

Now,

$$V_{avg} = \frac{\frac{1}{2}gt^2}{t} = 7$$

$$t = \frac{14}{g}$$

Therefore,

$$H = \frac{1}{2} g \left(\frac{14}{g} \right)^2$$

$$H = 9.8$$

So, the required height is 9.8m.

25. Correct option: (A) The particle starts with certain velocity but the motion is retarded and finally the particle stops.

The slope is initially decreasing, and it equals 0.

As a result, as time passes, there is retardation, and eventually the particle stops moving. It had to have gotten going with some speed, clearly.

26. Correct option: (D).

Since the direction of the velocity during rise and fall is opposite to each other (in the positive zone, velocity falls linearly during rise, and in the negative region, velocity increases linearly during fall), fall is depicted in the negative region.

The ball moves anticlockwise to gravity (which is constant), therefore as it rises, its velocity falls linearly until it reaches zero (at the highest point). The ball rises when it moves in the direction of gravity, but when it falls, it moves in the other direction and accelerates. As a result, the fall is depicted in the negative zone.

27. Correct option: (A) 8m, 16m.

Area under the v-t graph give the distance travelled by a body.

The distance = area under the curve

So, the distance $= (4 \times 2) + (2 \times 2) + (2 \times 2) = 16\text{m}$

The displacement is given by area under graph with positive velocity and subtract area enclosed with negative velocity.

The displacement $= = (8 \times 4) - (2 \times 2) = 8\text{m}$

So, the displacement of body is 8m and distance travelled by the body is 16m.

28. Correct option: (D) $6 \frac{\text{cm}}{\text{s}^2}$**.**

Acceleration in the first part,

$$a_1 = \frac{20 - 0}{20} = 1 \frac{\text{cm}}{\text{s}^2}$$

Acceleration in the second part,

In this part there is no change in velocity.

$$a_2 = 0$$

Acceleration in the third part,

$$a_3 = \frac{80 - 20}{40 - 20} = \frac{60}{10} = 6 \frac{\text{cm}}{\text{s}^2}$$

Acceleration in the fourth part,

$$a_4 = \frac{80 - 0}{80 - 40} = \frac{80}{40} = 2 \frac{\text{cm}}{\text{s}^2}$$

So, the maximum acceleration is $6 \frac{\text{cm}}{\text{s}^2}$.

29. Correct option: (A).

$v = \text{constant}$

When a body is moving uniformly, its speed remains constant over time.

If a body moves at a constant pace and covers the same distances in the same amounts of time, it is said to be in uniform motion. The speed-time graph and the distance-time graph, which both display a straight line parallel to the time axis, respectively, illustrate uniform motion.

30. Correct option: (D) Change in velocity.

The change in velocity is represented by the area under an acceleration graph. In other words, the change in velocity for a given time period is equal to the area under the acceleration graph for that time period.

Acceleration= $\frac{\Delta v}{\Delta t}$

Area between acceleration-time graph gives:

$a \times \Delta t = \frac{\Delta v}{\Delta t} \times \Delta t = \Delta v$

LEVEL-II

1. Correct option: (A) $1:\sqrt{2}$**.**

Height of ball one, $h_1 = h$

Height of ball two, $h_2 = 2h$

For free fall time is given by as,

$t = \sqrt{\frac{2S}{g}}$

Where, S is the vertical distance

Therefore, the ratio of time of these balls to reach the earth is given by as,

$\frac{t_1}{t_2} = \sqrt{\frac{2h_1}{2h_2}} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{h}{2h}} = \frac{1}{\sqrt{2}}$

$t_1 : t_2 = 1:\sqrt{2}$

So, , the ratio of time of two balls to reach the earth is $1:\sqrt{2}$.

2. Correct option: (A) $\sqrt{3}v_0$**.**

Maximum height indicates zero final velocity.

Therefore, Using the equation of motion,

$v^2 - u^2 = 2gh$

Now, put $h = H_{\max}, v = 0$

$v_0^2 = 2gH_{\max}$

$H_{\max} = \frac{v_0^2}{2g}$

If height is triple, then the maximum height is given by as,

$3H = \frac{u^2}{2g}$

$u^2 = 3v_0^2$

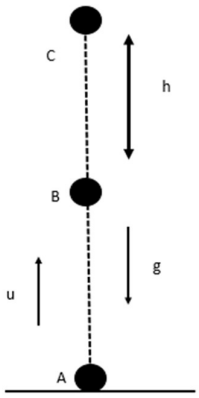
$u = \sqrt{3}v_0$

So, To triple the maximum height, ball should be thrown with velocity $\sqrt{3}v_0$.

3. Correct option: (B) $\frac{1}{2}gt^2$.

Let us consider the time of flight is T then, $T=\frac{u}{g}$.

Let h represent the distance travelled during the last 't' seconds of the object's ascension.



Velocity at point B=$= V_B = u - g\left(T - t\right)$

$= V_B = u - g\left(T - t\right)$

$\Rightarrow u - g\left(\frac{u}{g} - t\right) = gt$

$\Rightarrow h = V_B t - \frac{1}{2}gt^2$

$\Rightarrow h = gt^2 - \frac{1}{2}gt^2 = \frac{1}{2}gt^2$

So, the distance travelled during the last 't' seconds of the object's ascension is $\frac{1}{2}gt^2$.

4. Correct option: (A) 0.

Area under the curve give the displacement.

As is well known, the area beneath the velocity-time graph indicates an object's displacement. In this instance, the graph creates two distinct triangles with the same area but in opposite quadrants. The area of the two triangles cannot be added up because it is zero. (As the symbol is opposed to both, as stated).

We know that velocity is define as,

$$\text{Average velocity}=\frac{\text{Total distance}}{\text{Total time}}$$

The overall displacement is zero and the area under the v-t curve is zero for 10 seconds.

5. Correct options: (A) and (D).

The slope of the curve is constant, therefore the particle accelerates at a fixed rate.

At $t = 10$ seconds, the particle reverses direction.

Because the area above and below the $x-$ axis are not equal, the particle has a non-zero displacement.

Because of the symmetry, the average speed between the intervals of 0 and 10 seconds is identical to that between 10 and 20 seconds. slope and the size of the region under the curve are both the same.

So, Correct option: (A) and (D).

6. Correct options: (A), (C) and (D).

In this time velocity is positive till $t = 24\,\text{sec}$.

Therefore, Area till 24 sec,

$$A = 2\times10 + \frac{1}{2}(2+6)\times8 + \frac{1}{2}\times6\times6$$

$$A = 20\text{m}$$

At, $t = 0, x = -16\text{m}$

Maximum coordinate of position = $70 + (-16) = 54\text{m}$

Now position of particle till $t = 18\,\text{sec}$.

So,

$$\text{Area} = (2\times10 + \frac{1}{2}\times\left((2+6)\times8\right) - 16$$

$$\text{Area} = 36\text{m}$$

Position of the particle till $t = 30\,\text{sec}$ is,

$$\text{Area} = 2\times10 + \frac{1}{2}(2+6)\times8 + \frac{1}{2}\times6\times6 - \frac{1}{2}\times6\times6$$

$$\text{Area} = \left(20+32+18-18\right) - 16$$

$$\text{Area} = 36\text{m}$$

So, Correct option: (A),(C) and (D).

7. Correct option: (A), (D)

8. Correct option: (D)

9. Correct option: (A), (B) and (C).

The stone won't have any initially velocity towards the balloon, but after it is released, it will travel at a speed that is equal to the balloon's instantaneous speed relative to the earth. So, (A) and (B) are both true.

After being released, the stone will accelerate downward by g. Acceleration will therefore be $\sqrt{g+a}$ with regard to the balloon. (C) is therefore true.

10. Correct option: (B) $t_1 > t_2$.

Using equation of motion,

$$s = \frac{1}{2}gt^2$$

If the time taken for the first 50m is t_1,

$$t_1^2 = \frac{50\times2}{g} = \frac{100}{g}$$

$$t_1 = \frac{10}{\sqrt{g}}$$

Now, If body is fall from a height of 100m, then time is t,

$$100 = \frac{1}{2}gt^2$$

$$t = \frac{10\sqrt{2}}{\sqrt{g}}$$

Now, for remaining 50m time period is t_2,

Therefore,

$t_2 = t - t_1$

$t_2 = \frac{10}{\sqrt{g}}\left(\sqrt{2}-1\right)$

$t_2 = 0.4t_1$

$t_1 > t_2$

11. Correct option: (A) 5:2.

Using equation of motion,

$s = \frac{1}{2}gt^2$

If the time taken for the first 50m is t_1,

$t_1^2 = \frac{50\times 2}{g} = \frac{100}{10}$

$t_1 = \sqrt{10} = 3.162\,\text{sec}$

Now, If body is fall from a height of 100m, then time is t,

$100 = \frac{1}{2}gt^2$

$t = \sqrt{20}$

Therefore,

$t_2 = t - t_1$

$t_2 = \sqrt{20} - \sqrt{10}$

$t_2 = 1.309\,\text{sec}$

Now,

$\frac{t_1}{t_2} = \frac{3.162}{1.309}$

$\frac{t_1}{t_2} = 2.432\cdots\cdots(1)$

or

$\frac{t_1}{t_2} = \frac{5}{2} = 2.5\cdots\cdots(2)$

Form equation (1) and equation (2), the option (A) is correct.

12. Correct option: (B) $\sqrt{2}:1$**.**

Using equation of motion,

$s = \frac{1}{2}gt^2$

If the time taken for the first 50m is t_1,

$t_1^2 = \frac{50\times 2}{g} = \frac{100}{g}$

$t_1 = \frac{10}{\sqrt{g}}$

Now, If body is fall from a height of 100m, then time is t,

$$100 = \frac{1}{2}gt^2$$

$$t = \frac{10\sqrt{2}}{\sqrt{g}}$$

Therefore,

$$t : t_1 = \frac{10\sqrt{2}}{\sqrt{g}} \times \frac{\sqrt{g}}{10} = \sqrt{2}$$

$$t : t_1 = \sqrt{2} : 1$$

So, the ratio of times to reach the ground and to reach first half of the distance is $\sqrt{2}:1$.

13. Correct option: (B) 5s.

Given:

The average velocity= $= 20\frac{\text{m}}{\text{s}}$

From the graph time period is 25 sec.

Therefore distance is given by as,

$$\text{Average velcoity} = \frac{\text{Distnace}}{\text{time}}$$

$$\text{distnace} = \text{Average velcoity} \times \text{time}$$

$$\text{Distnace(s)} = 25 \times 20 = 500\text{m}$$

In order to get the area of the velocity-time graph, we add the areas of two triangles and a rectangle.

So,

$$Area = 2 \times \frac{1}{2} \times t \times 5t + (25 - 2t) = 500$$

$$t^2 - 25t + 100 = 0$$

Solving above equation we get,

$t = 5\,\text{sec}$

So, the value of t is 5s.

14. Correct option: (B) 25ms^{-1}**.**

The rate of acceleration and deceleration is constant and it is equal to 5ms^{-1}.

From the graph maximum velocity= 5t

$\because$ t=5sec

Therefor

Maximum velocity= $5 \times 5 = 25\text{ms}^{-1}$

So, the maximum velocity 25ms^{-1}.

15. Correct option: (A)

16. Correct option: (C) $\frac{1}{2}g_1t_1t_2$**.**

Total time of flight= $t_2 + t_1$

$$T = t_2 + t_1 = \frac{2u}{g}$$

$$u = \frac{Tg}{2} = \frac{(t_1 + t_2)g}{2}$$

If u is initial velocity.

Using equation of motion,

$$h = ut_1 - \frac{1}{2}gt_1^2$$

$$h = \frac{(t_2 + t_1)t_1 g}{2} - \frac{1}{2}gt_1^2$$

$$h = \frac{1}{2}gt_1(t_2 + t_2 - t_1) = \frac{1}{2}gt_1t_2$$

So, the height is $\frac{1}{2}g_1t_1t_2$.

17. Correct option: (A) $\frac{1}{2}g(t_1 + t_2)$**.**

Total time of flight= $t_2 + t_1$

$$T = t_2 + t_1 = \frac{2u}{g}$$

$$u = \frac{Tg}{2} = \frac{(t_1 + t_2)g}{2}$$

So, the velocity of projection is $\frac{1}{2}g(t_1 + t_2)$.

18. Correct option: (C) $\frac{1}{8}g(t_1 + t_2)^2$**.**

Total time of flight= $t_2 + t_1$

$$T = t_2 + t_1 = \frac{2u}{g}$$

$$u = \frac{Tg}{2} = \frac{(t_1 + t_2)g}{2}$$

Now using equation of motion,

$$v^2 = u^2 - 2gh_1$$

$$0 = \left[\frac{g}{2}(t_1 + t_2)\right]^2 - 2gh_{\max}$$

$$h_{\max} = \frac{g}{8}(t_1 + t_2)^2$$

So, the maximum height is $\frac{1}{8}g(t_1 + t_2)^2$.

19. a-S; b-R; c-Q; d-P

20. a-Q; b-S; c-R; d-R

3. Newton's Laws of Motion

Solutions

LEVEL – I

1. Correct option: (C) Both A and B.

The remark that follows alludes to the idea of forces in physics. A force is a push or a pull that can alter how an object moves. Depending on the kind and strength of the applied force, it has the ability to either move a stationary body or halt a moving one.

(A) A balanced force is a pair of forces that are opposing in direction and of equal magnitude as they act on an object. In this scenario, the object experiences no net force and either remains immobile or keeps moving in a straight path at a constant speed. The object will remain at rest, for instance, if you push it to the right with a force of 10 newtons and your friend pulls it to the left with a force of 10 newtons as well.

(B) Unbalanced Force: Two forces acting on an item that are not in opposition to one another or that are not equal in magnitude are referred to as unbalanced forces. Because there is a force acting on the object in this situation, its motion is altered. For instance, if you push something with a force of 15 newtons to the right and a buddy pulls it with a force of 10 newtons to the left, the forces are out of balance, and the thing will move to the right faster due to the net force of 5 newtons.

Accordingly, depending on their unique properties and the consequent net force exerted on the item, both balanced and unbalanced forces may be to blame for moving a stationary body or stopping a moving body.

2. Correct option: (A) balanced force.

A balanced force has the ability to alter size and shape. So a toothpaste tube changes shape when it is squeezed. An illustration of a balanced force is the force that caused this.

3. Correct option: (A) Balanced force act on the balloon.

A rubber balloon is held between the hands, and when pressed, it changes shape. This occurs because A) The balloon is subject to balanced force. B) The balloon is affected by an uneven force

C) The balloon experiences frictional forces. D) The balloon experiences the effects of gravity The balloon is compressed from two sides by unbalanced hand pressures. The balloon shape changes as a result.

As we All know, Newton's first law of motion states that unless an external force is applied, a body tends to be in a condition of inertia. Here, an uneven force is taken into account. This is due to the fact that only an uneven force can cause a body to move. Or, we may say that an unbalanced force tends to cause bodily motion because it isn't balanced by other forces.

The opposing force must counteract the balanced force in order to achieve balance. In other words, there is no motion caused by this force. But occasionally, it can cause an alteration in the body's shape. In the scenario under consideration, a rubber balloon encounters a balanced force when pressed between hands, which causes the rubber balloon to alter shape.

4. Correct option: (D) All the above.

A) Change in the body's speed: An unbalanced force might cause the body to accelerate or decelerate, changing its speed. The speed will rise if the force acts in the direction of the motion; it will decrease if the force works in the opposite direction.

B) Change in bodily shape: An imbalanced force can cause the body to deform or change its shape. This effect is especially important for deformable items like fluids or soft materials. The body's shape may change as a result of the force through compression, stretching, or bending.
C) Change in the body's motion direction: An unbalanced force has the power to alter the body's motion direction. The body may change its trajectory and move in a different direction when the force is applied at an angle to the original direction of motion.
In conclusion, all the effects listed in alternatives A, B, and C are caused by changes in the motion, speed, and shape of a body due to an unbalanced force. Therefore, D) All of the above is the right response.

5. Correct option: (B) Forward.

The passengers of a halted car are abruptly pushed forward when this happens.
Because of the inertia opposing the backward motion, when the car abruptly stops in the backward direction from a state of motion, the passengers feel a force applied to them in the forward direction.
As a result, when a moving car suddenly stops, the passengers are thrusting forward.

6. Correct option: (A) Backward.

The passengers of a moving car are abruptly yanked backwards as the car stops.
If a body is at rest or moving in a straight line at a constant speed, Newton's first rule of motion or inertia states that "unless an external force is applied to it, it will remain at rest or keep moving in a straight line at constant speed."
The backward jerk that occurs when a stationary car starts is felt by the passengers. The occupants experience a force being applied to them as the car begins and abruptly moves forward from a condition of rest due to inertia. The forward motion is resisted by the inertia.

7. Correct option: (B) inertia of motion.

The tangent to the wheel is the straight line along which the mud particles continue to move when they break free from the wheel. This is a result of motion's inertia.

8. Correct option: (C)mass of the object.

The first law of motion states that "A body continues to be in its state of rest or uniform motion along a given direction, until and unless an external force is applied on it." The law of inertia is another name for this.
A body's inability to alter its position of rest or uniform motion is known as inertia.
A body's mass can be thought of as its inertia in motion.
Inertia is therefore dependent on mass. The inertia of a body increases with mass, making movement of the body more challenging.

9. Correct option: (A) Newton's first law of motion.

Newton's First Law of Motion offers the foundation for the solution to this issue. At first, both the leaves and the tree are dormant. However, when a tree is shook severely, the tree moves while the leaves tend to remain still. As a result, if we shake a tree limb aggressively while it is in the position of rest, some of the leaves may come off the tree.

10. The correct option (B) move forward.

Newton's first law of motion states that a body in motion will stay in motion even in the absence of an external force. Since the water in the tanker is not a component of the tanker, it is not stopped when the tanker is put on brakes because there is no outside force acting on it. As a result, the water in the tanker will advance. As long as there is no external force acting on an object, it will continue to move in the same direction and at the same pace. In this scenario, using the brakes won't stop the water from going forward. If the tanker doesn't have a closed roof, the tanker will stop, but the water will keep flowing and could overflow.

11. Correct option: (D) mass.

The inertia of an object is determined by its mass. Therefore, inertia is measured by mass.

12. Correct option: (D) Keep it moving with uniform velocity.

According to Newton's first law, a body that is at rest or travelling in a straight path at a constant speed will continue to do so unless it is subjected to an external force. The law of inertia is the name given to this premise.

Therefore, a force is not needed to maintain the particle's uniform speed.

13. Correct option: (C) Inertia.

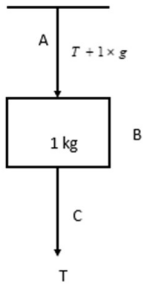
The body's inertia is described by Newton's first law of motion. According to this, everyone has a propensity to hold onto their state (whether it be rest or motion) because of their inertia.

According to the first law of motion, "a body at rest will stay at rest, and a body in motion will continue to move in a straight line until an external force acts on it."

14. Correct option: (A) Inertia of direction.

Due of directional inertia, passengers are thrown outward. Passengers will be thrown forward if the breaks are rapidly applied, and backward if the bus is suddenly accelerating.

15. Correct option: (B) The portion BC of the string will break.



Imagine that the string C experiences a force and develops a tension T.

The tension in piece AB for an equilibrium of a 1-kilogram block will be $T+1\times g$, hence the portion AB will always break first. When C receives an abrupt jolt, C creates an impulsive tension that is greater than the breaking tension initially, which breaks before this impulse can travel to A as a wave through a block.

16. Correct option: (A) (i) is true & (ii) is false.

Because the body of a bus passenger tends to travel forward due to inertia of motion, the passenger would jump off and run in the direction of the bus motion.

17. Correct option: (A) Inertia of rest.

When the bus accelerates rapidly, the passenger resists changing their condition of rest and continues to rest in the same manner, it will look like the person is regressing. The fact that No one can independently alter its state of motion.

18. Correct option: (B) Force.

The rate of change of momentum is directly inversely proportional to applied force, according to Newton's second law.

$F \propto \frac{m(v-u)}{t}$ from this , de deduce,

$F = ma$

So, Newton's 2nd law gives us measure of force.

19. Correct option: (C) Net force.

Mass is a scalar quantity; hence the direction of force and acceleration will be the same according to Newton's Second Law of Motion, which states that the net force on an object is equal to mass times its acceleration.

20. Correct option: (A) $2mv$**.**

The initial velocity of the body before collision $u = -V$

where the minus symbol denotes a movement towards the wall by the body.

Now initial momentum of the body is given by as, $p_i = mu = -mV$

The final velocity of the body after the collision, $v = V$

when the movement of the body away from the wall is shown positively.

Now, final momentum of the body is given as,

$p_f = mv = mV$

Change in momentum of body is given by as,

$\Delta P = P_f - P_i = mV - (-mV) = 2mV$

So, the change in momentum is $2mv$.

21. Correct option: (B) Proportional to the net force applied on it.

Newton's second law of motion states that "the change in momentum is in the direction of the applied force and the rate of change of momentum is directly proportional to the force applied." The force is expressed mathematically by Newton's second law of motion.

$$\text{Force} \propto \frac{dp}{dt}$$

22. Correct option: (B) Reduce the force exerted by the ball.

As the duration of the ball's slowing down increases, the impulse that the ball applies to the hand by moving the hand rearward decreases. As a result, the fielder uses less force to capture the ball with his hands.

23. Correct option: (B) $5\frac{\text{m}}{\text{s}^2}$**.**

Given:

$F = 5\text{N}$

$W = 9.8\text{N}$

Now we know that,

$W = mg$

$9.8 = m \times 9.8$

$m = 1\text{kg}$

Now force is given by as,

$F = ma$

$5 = 1 \times a$

$a = 5\frac{\text{m}}{\text{s}^2}$

So, the acceleration produced is $5\frac{\text{m}}{\text{s}^2}$.

24. The correct option: (B) $2\frac{\text{m}}{\text{s}^2}$**.**

Given:

$m = 2m$

$a = 4\frac{\text{m}}{\text{s}^2}$

Now force is given by as,

$F = ma$

$F = 4m$

If the car pulls another car of same mass,

Now force is given by as,

$F = 2ma'$

$a' = \frac{F}{2m}$

Now put the value of F in above equation,

$a' = \frac{4m}{2m}$

$a' = 2\frac{m}{s^2}$

So, the acceleration produced is $2\frac{\text{m}}{\text{s}^2}$.

25. Correct option: (C) South-west.

A force acting in the north-east is applied to an item. A second force should be applied in the south-west to counteract this one.

The second force should be applied in a 180° direction.

26. Correct option: (A) $10\sqrt{2}\text{kg}$**.**

Given:

Force vector, $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$

Now magnitude of the force is given by as,

$|F| = \sqrt{36 + 64 + 100} = 10\sqrt{2}N$

Now mass is given by as,

$m = \frac{F}{a} = \frac{10\sqrt{2}}{1} = 10\sqrt{2}\text{kg}$

So, the mass of the body is $10\sqrt{2}\text{kg}$.

27. Correct option: (C)2N.

Given:

$m = 10\text{kg}$

$a = 0.2\frac{\text{m}}{\text{s}^2}$

Now, force is given by as,

$F = ma$

$F = 10 \times 0.2$

$F = 2\text{N}$

So, the magnitude of force acting on the body is 2N

28. Correct option: (A) 1800 N.

Given:

$m = 1800\text{kg}$

$u = 10\frac{\text{m}}{\text{s}^2}$

$s = 50\text{m}$

Now using the equation of motion,

$v^2 - u^2 = 2as$

$0 - 10^2 = 2a \times 50$

$a = \frac{-100}{100} = -1\frac{\text{m}}{\text{s}^2}$

Now, force is given by as,

$F = ma$

$F = 1800 \times 1$

$F = 1800\text{N}$

So, the force acting on the ca is 1800 N.

29. Correct option: (B) 10 N.

Given:

$t = 4\text{sec}$

$s = 10\text{m}$

m=2kg

Now, using the second equation of motion,

$s = ut + \frac{1}{2}at^2$

$10 = 0 + \frac{1}{2}a \times 4$

$a = \frac{10}{2} = 5\frac{m,}{s^2}$

Now force is given by as,

$F = ma$

$F = 2 \times 5$

$F = 10\text{N}$

So, the magnitude of the force is given by as 10N.

30. Correct option: (A) $12500\hat{i} + 6250\hat{j}$ **.**

$m = 10g = 10 \times 10^{-3}\text{kg}$

F=$10\hat{i} + 5\hat{j}$

Now using the force formula,

$ma = 10\hat{i} + 5\hat{j}$

$10 \times 10^{-3} a = 10\hat{i} + 5\hat{j}$

$\frac{10}{1000} a = 10\hat{i} + 5\hat{j}$

$a = 1000\hat{i} + 500\hat{j}$

$\frac{dv}{dt} = 1000\hat{i} + 500\hat{j}$

$dv = \left(1000\hat{i} + 500\hat{j}\right) dt$

$v = 1000t\hat{i} + 500\hat{j}$

Now,

Velocity is $x-$ direction,

$\frac{dx}{dt} = 100t$

$x = \int_0^5 100t dt$

$x = \frac{100(5)^2}{2}\hat{i}$

$x = 12500\hat{i}$

Velocity is $y-$ direction,

$\frac{dy}{dt} = 500t$

$x = \int_0^5 500t dt$

$x = \frac{500(5)^2}{2}\hat{i}$

$x = 6350\hat{j}$

So, the position vector is $12500\hat{i} + 6250\hat{j}$.

31. Correct option: (B) 6250 N.

Given:

$m = 10\text{g}=0.01\text{kg}$

$\text{u}=250\frac{\text{m}}{\text{s}}$

$s = 5\text{cm} = 0.05m$

$v = 0\frac{\text{m}}{\text{s}}$

Using the equation of motion,

$v^2 - u^2 = 2as$

$0^2 - 250^2 = 2 \times 0.05 \times a$

$a = \frac{62500}{0.1} = -625000\frac{\text{m}}{\text{s}^2}$

The magnitude of force is given by as,

$F = ma$

$F = 0.01 \times 625000$

$F = 6250\text{N}$

So, the magnitude of the force is 6250 N.

32. Correct option: (A) 0°**.**

The weight (W) and the action (A) will both move in the same direction and side by side. As a result, there should be no angle between them.

On a table, a book is said to be lying. It is necessary to determine the angle between the weight of the book and how it acts on the table. The phrase "action of the book on the table" refers to the force that the book is exerting on the surface.

Action forces are often defined as forces that can operate independently, in response to a field, or as a result of biological beings like humans and animals. The weight of a body, electrostatic attraction and repulsion, and other forces are examples of action forces.

Reaction forces, on the other hand, can be thought of as forces that do not exist in the absence of an action force. They are there to counter whatever that a force of action has acted upon. The normal force, friction, tension, and other forces can be reaction forces.

The only force exerted on the table by the book now that it is lying (and not moving) is its weight. Consequently, this might be regarded as the book's action. The angle is zero since the book's weight and motion on the table both move in the same direction.

33. Correct option: (D) $2\frac{\text{m}}{\text{s}}$**.**

Acceleration of mass m_1, $a_1 = 6\frac{\text{m}}{\text{s}^2}$

Acceleration of mass m_2, $a_2 = 3\frac{\text{m}}{\text{s}^2}$

Now we know that force is given by as,

$F = ma$

$m = \frac{F}{a}$

$m_1 = \frac{F}{6} \quad or \quad m_2 = \frac{F}{3}$

If the two masses are fixed together, and the same force is applied to the combination.

So, the net force is given by as,

$F = (m_1 + m_2)a$

$F = \left(\frac{F}{6} + \frac{F}{3}\right)a$

$F = \left(\frac{F + 2F}{6}\right)a$

$a = \frac{6}{3}$

$a = 2\frac{\text{m}}{\text{s}^2}$

So, the acceleration is $2\frac{\text{m}}{\text{s}}$.

34. Correct option: (C) $50\frac{\text{kg m}}{\text{s}}$**.**

Given:
Mass of the body, $m = 10\text{kg}$

Velocity, $v = 5\frac{\text{m}}{\text{s}}$

So, the linear momentum is given by as,
$p = mv$

$p = 10 \times 5$

$p = 50\frac{\text{kg m}}{\text{s}}$

So, the linear momentum is $50\frac{\text{kg m}}{\text{s}}$.

35. The correct option (B) $100\frac{\text{kg m}}{\text{s}}$**.**

Initial velocity, $u = 0\frac{\text{m}}{\text{s}}$

Acceleration, $a = 4\frac{\text{m}}{\text{s}^2}$

Time, $t = 5\text{sec}$
Mass, $m = 5\text{kg}$
Now using the equation of motion,
$v = u + at$

$v = 0 + 4 \times 5$

$v = 20\frac{\text{m}}{\text{s}}$

Now , momentum is given by as,
$p = mv$
$p = 5 \times 20$
$p = 100\frac{\text{kg}m}{s}$
So, the momentum is $100\frac{\text{kg}m}{s}$.

36. Correct option: (A) Backward

According to Newton's third law, "Every action has an equal and opposite reaction," a forward jump causes a boat to experience backward force. As a result, the boat is moving backward.

37. Correct option: (D) All of the above.

According to Newton's Third Law of Motion, "There is always an equal and opposite reaction to every action."
Following are interpretations of Newton's Third Law of Motion.

1. There are always two forces at work. The forces exerted on a body A by a body B are equal and in opposition to each other.
2. The forces exerted on A by B and B by A act simultaneously.
3. Forces of action and reaction act on various bodies rather than the same body

38. Correct option: (B) (i) is false & (ii) is true.

The horse experiences some force from the ground as it applies some force to it. The horse moves as a result of this response force. As the horse advances, the force in turn pulls the cart. Therefore, the force of the earth is what causes the horse to move. Friction force is the name given to this force.

39. Correct option: (D) all the above .

According to Newton's third law of motion, every action has a corresponding and opposing reaction.

The third rule of motion of Newton is the physical concept that governs how a jet aircraft operates. There is an equal and opposite reaction to every action, according to Newton's third law of motion. A jet plane pushes air backward, and air pushes the jet plane backward in the other direction.When

someone discharges a gun, a reaction force forces him to go backward.

The ground is pushed by the rocket during rocket lunching, and is then pushed back by the opposing force.

40. Correct option: (C) Third law of motion.

Swimming is based on Newton's third rule of motion since we move forward and experience equal forces from the water as we push it.

41. Correct option: (D) linear momentum.

The forces acting on the two objects are equal in size and react in opposition, according to Newton's third law. Such pressures result in one object gaining momentum while the other loses momentum, preserving the system's overall momentum in the absence of any external forces. As a result, the law of conservation of momentum follows from Newton's third law of motion. According to Newton's third law, an object B will exert a force on an object A that is identical in magnitude but directed in the opposite direction. Newton derived the law of conservation of momentum using this concept.

42. Correct option: (C) Gravitational force

Contact forces are the forces that apply to bodies when they come into physical contact. For instance, frictional force and the force applied when two bodies collide.

Non-contact forces are those that affect bodies without actually making physical contact with them. Examples include the gravitational and electrostatic forces.

So, the friction, Spring force and tension force are contact force and the gravitational force is non-contact force.

LEVEL -II

1. Correct option: (C) Upper part of the body continues to be in the state of motion, whereas the lower part of the body in contact with the seat comes to rest.

As a result of inertia. The lower body tends to come to rest when a fast bus abruptly stops, but the upper body typically continues to move uniformly. The passengers are consequently propelled forward. When a train suddenly stops, passengers in the moving train experience an instantaneous jolt forward because the upper half of their bodies is still in motion while the lower part of their bodies that are in contact with the seat are still at rest.

2. Correct option: (C) Newton's first law of motion.

An object's inertia tends to oppose any modification to either its condition of rest or state of motion. A carpet starts moving when it is struck with a stick. However, dust particles make an effort to fight off changes to their resting condition. Newton's first law of motion states that as the carpet moves, the dust particles remain at rest. As a result, due to inertia, the dust particle emerges through the carpet.

3. Correct option: (D) $2\times10^{7}\,\frac{\mathrm{N}}{\mathrm{m}^2}$.

Number of balls, $N=10000$

Mass of each ball, $m=1g=1\times10^{-3}\mathrm{Kg}$

Area, $A=1\mathrm{cm}^2=10^{-4}\mathrm{m}^2$

Velocity, $v=100\,\frac{\mathrm{m}}{\mathrm{s}^{-1}}$

Pressure is given by as,

$$P=\frac{F}{A}=\frac{ma}{A}$$

$$P=\frac{10000\times10^{-3}\left(v-(-v)\right)}{10^{-4}}$$

$$P=2\times10^{7}\,\frac{\mathrm{N}}{\mathrm{m}^2}$$

So, the pressure on the surface is $2\times10^{7}\,\frac{\mathrm{N}}{\mathrm{m}^2}$.

4. Correct option: (A) 5000 N.

Given:

The mass of the vehicle, $m=100\mathrm{kg}$

Initial velocity, $u=5\,\frac{\mathrm{m}}{\mathrm{s}}$

Time required to stop the vehicle, $t=\frac{1}{10}=0.1\mathrm{sec}$

Final velocity, $v=0$,

From the newton second law of motion,

$$F=\frac{dp}{dt}$$

Here, $p=$momentum

Now,

$$F=m\frac{v-u}{t}$$

$$F=100\frac{[0-5]}{0.1}$$

$$F=-5000N$$

Negative sign indicates the force is in the opposite direction.

So, the value of force is 5000 N.

5. Correct option: (B) $2mnu$.

Maas of each ball= $m$kg

Velocity, $=u\,\frac{\mathrm{m}}{\mathrm{s}}$

In elastic collision we have ,

Change in momentum = $mu-(mu)=mu+mu=2mu$

Now for N balls change in momentum is given by as,

$$\Delta p=N\times2mu$$

Rate change of momentum = $\frac{\Delta p}{t} = \frac{N2mu}{1\sec} = N2mu$

Now according to Newton second law of motion,

$F = \frac{\Delta p}{t}$

$F = 2Nmu$

So, the force experienced by the surface is $2Nmu$.

6. Correct option: (D) $\sqrt{2}m_1 = m_2$.

We know that ,

$v \propto \sqrt{s}$

$\frac{v_1}{v_2} = \sqrt{\frac{2h}{h}} = \frac{\sqrt{2}}{1}$

For momentum,

$\frac{P_1}{p_2} = \frac{m_1 v_1}{m_1 v_2}$

$\frac{P_1}{p_2} = \frac{m_1}{m_2} \times \frac{\sqrt{2}}{1}$

$\sqrt{2}m_1 = m_2$

So, the required ratio is $\sqrt{2}m_1 = m_2$.

7. Correct option: (C) $\frac{a_1 a_2}{a_1 + a_2}$.

Now using the Newton law of motion,

Force on first body, $F_1 = m_1 a_1$

Force on the second body, $F_2 = m_2 a_2$

If the two bodies are combined, then force is,

$F = (m_1 + m_2)a$

$F = \left(\frac{F}{a_1} + \frac{F}{a_2}\right)a$

$F = \frac{a_1 a_2}{a_1 + a_2}$

So, the acceleration produced is $\frac{a_1 a_2}{a_1 + a_2}$.

8. Correct option: (C) $1\frac{\text{m}}{\text{s}}$.

Maas of the bullet, $m_b = 200g = 200 \times 10^{-3}\text{kg}$

Velocity of the bullet, $v = 5\frac{\text{m}}{\text{sec}}$

Maas of the gun, $m_g = 1\text{kg}$

So, the recoil velocity is given by as,

$$v = \frac{m_b v_b}{m_g}$$

$$V = \frac{0.2 \times 5}{1}$$

$$V = 1\frac{\text{m}}{\text{s}}$$

So, the velocity of the gun is $1\frac{\text{m}}{\text{s}}$.

9. Correct options: (B) and (C).

Both kg-m/s (kilogram-metre per second) and g-cm/s (gram-centimetre per second) are units of momentum. This demonstrates that momentum is a vector quantity with both magnitude and direction. It also displays the product of mass and velocity.

Momentum is the amount of motion a body can hold. Due to the object's mass and velocity, it is a measure of its motion. A body will have more momentum if its mass is greater and its speed is higher.

10. Correct options: (A) And (B).

The International System of Units (SI) uses the Newton (N) as the primary unit of force. The force needed to accelerate a mass of one kilogramme by one metre per second squared is known as a Newton ($N = \frac{\text{kg m}}{\text{s}^2}$).

A force measurement in the centimeter-gram-second (CGS) system is called a dyne (dyn). The force needed to propel a mass of one gramme by one centimetre per second squared is known as a dyne ($N = \frac{\text{kg m}}{\text{s}^2}$).

11. Correct option: (A), (B), and (D)

12. Correct options: (A) and (D).

Forces in the vertical direction are zero because the mass is at rest and the resultant of the forces acting on it is equal to zero.

From the diagram, In the vertical direction:

$$T_1 \sin 30^\circ + T_2 \sin 60^\circ - mg = 0$$

$$\frac{T_1}{2} + \frac{\sqrt{3}T_2}{2} = 5g$$

$$T_1 + \sqrt{3}T_2 = 100$$

Similarly in horizontal direction:

$$T_1 \cos 30^\circ - T_2 \cos 60^\circ$$

$$\frac{T_1}{T_2} = \frac{1}{\sqrt{3}}$$

Solving above equation we get,

$$T_1 + \sqrt{3} \times \sqrt{3}T_1 = 100$$

$$T_1 = 25\text{N}$$

$$T_2 = 25\sqrt{3}\text{N}$$

So, the $T_1 = 25\text{N}$ and $T_2 = 25\sqrt{3}\text{N}$.

So, Correct options: (A) and (D).

13. Correct option: (B) 0.5 kg.

Using the newton law of force,

For mass m_1,

$F = m_1 a_1$

$5 = m_1 \times 10$

$m_1 = 0.5\text{kg}$

So, the mass of the first body is 0.5 kg.

14. Correct option: (C) 0.25 kg.

Using the newton law of force,

For mass m_2,

$F = m_2 a_2$

$5 = m_2 \times 20$

$m_1 = 0.25\,\text{kg}$

So, the mass of the second body is 0.25 kg.

15. Correct option: (D) $6.67\frac{\text{m}}{\text{s}^2}$**.**

For mass m_1,

Using the newton law of force,

$F = m_1 a_1$

$5 = m_1 \times 10$

$m_1 = 0.5\text{kg}$

For mass m_2,

Using the newton law of force,

$F = m_2 a_2$

$5 = m_2 \times 20$

$m_1 = 0.25\,\text{kg}$

Now, the total mass= $m_1 + m_2 = 0.5 + 0.25 = 0.75\text{kg}$

Now acceleration when both the masses were tied,

$F = m_T a_T$

$a_T = \frac{5}{m_T} = \frac{5 \times 4}{3} = \frac{20}{2}\frac{\text{m}}{\text{s}^2} = 6.67\frac{\text{m}}{\text{s}^2}$

So, the acceleration is $6.67\frac{\text{m}}{\text{s}^2}$.

16 Correct option: (A)

17 Correct option: (B)

18 Correct option: (B)

19. Correct Matching is: (a)-(S), (b)-(R), (c)-(Q), (d)-(P).

Column-I	Column-II
a) Inertia	S) Measurement of mass
b) Linear momentum	R) Quantity of motion contained in the body
c) Newton's 2nd law	Q) Rate of change of momentum
d) Newton's 3rd law	P) Action = -reaction

The outcome of the forces acting on the mass while it is at rest is equal to zero, hence the forces acting in the vertical direction are.
A body's momentum is the amount of motion it is capable of. A body's momentum is calculated by multiplying its mass by its velocity. It's indicated by. When a body is at rest, its momentum is also zero because its velocity is zero.
A body's rate of momentum change occurs in the same direction as the applied force and is directly proportional to the applied force. The right way to phrase Newton's second law is as stated above.
According to the third rule, there is an equal and opposite reaction to every action (force) in nature. When object A pulls on item B, object B pulls back on object A with an equal and opposing force. In other words, interactions produce forces.
Correct option: (a)-(S), (b)-(R), (c)-(Q), (d)-(P).

20. Correct Matching is (a)-(R), (b)-(R),(c)-(P),(d)-(Q).

Column-I	**Column-II**
a)Twice the mass and was moving at the same speed	R) $48\frac{\text{kg m}}{\text{s}}$
b) The same mass and was moving with twice the speed	R) $48\frac{\text{kg m}}{\text{s}}$
c) One – half the mass and was moving with twice the speed	P) $24\frac{\text{kg m}}{\text{s}}$
d) The same mass and was moving with one half the speed	Q) $12\frac{\text{kg m}}{\text{s}}$

Given: The momentum of the bicycle, $P = 24\frac{\text{Kg m}}{\text{s}}$.

The momentum is given by as,

$P = mv$

$24 = mv$

Twice the mass and was moving at the same speed:

(A) The momentum of the bicycle doubles if its mass doubles and it continues to move at the same speed. $24\frac{\text{kg m}}{\text{s}}$ is the bicycle's initial momentum , thus let's say that it has a 24 kg mass. Its mass is 24 k g and its speed is 1. M / s . The bicycle has a momentum of $48\frac{\text{kg m}}{\text{s}}$ when the mass is doubled while maintaining the same speed.

(B) The momentum of the bicycle will be $48\frac{\text{kg m}}{\text{s}}$ the mass of the bicycle remains the same and the velocity doubles. The mass of the bicycle will remain 24 kg and the velocity is now $2\frac{\text{m}}{\text{s}}$.

$$p = mv = 24 \times 2 = 48 \frac{\text{kg m}}{\text{s}}$$

(C) The bicycle will have a momentum $24 \frac{\text{kg m}}{\text{s}}$ of if its mass is halved and velocity doubled.

$$p = \left(\frac{24}{2} \right) (1 \times 2)$$

$$p = 24 \frac{\text{kg m}}{\text{s}}$$

(D) The bicycle will have a momentum of $12 \frac{\text{kg m}}{\text{s}}$ if its mass is kept the same, but speed is halved.

$$p = (24) \left(1 \times \frac{1}{2} \right)$$

$$p = 12 \frac{\text{kg m}}{\text{s}}$$

So, the correct matching is (a)-(R), (b)-(R),(c)-(P),(d)-(Q).

21. Correct Answer is: 2

$F = 100 \text{dyn}$

$m = 5 \text{g}$

$t = 10 \text{sec}$

$u = 0$

u is the initial velocity.

Now using the newton's law of force,

$F = ma$

$$a = \frac{100 \times 10^{-5}}{5 \times 10^{-3}}$$

$$a = 0.20 \frac{\text{m}}{\text{s}^2}$$

Now, using the equation of motion,

$v = u + at$

$v = 0 + 0.20 \times 10$

$v = 2 \frac{m}{s}$

$v = 200 \frac{cm}{s}$

So, the velocity produced is $2 \times 100 \frac{\text{cm}}{\text{s}}$.

Forces On a Rigid Body or Turning Forces:

LEVEL-I

1. Correct option: (A) Force $\times$ Perpendicular distance.

Moment = force (in N) $\times d$ (perpendicular distance in meters)

2. Correct option: (B) Nm.

The SI unit for moment of force is Newton-meter (Nm), which is the product of force and perpendicular distance from the axis.

3. Correct option: (B) 10^{-5} dynes **.**

The CGS unit of force is the dyne, while the SI unit is the newton.

$1N=10^{-5}$dynes

4. Correct option: (A) 3750dyne cm **.**

$F = 50\text{dynes}$

Perpendicular distance= 75cm

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

Moment of force= $50\text{dynes} \times 75\text{cm}$

Moment of force= 3750dyne-cm

Therefore, the moment of force is 3750 dyne-cm.

5. Correct option: (B) 0.2m

Given:

Force, $F = 50\text{N}$

Moment of force= 10N-m

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

$10=50\times$Perpendiculardistnace

Perpendiculardistnace = $0.2m$

So, the perpendicular distance is 0.2m.

6. Correct option: (A) 140 N-m

Given:

$F = 80\text{N}$

Perpendicular distance=1.75m

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

Moment of force= $80\text{N} \times 1.75\text{m}$

Moment of force= 140N-m

So, the moment of force is 140n-m.

7. Correct option: (A) Zero.

The algebraic sum of the moments about the turning point is zero when a body is in equilibrium with two or more forces at work.

If it is not zero, a turning moment will occur, causing the body to rotate either clockwise or anticlockwise.

Therefore, for rotational equilibrium, the moment of force should be zero.

8. Correct option: (B) 35 dynes.

Perpendicular distance, $= 45\text{cm}$

Moment of force= $= 45\text{cm}$

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

$1575\text{dyne-cm}=F\times 45\text{cm}$

$F = 35\text{dyne}$

So, the applied force will be 35 dynes.

9. Correct option: (A) moment of couple.

The magnitude of one of the forces and the angle between their lines of action are multiplied to create the moment of a couple. $M = F \times d$. It uses kip-feet, pound-inches, KN-meter, etc. as units. All locations in a couple's plane have the same magnitude of the moment of the pair.

An item experiences a couple's action as a rotational force rather than a net translational motion. This is so because pure rotation about the fixed point results from equal and opposite forces producing clockwise and anticlockwise moments that balance each other out.

10. Correct option: (B) 0.206 Nm.

Force, $F = 125\text{N}$

Moment of force= 25.75N-m

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

25.75=125×Perpendiculardistnace

$$\text{Perpendiculardistnace} = \frac{25.75}{125}$$

$$\text{Perpendiculardistnace} = 0.206m$$

So, the minimum distance between point of application of force and axis of rotation of rigid body is 0.206m.

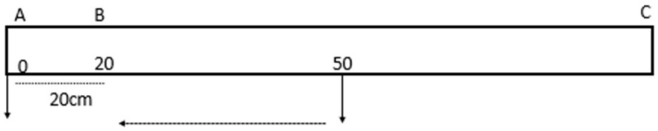
11. Correct option: (C) both A and B.

(A) Pair- pair is defined as two forces that have the same magnitude, are directed in the opposite directions, and have the same direction of action. When two forces match these requirements, they form a pair, and their combined impact may cause the item they are acting upon to move in a direction that is called translational motion (or linear motion).

(B) A couple- A couple is a system comprising two forces that are equal in magnitude, directed in the opposite direction, and acting parallel to one another, as was said in the preceding response. The arm, also referred to as the "lever arm," is the distance between the lines of action. The couple's effect produces turning motion (rotational motion) around a given point or axis without generating any net translational motion.

They meet the requirements for both a pair and a couple in the situation indicated in the question, when the two forces are equal, unlike, and parallel but their courses of action are not the same.

12. Correct option: (B) 100gf.



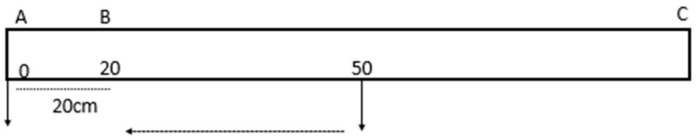
Let represent the weight in metres. A 100gf weight is hanged from one end of the scale, which is balanced at 20 cm (0 mark). BC's longer arm's weight is balanced by 100gf.

Clock wise moment = Anti clock wise moment

$$w \times 30 = 20 \times 100 w = \frac{2000}{30} = 66.66\text{gf}$$

Weight of 100gf should be suspended at zero mark.

13. Correct option: (C) 66.66gf.



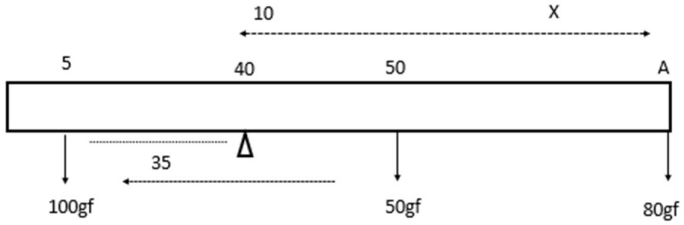
Let represent the weight in metres. A 100gf weight is hanged from one end of the scale, which is balanced at 20 cm (0 mark). BC's longer arm's weight is balanced by 100gf.
Clock wise moment = Anti clock wise moment

$$w\times 30 = 20\times 100 w = \frac{2000}{30} = 66.66\text{gf}$$

Weight of 66.66 gf should be suspended at zero mark.

14. Correct option: (B) 77.5 cm.

Force, $F = 50\text{N}$
Weight of meter scale=50gf
Balanced marked=40cm
Weight of 100gf at 5cm mark.



Now, let x is the distance of 80gf load. From 40cm
Now the moment of force is given by as,

$$\text{Moment of force} = \text{force}\times\text{Perpendicular distnace}$$

$$(80\times x)+(50+10)=100\times 35$$

$$80x = 3000$$

$$x = \frac{3000}{80}$$

$$x = 37.5\text{cm}$$

Distance in term of mater scale=40+37=77.5cm.
So, the required distance is 77.5 cm.

15. Correct option: (D) 64000dyne-cm **.**

Given: Force,
Perpendicular distance = 40cm
Moment of force is given by as,

$$\text{Moment of force}=F\times\text{Perpendicular distnace}$$

$$\text{Moment of force}=1600\times 40$$

$$\text{Moment of force}=64000\text{dyne-cm}$$

So, the moment of force is 64000dyne-cm .

16. Correct option: (A)

Force, $F = 525\text{N}$

Moment of force=420Nm

Moment of force is given by as,

$$\text{Moment of force}=F\times\text{Perpendicular distnace}$$

$$420=525\times\text{Perpendicular distnace}$$

$$\text{Perpendicular distnace}=\frac{420}{525}$$

$$\text{Perpendicular distnace} = 0.8$$

So, the shortest distance between the turning point and the force application point is 0.8m.

17. Correct option: (B) 9 Nm Anticlockwise.

From the diagram:

Moment of force is given by as,

$$\text{Moment of force}=F_1\times\text{Perpendicular distnace}$$

$$\text{Moment of force} = 3\times 3 = 9\text{Nm}$$

So, the moment due to F_1 is 9 Nm (Anticlockwise) .

18. Correct option: (A) 36 Nm Anticlockwise.

Moment of force is given by as,

$$\text{Moment of force}=F_2\times\text{Perpendicular distnace}$$

$$\text{Moment of force} = 3\times 12 = 36\,\text{Nm}$$

So, the moment due to F_2 is 36 Nm (Anticlockwise

19. Correct option: (B) 66Nm.

Moment of force is given by as,

$$\text{Moment of force}=F\times\text{Perpendicular distnace}$$

$$\text{Moment of force} = 3\times 22 = 66\,\text{Nm}$$

So, the moment of force is 66 Nm.

20. Correct option: (A) 48 Nm Anticlockwise.

From the diagram:

Moment of force is given by as,

$$\text{Moment of force}=F_1\times\text{Perpendicular distnace}$$

$$\text{Moment of force}=12\times 4 = 48\,\text{Nm}$$

So, the moment due to F_1 is 48 Nm (Anticlockwise) .

21. Correct option: (A)

From the diagram:

Moment of force due to F_1:

Moment of force is given by as,

$$\text{Moment of force}=F_1\times\text{Perpendicular distnace}$$

$$\text{Moment of force}=12\times 3 = 36\,\text{Nm(Anticlockwise)}$$

Moment of force due to F_2:

Moment of force is given by as,

$$\text{Moment of force}=F_2\times\text{Perpendicular distnace}$$

$$\text{Moment of force}=5\times 6 = 30\,\text{Nm(clockwise)}$$

Moment of force due to F_3:

Moment of force is given by as,

$\text{Moment of force} = F_3 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 22 \times 3 = 66\ \text{Nm(clockwise)}$

So, the total cock-wise moment = 30+66=96 Nm.

22. Correct option: (B) 220 Nm.

Moment of force due to F_2:

Moment of force is given by as,

$\text{Moment of force} = F_2 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 11 \times 20 = 220\ \text{Nm(Anti-clockwise)}$

So, the moment of force due to F_2 is 220 Nm.

23. Correct option: (B) 36 Nm.

Moment of force due to F_1:

Moment of force is given by as,

$\text{Moment of force} = F_1 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 12 \times 3 = 36\ \text{Nm(Anticlockwise)}$

So, the total anticlockwise is 36 nm.

24. Correct option: (C) 320 Nm.

Moment of force due to F_3:

Moment of force is given by as,

$\text{Moment of force} = F_3 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 20 \times 16 = 320\ \text{Nm(clockwise)}$

So, the moment of force due to F_3 is 320 Nm.

25. Correct option: (D)

Moment of force due to F_4:

Moment of force is given by as,

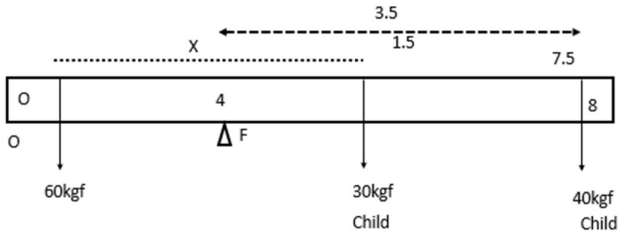
$\text{Moment of force} = F_4 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 3 \times 23 = 69\ \text{Nm(clockwise)}$

So, the moment of force due to F_4 is 69 Nm(clockwise).

LEVEL-II

1. Correct option: (A) 3.08m.



Let lady of 60kgf sits at a.

Distance from 4m mark.

Now clockwise moment,

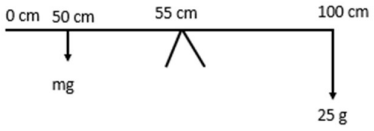
$60 \times x = (30 \times 1.5) + (40 \times 3.5)$

$60x = 45 + 140 = 185$

$x = \frac{185}{60} = 3.08\text{m}$

So, the required distance is 3.08m.

2. Correct option: (D) 225 gm.



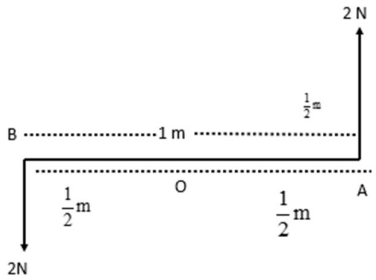
The moment of forces on the knife edge should be equal on both sides for equilibrium.
Therefore,

$mg(55-0)=25(100-25)$

$m=\frac{25 \times 45}{5} = 225\text{g}$

So, the mass of the scale is 225g

3. Correct option: (A)



Moment of force at A about O ,

$= 2 \times \frac{1}{2} = 1\text{Nm}$

Moment of force at B about O ,

$= 2 \times \frac{1}{2} = 1\text{Nm}$

So, the resultant force is =1+1=2N

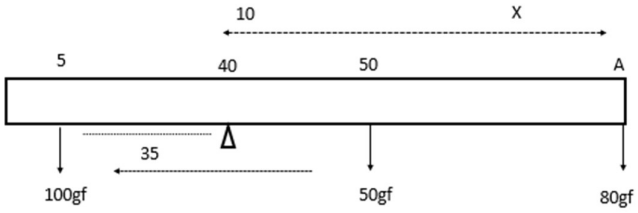
4. Correct option: (B) 77.5 cm.

Force, $F = 50\text{N}$

Weight of meter scale=50gf

Balanced marked=40cm

Weight of 100gf at 5cm mark.



Now, let x is the distance of 80gf load. From 40cm

Now the moment of force is given by as,

$\text{Moment of force} = \text{force} \times \text{Perpendicular distnace}$

$(80 \times x) + (50+10) = 100 \times 35$

$80x = 3000$

$x = \frac{3000}{80}$

$x = 37.5\text{cm}$

Distance in term of mater scale=40+37=77.5cm.

So, the required distance is 77.5 cm.

5. Correct option: (C)

6. Correct option: (C) 66Nm clockwise.

Moment of force due to F_3:

Moment of force is given by as,

$\text{Moment of force} = F_3 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 22 \times 3 = 66\ \text{Nm(clockwise)}$

7. Correct option: (A) 420 N cm.

Moment of force due to 6N:

Moment of force is given by as,

$\text{Moment of force} = 6\text{N} \times \text{Perpendicular distnace}$

$\text{Moment of force} = 6\text{N} \times 70\text{cm} = 420\ \text{Ncm}$

So, the moment due to 6N is 420 Nm.

8. Correct option: (B) 60 Nm Cw.

From the diagram:

Moment of force due to F_1:

Moment of force is given by as,

$\text{Moment of force} = F_1 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 3 \times 12 = 36\ \text{Nm(Anticlockwise)}$

Moment of force due to F_2:

Moment of force is given by as,

$\text{Moment of force} = F_2 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 5 \times 6 = 30\ \text{Nm(clockwise)}$

Moment of force due to F_3:

Moment of force is given by as,

$\text{Moment of force} = F_3 \times \text{Perpendicular distnace}$

$\text{Moment of force} = 22 \times 3 = 66\ \text{Nm(clockwise)}$

So, the total resultant moment= Clockwise moment- Anticlockwise moment

Resultant moment = $(66+30) - 36 = 60\text{Nm}$

So, the total resultant moment is 60Nm Cw.

9. Correct option: (A) $\frac{1}{5}$.

Force, $F = 50N$

Moment of force= 10N-m

Moment of force is given by as,

Moment of force = $F \times$ perpendicular distance

10=50×Perpendiculardistnace

Perpendiculardistnace $= \frac{10}{50} = \frac{1}{5} m$

So, the perpendicular distance is $\frac{1}{5}$m.

10. Correct option: (B) 3m.

Given:

Moment of couple =12 N-m

Force, $F = 4\text{N}$

Moment couple is given by as,

Moment of couple= $= F \times \text{Arm of the couple}$

$12 = 4 \times \text{Arm of the couple}$

$\text{Arm of the couple} = 3\text{m}$

So, the arm of couple is 3m.

11. Correct options: (A) and (B) .

Moment of couple =14.3 N-m

Force, $F = 13\text{N}$

Moment couple is given by as,

Moment of couple= $= F \times \text{Arm of the couple}$

$14.3 = 13 \times \text{Arm of the couple}$

$\text{Arm of the couple} = 1.1\text{m}$

So, the length of the arm of the couple 1.1m.

We know that

$1\text{m} = 100\text{cm}$

So, 1.1m in cm is 110 cm

So, Correct option: (A) and (B).

12. Correct option: (A) and (C).

Moment of couple =21 N-m

Arm of couple =84 cm =0.84m

Moment couple is given by as,

Moment of couple= $= F \times \text{Arm of the couple}$

$21 = F \times 0.84$

$\text{Arm of the couple} = 25\text{N}$

We know that,

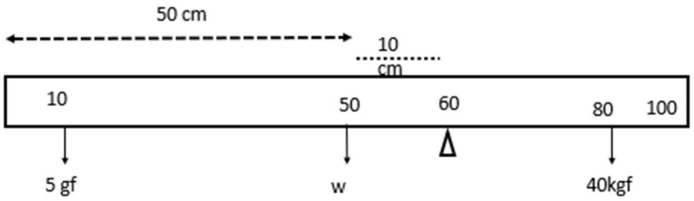
$1\text{N} = 1\frac{\text{kg}}{\text{ms}^2}$

Therefore,

$1\text{N} = 1\frac{\text{kg}}{\text{ms}^2}$

So, Correct option: (A) and (C).

13. Correct option: (A) 55 gf.



Let w represent the mass of the metre scale acting at 50 cm.

Now, clockwise moment= $40\times(80-60)$

clockwise moment $=40\times(20)=800g\text{fcm}$

Now,

Anticlockwise moment $=5\times(60-10)+w(60-50)=(250+10w)g\text{fcm}$

We know that,

Clockwise moments = Anticlockwise moments

$800 = 250 + 10w$

$w = 55\text{gf}$

So, the weight of meter scale is 55 gf.

14. Correct option: (B) 3750dyne cm **.**

$F=50\text{dynes}$

Perpendicular distance= $75\,\text{cm}$

Moment of force is given by as,

Moment of force = $F\times$ perpendicular distance

Moment of force= $50\text{dynes}\times75\text{cm}$

Moment of force= 3750dyne-cm

Therefore, the moment of force is 3750 dyne-cm

15. Correct option: (C) 140 N-m

Given:

$F=80\text{N}$

Perpendicular distance=1.75m

Moment of force is given by as,

Moment of force = $F\times$ perpendicular distance

Moment of force= $80\text{N}\times1.75\text{m}$

Moment of force= 140N-m

So, the moment of force is 140n-m.

16. Correct option: (B)100 N.

Clockwise moments = Anticlockwise moments

Moment of force due to 100N:

Clockwise moment:

Moment of force is given by as,

$\text{Moment of force}=100\times\text{Perpendicular distnace}$

$\text{Moment of force}=100\times30=3000\,\text{Nm(clockwise)}$

Anticlockwise moment,

$3000 = w \times 20$

$w = 150\text{N}$

Moment of force=w×Perpendicular distnace

Moment of force=$\text{w} \times 20$

We know that ,

Clockwise moments = Anticlockwise moments

$3000 = w \times 20$

$w = 150\text{N}$

So, the value of is w 150 N.

17. Correct option: (A) 3000 N-cm

Moment of force due to 100N:

Clockwise moment:

Moment of force is given by as,

Moment of force=$100 \times$Perpendicular distnace

Moment of force=$100 \times 30 = 3000$ Nm(clockwise)

So, the moment of force due 100N is 3000 N-cm.

18. Correct Matching is: (a)-(R), (b)-(P), (c)-(S), (d)-(Q).

(a) $1\text{N}=10^{-3}$dynes

(b) Moment of force is the name for the force's turning effect. It is the result of multiplying the force by the separation between the force's line of action and the pivot point or point at which the item will turn. Newton metre (Nm) is the S.I. unit.

(c) Dyne and cm are CGS units of force and distance, respectively. Moment of force is calculated as the product of force and the angle at which the force is perpendicular to the axis. In the CGS system, the moment of force is measured in dyne-cm.

(d) The product of the force and the angle between the lines of action of the two forces is known as the couple's moment. The arm of the Couple is another name for the distance that separates two forces' directions of action perpendicularly in this context.

4. Pressure
Solutions

LEVEL-I

Thrust and Pressure & Atmospheric Pressure:

1. Correct option: (B) Thrust.

Thrust is the term for the force applied to an item perpendicular to the surface.
Newton is the SI unit for thrust.

2. Correct option: (A) Newton.

The International System of Units (SI) uses the newton (symbol: N) as its unit of force.

3. Correct option: (B) Dyne.

The C.G.S. unit is dyne, while the S.I. unit of thrust is newton (N).

4. Correct option: (B) Vector.

The force perpendicular to a body's surface is known as a thrust. Since thrust is described in terms of both magnitude and direction (since it only applies to bodies with surfaces perpendicular to them), it is regarded as a vector quantity.

5. Correct option: (A) Pressure.

Pressure is defined as force per unit of area. The force applied perpendicularly to an object's surface per unit area across which that force is dispersed is referred to as pressure.
When compared to atmospheric pressure, gauge pressure is the pressure.

6. Correct option: (A) Pascal.

One newton per square meter is equivalent to one pascal (abbreviated as Pa), the SI unit of pressure. $\left(\frac{N}{m^2}\right)$.

7. Correct option: (C) Hydrostatic pressure.

The pressure that a fluid is exerting at any one time due to gravity is known as hydrostatic pressure. As the weight of the fluid increases when a downward force is applied, hydrostatic pressure is proportional to the depth measured from the surface. Fluids apply pressure uniformly in all directions.

8. Correct option: (D) Mercury.

The substance used in a barometer is mercury. Below is a list of some of the qualities that make it a good option:

- a lot of density Mercury with a shorter length (76 cm) can balance the atmosphere's pressure.
- It reacts to pressure changes more quickly.
- Its freezing point is low.
- It is difficult to evaporate.
- Because it is bright, it is simple to read the mercury.

9. Correct option: (B) 1:10.

The SI measure of pressure is $\frac{N}{m^2}$, while the CGS unit is $\frac{dyne}{cm^2}$.

$1N=10^5 dyne$

$1m=10^0 cm$

So, the ratio of CGS unit to SI unit is given by as,

$$\frac{1\frac{\text{dyne}}{\text{cm}^2}}{\frac{\text{N}}{\text{m}^2}} = \frac{1\frac{\text{dyne}}{\text{cm}^2}}{\frac{10^5\ \text{dyne}}{(100)^2\text{cm}^2}} = \frac{1}{10}$$

So, the ratio of CGS unit to SI unit is given is 1:10.

10. Correct option: (A) $1.01\times10^5\text{Nm}^{-2}$**.**

The Standard atmospheric pressure is $1.01\times10^5\text{Nm}^{-2}$.

11. Correct option: (C) 10m^2**.**

Given:

F=40N

$$P = 4\frac{\text{N}}{\text{m}^2}$$

The pressure is given by as,

$$P = \frac{F}{A}$$

$$A = \frac{F}{P}$$

$$A = \frac{40}{4}$$

$$A = 10\text{m}^2$$

So, the area on which the force of water is acting is 10m^2.

12. Correct option: (B) 2 atm.

Given:

atmospheric pressure = 1 atm

Pressure at the bottom = pressure due to water + pressure due to the atmosphere.

3atm = pressure due to water column +1atm

pressure due to water column =(3-1) atm

= 2atm.

So, calculate the pressure due to the liquid column in the tank is 2atm.

13. Correct option: (A) 3200 Pa.

Given:

$F = 16N$

$$A = 50\text{cm}^2 = \frac{50}{10000}m^2$$

Now pressure is given by as,

$$a \quad P = \frac{F}{A}$$

$$P = \frac{16\times10000}{50} = 3200\text{Pa}$$

So, the pressure in Pascal is 3200 Pa.

Pressure Due to Liquid Column:

14. Correct option: (B) hdg **.**

Only the liquid's density and vertical depth have any bearing on the pressure that it exerts at a given spot. The container's shape has no bearing on it. The same liquid is contained in three jars of varying forms, and the pressure at the bottom of each vessel operates equally in all directions.

Thus, $P = hdg$, where h is the height, d is the density, and g is the acceleration due to gravity, is the formula for the pressure that a liquid of height h exerts.

15. Correct option: (C) 6400 Pa.

The density of alcohol column, $d = 0.8\text{gcm}^{-3} = 0.8 \times 1000 = 800\text{kgm}^{-3}$.

Size of the alcohol column's vertical column, $h = 0.8\text{m}$

Now pressure is given by,

$P = hdg$

$P = 0.8 \times 800 \times 10$

$P = 6400\text{Pa}$

So, the pressure is 6400 Pa.

16. Correct option: (A) 34 m.

The pressure exerted by water, $P = 333200\,\text{Pa}$

Density of water, $\rho = 1000\frac{\text{kg}}{\text{m}^3}$

Acceleration due to gravity, $g = 9.8\frac{\text{m}}{\text{s}^2}$.

Now, pressure is given by,

$P = h\rho g$

$333200 = h \times 1000 \times 9.8$

$h = \frac{333200}{1000 \times 9.8}$

$h = 34m$

Therefore, 34 metres of water column exert 333200 Pa of pressure.

17. Correct option: (D) 3920 Pa.

Height of the water column , $h = 40\text{cm} = \frac{40}{100}\text{m}$

Density of water, $\rho = 1000\frac{\text{kg}}{\text{m}^3}$

Acceleration due to gravity, $g = 9.8\frac{\text{m}}{\text{s}^2}$

Now, pressure is given by,

$P = h\rho g$

$P = \frac{40}{100} \times 1000 \times 9.8$

$P = 3920\,\text{Pa}$

So, water exerting hydrostatic pressure in a beaker's bottom is 3920 Pa.

18. Correct option: (C) 10.3 m.

There is just a change in the density and height of the fluid, therefore the pressure will be the same whether mercury or water is employed.

Density of mercury is $13.6\frac{\text{g}}{\text{cm}^2}$.

For, mercury, the height is 76 cm.

Pressure is given by,

$P = h\rho g$

$P = 76 \times 13.6 \times g$

Let's consider the height of the water column is $h\,\text{m}$.

Now pressure is given by,

$P = h\rho g$

$h \times 1 = 76 \times 13.6$

$h = 1033.6$

$h = 10.3\text{m}$

So, the height of the water column is 10.3 m.

19. Correct option: (A) $33.6 \times 10^3 \text{Pa}$ **.**

Given:

Area under the foot is, $A = 180\text{cm}^2 = \dfrac{180}{10000}\text{m}^2$

Force is given by,

$F = 60\text{kgwt} = 60 \times 9.8\text{N}$

$F = 60 \times \dfrac{98}{10} = 6 \times 98\text{N}$

The pressure is given by,

$P = \dfrac{F}{A}$

$P = \dfrac{6 \times 98 \times 10000}{180} = \dfrac{98}{3} \times 10^3 \text{Pa}$

$P = 32.6 \times 10^3 \text{Pa}$

So, the pressure exerted on the ground by the person is $32.6 \times 10^3 \text{Pa}$.

20. Correct option:(B) $2 \times 10^5 \text{Pa}$ **.**

$1\text{atm} = 10^5 \text{Pa}$

The required water column height to equal one atmosphere is: $\rho g h = 10^5$

$1000 \times 10 \times h = 10^5$

$h = 10\text{m}$

1 atm of pressure is equal to 10 m of water column.

2 atm of pressure is produced by 20 m of water.

As we know that, $1\text{atm} = 10^5 \text{Pa}$.

Therefore 2 atm in Pascal is,

$2\text{atm} = 2 \times 10^5 \text{Pa}$

So, the pressure on his body is nearly $2 \times 10^5 \text{Pa}$.

21. Correct option: (A) Barometer.

The barometer is the name of the device used to gauge atmospheric pressure. A manometer is used to gauge gas pressure. Temperature is measured with a thermometer.

22. Correct option: (A) $1.0129 \times 10^5 \text{Pa}$ **.**

Density of mercury is $13.6 \dfrac{\text{g}}{\text{cm}^2}$.

Acceleration due to gravity, $\text{g} = 980 \dfrac{\text{cm}}{\text{s}^2}$

For, mercury, the height is 76 cm.

Pressure is given by,

$P = h\rho g$

$P = 76 \times 13.6 \times 980$

$P = 1.0129 \times 10^5 \text{ Pa}$

So, the pressure is $1.0129 \times 10^5 \text{Pa}$.

23. Correct option: (D) $1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$

The water column's bottom has a pressure of,

$P = P_0 + \rho gh$

$P = 10^6 + 1 \times 980 \times 12.5$

$P = 1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$

So, the pressure is $1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$.

Pascal's law:

24. Correct option: (A) Pressure.

According to Pascal's law, pressure applied anywhere within a confined liquid propagates equally and undiminished in all directions.

25. Correct option: (D) Pascal's law.

The Pascal's law theory underlies the operation of every hydraulic device. Every hydraulic device works on the basis that a tiny force given to the smaller piston will cause a larger force to be applied to the larger piston.

26. Correct option: (A) Hydraulic press.

Hydraulic lifts are those that employ Pascal's Law to raise items with the aid of fluid. It utilises the law of Pascal.

27. Correct option: (A) $\frac{F_1}{A_1} = \frac{F_2}{A_2}$**.**

The pressure is given by,

$P = \frac{F}{A}$

So, the correct mathematical expression for Pascal's law is given by as,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

28. Correct option: (C) 20 N.

Given:

$P_1 = 10\text{N}$

$A_1 = 5\text{m}^2$

$A_2 = 10\text{m}^2$

Now, we know that,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{10\text{N}}{5\text{m}^2} = \frac{F_2}{10\text{m}^2}$

$F_2 = 20\text{N}$

So, the upward thrust exerted by the piston is 20 N.

29. Correct option: (D) 10.

Given:

$F_1 = 1250\text{N}$

$A_2 = 625\text{cm}^2$

$A_2 = 5\text{cm}^2$

Now, we know that pressure is constant,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{1250}{625} = \frac{F_2}{5}$

$F_2 = 10N$

So, the force 10 N on the smaller piston will support a load of 1250 N on the larger piston.

30. Correct option: (A) 7200 kg wt.

Pressure in a hydraulic press is transformed equally:

$P_1 = P_2$

Force per unit area is used to measure pressure.

Therefore,

$\frac{50 \times g}{A_1} = \frac{m \times g}{A_2}$

$\frac{50 \times 10}{(1.25)^2 x \times \pi} = \frac{m \times g}{(15)^2 x \times \pi}$

$m = 7200\text{kg}$

The larger piston will exert a force of 7200 kg wt.

LEVEL-II

1. Correct option: (C) $10^7\,\text{Pa}$**.**

The pressure is given by,

$P = \frac{F}{A}$

$P = \frac{500}{0.5 \times 10^{-4}}$

$P = 10^7\,\text{Pa}$

So, the pressure acting on the area is $10^7\,\text{Pa}$.

2. Correct option: (D) Both B & C

Fluids are substances that have the ability to flow and don't have a set shape. Unless contained in a container, gases and liquids do not have a fixed shape and continue to flow. This implies that fluids can refer to both gases and liquids. Solids aren't fluids due of their inherent rigidity.

3. Correct option: (C) 2.5 m.

Given:

$\text{Density}, \rho = 4\text{kgm}^{-3}$

Pressure, $P = 100\text{Pa}$

Now pressure is given by as,

$P = h\rho g$

$h = \frac{P}{\rho g}$

$h = \frac{100}{4 \times 10} = 2.5\text{m}$

So, the height of liquid column above that point is 2.5 m.

4. Correct option: (A) hdg **.**

Only the liquid's density and vertical depth have any bearing on the pressure that it exerts at a given spot. The container's shape has no bearing on it. The same liquid is contained in three jars of varying forms, and the pressure at the bottom of each vessel operates equally in all directions.

Thus, $P = hdg$, where h is the height, d is the density, and g is the acceleration due to gravity, is the formula for the pressure that a liquid of height h exerts.

5. Correct option: (B) 5 m.

Given:

$\text{Density}, \rho = 12\text{kgm}^{-3}$

Pressure, $P = 600\text{Pa}$

Now pressure is given by as,

$P = h\rho g$

$h = \frac{P}{\rho g}$

$h = \frac{600}{12 \times 10} = 5\text{m}$

So, the height of liquid column above that point is 5 m.

6. Correct options: (A) and (B).

An aircraft moves through the air with the use of thrust. The force per area is known as pressure. The force of gravitational attraction is weight.

Thus, weight and thrust both symbolise force. Therefore, they will have same unit. .

7. Correct options: (A),(B), (C) and (D).

(A) In petrol stations, lifting cars is done using the hydraulic press technique. Based on Pascal's concept, a car hydraulic lift is an illustration of a force doubled by hydraulic press. The little cylinder's fluid must be pushed far farther than the height of the car.

(B) Oil is extracted from oil seeds using a hydraulic press, among other uses.

(C) Straw and cotton bales are compressed using hydraulic presses. Metals can also be punched with holes with it.

(D) A hydraulic press is a mechanical tool that shapes, deforms, and arranges various kinds of metals, plastics, rubber, and wood using the static pressure of a liquid, as specified by Pascal's principle. The mainframe, power system, and controls make up a hydraulic press's mechanism.

8. Correct options: (A) and (B).

The pressure is given by,

$P = \frac{F}{A}$

$P = \frac{5\text{N}}{5\text{m}^2}$

$P = 1\frac{N}{m^2}$

$\because 1\text{Pa} = 1\frac{N}{m^2}$

$P = 1\text{Pa}$

So, the correct options are (A) and (B).

9. Correct options: (B) and (C).

The pascal (Pa), which is equal to one Newton per square metre of force, is the unit of pressure in the SI system.

$1\text{Pa}=1\frac{\text{N}}{\text{m}^2}$

10. Correct options: (B) and (C).

The pressure is given by,

$P = \frac{F}{A}$

$P = \frac{400\text{N}}{0.04\text{m}^2}$

$P = 10^4 \frac{\text{N}}{\text{m}^2}$

$\because 1\text{Pa} = 1\frac{\text{N}}{\text{m}^2}$

$P = 10^4\,\text{Pa}$

So, the correct options are (B) and (C).

11. Correct option: (C) 550N.

Given:

$P = 27500\text{Pa}$

$A = 200\text{cm}^2 = 2\times10^{-2}\,\text{m}^2$

The pressure is given by,

$P = \frac{F}{A}$

$F = P \times A$

$F = 27500\times2\times10^{-2}$

$F = 550\text{N}$

So, the magnitude of force is 550N.

12. Correct option: (D) 2000cm^2 **.**

Given:

$P = 1500\text{Pa}$

$F = 330\text{N}$

The pressure is given by,

$P = \frac{F}{A}$

$A = \frac{F}{P}$

$A = \frac{300}{1500}$

$A = 0.2\text{m}^2 = 2000\text{cm}^2$

So, the magnitude of A is 2000cm^2 .

13. Correct option: (A) 10,000 N.

$P = 50{,}000\text{Pa}$

$A = 0.20\text{m}^2$

The pressure is given by,

$$P = \frac{F}{A}$$

$$F = P \times A$$

$$F = 50{,}000 \times 0.2$$

$$F = 10000\text{N}$$

So, So, the magnitude of force is 10000 N.

14. Correct option: (B) 9 m.

Difference in pressure of water at ground floor and third floor= =120000-30000=90000Pa

We know that density of water is, d=1000kgm^{-3}

The pressure is given by,

$$P = hdg$$

$$h = \frac{P}{dg}$$

$$h = \frac{90000}{1000 \times 10}$$

$$h = 9\text{m}$$

So, the height of third floor is 9 m.

15. Correct option: (C)12 m.

We know that density of water is, ρ=1000kgm^{-3}

We know that, pressure is,

$$P = P_0 + \rho gh$$

Now,

$$P_0 + \rho gh_1 = 150000 \cdots\cdots (1)$$

$$P_0 + \rho gh_2 = 30{,}000 \cdots\cdots (2)$$

Now from the equation (1) and (2), we get,

$$\rho g(h_1 - h_2) = 120000$$

$$h_1 - h_2 = \frac{120000}{10 \times 1000}$$

$$h_1 - h_2 = 12m$$

So, height of fourth floor is 12 m.

16. Correct option: (B) 200 Pa.

We know that density of liquid is, d=20 kgm^{-3}

The pressure is given by,

$$P = hdg$$

$$P = 1 \times 20 \times 10$$

$$P = 200\text{Pa}$$

So, the pressure exerted by the liquid on the bottom surface of vessel is 200 Pa.

17. Correct option: B) Both statements are correct.

Statement I: 1 Pascal is equal to 1Nm^{-2} :

This statement is true. The SI unit of pressure is the Pascal (Pa). Its unit of measurement is Newtons per square metre (Nm^{-2}), or one Newton of force per square metre. The fundamental idea of pressure as force dispersed over an area is reflected in this connection.

Statement II: Pressure is directly proportional to force applied:
This statement is also true. Physics dictates that pressure is, in fact, inversely proportional to applied force. If the area stays constant, the pressure increases as the force exerted on a surface increase. The equation $\text{Pressure}=\frac{\text{Force}}{\text{Area}}$ captures this relationship. The pressure exerted on the same area rises proportionately as the force does.

18. Correct option: (B) Both statements are correct.

Statement-I: Liquid pressure at a point in a liquid does not depends on shape of the vessel in which liquid is kept.
This statement is true. Pascal's principle states that a confined fluid's pressure at a specific depth is distributed equally in all directions. This means that the depth of the point and the liquid's density are the only factors affecting the pressure at a given location within a liquid, independent of the shape of the container.
The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
As a result, the shape of the vessel in which the liquid is stored has no bearing on the pressure in the liquid at a given position.

Statement-II: Liquid pressure at point in a liquid on the density of liquid
The density of the liquid directly affects the pressure in a liquid at a given depth. This is because of the fundamental hydrostatic principle, according to which the weight of the fluid column above a given depth determines the pressure there. This assertion is true since pressure is a function of the fluid's weight, which is influenced by its density.

19. The correct matching is (a)-(P),(b)-(R),(c)-(Q),(d)-(S)

(a) The air pressure acting on the liquid's free surface must be included when determining the total pressure in a liquid at depth. Total pressure within a liquid at depth $'h'$ is equal to the sum of atmospheric pressure and liquid column pressure.
Total pressure inside a fluid at a depth $'h'$ is $P_0 + \rho gh$.

(b) The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
Thus, Thus, the pressure exerted by a liquid depends on density of the liquid, depth below the surface and acceleration due to gravity

(c) The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
Thus, the pressure exerted by a liquid depends on density of the liquid, depth below the surface and acceleration due to gravity.
The static fluid pressure is independent of the liquid's volume, mass, total surface area, and container form.

(d) Pressure exerted sideways is known as lateral pressure.
The parallel force per unit area produced when fluid molecules collide with a surface is referred to as lateral pressure, also known as shear pressure. In addition to acting vertically,

it also exerts a sideward force that affects buoyancy, fluid flow, and the stability of objects submerged in fluids, which helps to explain complex fluid dynamics phenomena.

20. The correct matching is (a)-(R)(Q),(b)-(S),(c)-(P),(d)-(T).

(a) One newton per square metre is equivalent to one pascal (abbreviated as Pa), the SI unit of pressure.

(b) The barye (Ba), which is equivalent to $1\frac{\text{dyne}}{\text{cm}^2}$ or 0.1 Pa, is the CGS unit of pressure.

(c) $1\text{bar}=10^5\text{Pa}$

(d) $1\text{millibar}=10^2\text{Pa}$

The correct matching is (a)-(R)(Q),(b)-(S),(c)-(P),(d)-(T).

5. Friction

Solutions

LEVEL -I

1. Correct option: (C) Frictional force.

A force known as the force of friction or frictional force is activated when the surface of one body is caused to slide over the surface of another body.

The relative motion between the surfaces is often opposed by this force.

2. Correct option: (B) Static friction.

Static friction is the amount of friction that is activated and prevents one body from sliding over another. Its nature is one of self-adjustment. Only as much of it is brought into action as is required to halt the movement of one body over another.

3. Correct option: (A) Friction.

Friction between the ground and the ball causes a ball rolling on it to slow down and eventually halt. A type of force called friction opposes a body's relative motion on a solid or liquid surface.

4. Correct option: (C) Smoothness.

A force called friction works against the relative velocity of two surfaces coming into contact. The interlocking of the two surfaces' imperfections is what causes friction. Friction is present on these types of surfaces because interlocking occurs more frequently there. Thus, friction is a function of surface roughness.

The nature of the two surfaces affects friction. It depends on how smooth or rough the two surfaces are that are in contact. When two smooth surfaces come into contact with one another, there is less friction between them as a result of the smooth surfaces' reduced interlocking.

5. Correct option: (B) Rolling friction.

When an object (the tyre) rolls over another surface (the road) without slipping or skidding, it creates rolling friction. Rolling friction improves the efficiency of vehicle movement on highways because it is often significantly lower than sliding friction.

6. Correct option: (B) Area of cross-section

The type of surface in contact and its polish determines the friction that cannot be exceeded. It acts perpendicular to the two surfaces making contact and in the opposite direction to the body's motion. The amount of limiting friction is then independent on the contact area.

7. Correct option: (C) Newton.

Frictional force can be expressed in force units or Newtons.

8. Correct option: (D) When sliding.

In comparison to rolling over the same terrain, a body sliding over it will experience more friction. The entire body's weight is focused on a smaller area of contact with the ground when sliding, which increases the frictional force.

The region over which the frictional force works is less while a body is rolling because only a portion of its circumference is in contact with the ground at any given time. As a result, the frictional force is less than in the sliding situation.

9. Correct option: (B) 0.1.

Given: $f = 100\text{wt}$

We know that, $1g\ wt = 980 dyne$

$f = 100\text{wt} = 100 \times 980\text{dyne}$

Now, $1g\ wt = 980 dyne$

$f = 100\text{wt} = 100 \times 980\text{dyne}$

So, the coefficient of friction is 0.1.

10. Correct option: (A) Friction exerted by water is very small.

Between the ship's wheel and the water, the coefficient of force is lower. Thus, there is also less friction. As a result, the ship travelling through the sea cannot accelerate after applying the brakes since there is less friction.

11. Correct option: (C) Fill the irregularities in the surface.

Moving surfaces don't directly rub against one other when lubricant is used to provide a thin layer between them in a machine. Movement becomes smooth and interlocking of imperfections is minimised to a large amount. Thus, there is less friction.

12. Correct option: (D) None of them.

Rolling friction is the least and static friction is the most of the three types of friction that can exist between two surfaces in contact. Because of this, pushing a still item is more challenging than pushing one that is moving. Furthermore, pushing or pulling something on wheels is simpler than sliding it since rolling friction is smaller than sliding friction.

13. Correct option: (D) No units.

The coefficient of friction has no units because it is the ratio of force to force.

14. Correct option: (D) All of these.

There are a number of forces at work when a body is pushed to the ground:

(A) Frictional force: This force works against the force that is being applied by acting in the opposite direction. It enables the body to be moved in the desired direction while preventing it from freely sliding on the floor.

(B) Normal force: The force applied to the body by the floor that is perpendicular to the surface of the ground is known as the normal force. It maintains the body's balance and keeps it from collapsing to the ground.

(C) Weight of the body: The force exerted on the body vertically downward by gravity is known as the weight of the body. To keep the body in balance, the floor must withstand a force.

15. Correct option: (D) Both (B) and (C).

The body moves (B) if the applied force is greater than the frictional force. The frictional force in this situation is overcome by the applied force, and the body begins to move in the direction of the applied force. The body does not move if the applied force is less than the frictional force (C). Since the frictional force resists motion in this case because it is stronger than the applied force, the body remains at rest. The right response is (D) Both (B) & (C). If the applied force is greater than the frictional force, the body moves; if it is less, the body remains stationary.

16. Correct option: (B) Sliding friction.

The friction that is present when pulling some luggage without rollers called sliding friction. When two solid surfaces slide against one another, sliding friction develops and resists the motion of the surfaces. Sliding friction would be there in this situation between the bottom of the suitcase and the surface it is being pulled on.

17. Correct option: (A) Limiting friction.

The maximum value of the static friction that will put the body in motion is called Limiting friction. It is the threshold value of the force that put the object in motion.

18. Correct option: (A) $\mu_s > \mu_k > \mu_R$**.**

This suggests that the wheel's rotating influence causes rolling friction to be the least, kinetic friction to be in the centre of the series due to its moving qualities, and static friction to be the highest due to the limiting movement properties.

When two surfaces' surface irregularities interlock, it is initially harder (with higher frictional resistance) to overcome them; however, once the interlocking is broken or overcome, it is simpler (with lower frictional resistance) to glide the surfaces.

19. Correct option: (A) 15 N.

The static friction is given by as,

Static friction= $= \mu_s mg$

Static friction= $= 0.3 \times 5 \times 9.8 = 14.7 \approx 15\text{N}$

20. Correct option: (B) Normal force.

There is a force acting between two bodies when they are in contact with one another. This force is known as the contact force since it is caused by the contact between the two bodies. It is possible to divide this contact force into two halves. The frictional force is referred to as the horizontal component of the contact force, while the normal reaction force is the vertical component. A reaction force that acts perpendicular to the surfaces in contact is the typical reaction force. The force that prevents motion between two surfaces is known as friction.

The frictional force can be calculated using the formula $f = \mu N$, μ where is the coefficient of friction, which varies depending on the two interacting objects.

Therefore, μ since is a constant in this calculation, we can disregard it for the proportionality as $f \propto \mu$

It follows that the frictional force and the typical reaction force are directly related.

21. Correct option: (D) Pulling decreases the normal reaction.

Pushing involves exerting force in a downwardly slanted direction. This significantly improves the lawn roller's typical response. A greater frictional force that you must overcome results from the enhanced normal response. Pulling minimises frictional force since the pulling force is applied upward, which also lessens the typical response force. Pulling is easier as a result of the decreased force opposing the action.

22. Correct option: (A) Less than one.

The frictional force between two surfaces and the normal force forcing them together is expressed as a dimensionless number known as the coefficient of frictional force. Although it can range from 0 (for no friction) to 1 (for maximum friction), it typically falls below one. The nature of the surfaces in touch and how they interact determine the precise value.

23. Correct option: (B) 0.40.

Given:

$f = 200\text{N}$

$N = 500\text{N}$

The formula of friction is given by as,

$f = \mu N$

$200 = \mu(500)$ So, the coefficient of friction is 0.40.

$\mu = \frac{200}{500} = 0.40$

24. Correct option: (C) Both (A) & (B).

The force that a surface applies to support the weight of an object sitting on it is known as the normal force (N). In order to balance the weight of an object when it is placed on a surface, the surface applies an upward force known as the normal force.

(A) Body weight: The normal force has the same magnitude as the body weight but acts in the opposite direction. When a body has a weight of mg (where m is its mass and g is its gravitational acceleration), the normal force has a similar magnitude of mg but acts in the opposite direction of gravity—upward.

25. Correct option: (D) 20 N.

Given: Weight of the body, $m = 5\text{kg}$

Coefficient of friction, $\mu = 0.4$

Normal reaction of force is given by as,

$N = mg$

$N = 5 \times 9.8$

$N = 49$

Noe friction force is given by as,

$f = \mu N$

$f = 0.4 \times 49 = 19.6 \approx 20\text{N}$

So, the force is 20N.

26. Correct option: (D) Both B and C.

In this case, the limiting friction between the body and the surface is 15 N, while the external force applied to the body is 10 N. As a result, the force applied alone won't cause the body to begin moving. The "Does not move" option (B) is used to illustrate this.

However, because the applied force is attempting to overcome the friction and start motion, there is a propensity for the body to move in the direction of the external force. This is shown by choice (C), "Has a tendency to move in the direction of external force."

Therefore, both (B) and (C) are valid choices. The body initially does not move, but the force being applied has a tendency to cause it to do so.

27. Correct option: (A) $\mu_k = \frac{f_k}{N}$.

The ratio of the friction force to the normal force felt by a body moving on a dry, uneven surface is known as the kinetic friction coefficient, or μ_k. $\mu_k = \frac{f_k}{N}$

28. Correct option: (A) Limiting friction.

The maximum amount of static friction, or limiting friction, is reached when the body overcomes the force of static friction. The frictional force will stop growing after the limiting friction. The object advances past the frictional force, which is now constant, at this point.

29. Correct option: (B) Dynamic friction.

The objects move in relation to the surfaces in one of three ways: over a stationary surface, on a moving surface, or both moving but at various relative speeds. Dynamic friction is the word used to describe this situation, which is related to the term kinetic energy.

30. Correct option: (B) Decreases if the surface is inclined.

When a surface is inclined, the component of gravity operating parallel to the surface, increases. As a result the force of limiting friction which opposes its motion will decrease.

If the surfaces is less inclined, then the gravitational component parallel to the surface decrease. More forces is requesred to move the body. Means, the force of limiting friction increases.

31. Correct option: (C) It remains unchanged.

Coefficient of friction is given by as, $\text{Coefficient of friction}(\mu) = \frac{\text{Frictional force}(f)}{\text{Normal force}(N)}$

The two surfaces' respective material compositions and degree of roughness have an impact on the coefficient of friction. The coefficient of friction will increase if the two surfaces coming into contact are rough, and vice versa. Essentially, the normal force is a supporting force applied to a body in contact with another stable body. The friction coefficient is unaffected by the normal force. Because of this, the coefficient of friction does not change when the normal force is doubled.

32. Correct option: (C) Remains unaffected due to a change in velocity.

Changes in velocity have no impact on the force of limiting friction between two surfaces, which remains constant. It doesn't matter how quickly or slowly the surfaces are moving in relation to one another. As long as the external applied force does not go above the maximum limit of friction, the force of limiting friction, which is determined by the nature of the surfaces, the coefficient of friction, and the normal force, remains constant.

Therefore, (C) Remains unaltered due to a change in velocity is the right response. With a change in velocity, the force of limiting friction does not change.

33. Correct option: (B) Unchanged.

Nature determines the coefficient. Normal reaction has no bearing on the coefficient of friction. Therefore, the coefficient of friction remains unchanged if the typical reaction is cut in half.

34. Correct option: (B) Decreases.

The temperature rises as the frictional force increases, but the reverse is not true.

The distance between body particles rises with temperature due to thermal expansion, smoothing the surface as a result, and decreasing the frictional force between them.

35. Correct option: (C) $\mu_s = \tan\theta$**.**

Angle of friction is given by as, $\mu_s = \tan\theta$

Angle of friction: This is the angle formed between the normal reaction and the resultant force of friction.

36. Correct option: (A) $\mu_s N$ **.**

The block just begins to slide when the applied force (F) equals the static friction force (f_s). Right now, the following formulas can be used to determine the force of static friction:

$f_s = \mu_s N$

The nature of the surfaces in contact determines the dimension lessness of the coefficient of static friction μ_s. The force that the surface uses to maintain the weight of the block acts perpendicular to the surface and is known as the normal force N.

37. Correct option: (C) Rubber provides more friction than leather.

Rubber soles are used in tennis and gym shoes instead of leather because rubber offers more friction.

38. Correct option: (C) Because normal force decreases.

Pushing a thing increases the normal force between it and the surface, which makes it more difficult to move the object because of the increased friction. On the other hand, tugging an object reduces the normal force, which causes less friction and makes moving the object simpler.

39. Correct option: (A) Zero.

The frictional force will be 0 if there is no applied force (option A).

When two surfaces come into contact, frictional force develops as a reaction to an applied force. There is no force to cause friction between the body and the surface when no external force is applied to the body on a horizontal surface. As a result, there will be no frictional force.

Due to its inertia, the body stays at rest or moves at a steady speed (if it was already moving) in the absence of any external force. Only when a force is applied to try to move the body relative to the surface does friction come into play.

LEVEL - II

1. Correct option: (B) 240N.

Given:

$m = 120\text{kg}$

$g = 10\text{ms}^{-2}$

$\mu = 0.2$

The friction force is given by as,

$f = \mu N$

$\therefore N = mg$

$f = \mu mg$

$f = 0.2 \times 120 \times 10$

$f = 240\text{N}$

So, the force exerted by the engine is 240N.

2. Correct option: (C) 0.4.

$m = 100\text{kg}$

$g = 10\text{ms}^{-2}$

$f = 400\text{N}$

The friction force is given by as,

$f = \mu N$

$\therefore N = mg$

$f = \mu mg$

$400 = \mu \times 100 \times 10$

$\mu = \dfrac{400}{1000}$

$\mu = 0.4$

So, the coefficient of friction is 0.4.

3. Correct option: (C) 428.N.

Given: $f = 150\text{N}$, $\mu = 0.35$

Friction formula is given by as,

$f = \mu N$

$150 = 0.35N$

$N = \dfrac{150}{0.35}$

$N = 428.5\text{N}$

So, the normal reaction is 428.5N.

4. Correct option: (B) $5ms^{-2}$**.**

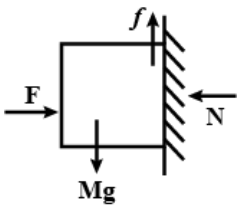
Given: Force of friction is given by as, $f = 0.5 \times 10 \times 10 = 50N$

Force that produces is given by as, $F = F - f = 100N\text{-}50N = 50N$

So, the acceleration is given by as, $a = \frac{F}{m} = \frac{50}{10} = 5ms^{-2}$

So, the acceleration of the block is $5ms^{-2}$.

5. Correct option: (C) $\boldsymbol{F = \frac{mg}{\mu}}$**.**



If the contact force is not perpendicular to the surface, a body in contact with a vertical surface may experience a horizontal force. The frictional force is equal to the horizontal force, which is the portion of the contact force parallel to the surface. The source of the frictional force is, $mg = \mu N$

$mg = \mu F$ Since, $N = F$

So, the horizontal force $F = \frac{mg}{\mu}$

6. Correct option: (A) $\boldsymbol{a = \frac{F - F_K}{m}}$

Let F is the external force applied on the body.

F_k is kinetic frictional force.

Then, the net force $f = F\text{-}F_k$

$ma = F\text{-}F_k$,

So, the acceleration of the body is, $a = \frac{F - F_K}{m}$

7. Correct option: (D) 50 N.

Friction force is given by as,

$F = \mu_s N$

$mg = \mu_s N$

$\frac{mg}{\mu_s} = N$

$N = \frac{20}{0.4} = 50N$

So, the horizontal force is 50 N.

8. Correct option: (A) 40 N $a = \frac{F - F_K}{m}$

9. Correct option: (A) Equal to each other.

Angle of friction is given by as, $\tan\alpha = \mu$

Where, α is the angle of friction and μ is the coefficient of friction.

Angle of repose is given by as, $\tan\theta = \mu$

Where, θ is the angle of friction and μ is the coefficient of friction.

Therefore, $\tan\theta = \tan\alpha$

$\theta = \alpha$

So, the angle of friction is equal to angle of repose.

10. Correct option: (B) 5ms^{-2}**.**

Given:

$m = 10\text{kg}$

$g = 10\text{ms}^{-2}$

$\mu = 0.5$

$F = 100\text{N}$

Force of friction is given by as, $f = 0.5 \times 10 \times 10 = 50\text{N}$

Force that produces is given by as, $F = F - f = 100\text{N-}50\text{N}=50\text{N}$

So, the acceleration is given by as, $a = \frac{F}{m} = \frac{50}{10} = 5\text{ms}^{-2}$

So, the acceleration of the block is 5ms^{-2}.

11. Correct option: (D) All of these.

Several forces act on a body at once while it is on an inclined plane. These forces consist of:

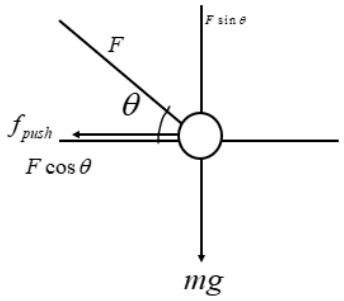
(A) Weight (gravitational force): The force of gravity causes the body's weight to act vertically downward. The body is being drawn towards the Earth's centre by this force.

(B) Normal reaction: The normal response is the force applied to the body by an inclined plane that acts perpendicular to the surface of the plane. It stops the body from entering the plane or piercing it.

(C) Frictional force: If a body tends to slide down an inclined plane, frictional force will act to counteract that tendency. The friction coefficient between the body and the plane determines the size of the frictional force.

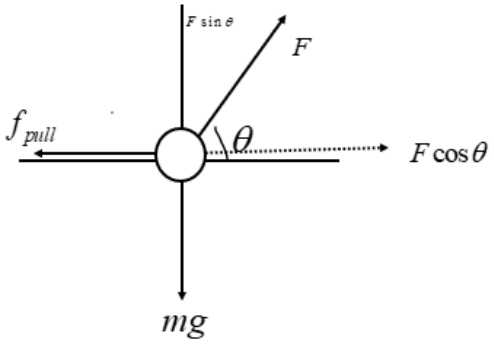
12. Correct option: (C) $N = mg - F\sin\theta$**.**

(a) Pulling:



From the diagram normal reaction is given by as, $N = mg - F\sin\theta$

13. Correct option: (A)



Normal reaction in case of pushing is given by as, $N = mg + F\sin\theta$

14. Correct option: (A) in case of S_1

When the same body placed on two surfaces which are inclined at 30o and 45o, he vertical component of the force acting on the body will decide the time in which it can reach the bottom.

The vertical component of the force acting on the first surface, $F_1 = \mu_1 mg\ sin30^o$

The vertical component of the force acting on the second surface, $F_2 = \mu_2 mg\ sin45^o$

If the block just started sliding on both the surfaces, $F_1 = F_2$

$\mu_1 mg\ sin30^o = \mu_2 mg\ sin45^o$

$\frac{\mu_1}{2} = \frac{\mu_2}{\sqrt{2}}$; $\mu_1 = \frac{2\mu_2}{\sqrt{2}}$ which means, $\mu_1 < \mu_2$

15. Correct option: (B) B reaches the bottom earlier

A and B are identical bodies and placed on similar inclined surfaces. Means the coefficient of friction is same on both the inclined surfaces.

A is on the inclined surface with an angle 30°.

B is on the inclined surface with an angle 60°

Since the weight of the two objects is same and the coefficient of friction is same, the body which is on more inclined surface will reach the bottom earlier. Because, the component of force due to gravity along the inclined surface is more.

16. Correct option: (A) 550N.

Given: $F = 165\text{N}$

$\mu = 0.30$

Friction is given by as, $F = \mu mg$

$165 = 0.3 \times mg$

Weight of the metal block, $mg = \frac{165}{0.3} = 550\ N$

17. Correct option: (C) $\mu_k = \frac{f_k}{N}$**.**

The ratio of the friction force to the normal force felt by a body moving on a dry, uneven surface is known as the kinetic friction coefficient, or μ_k . $\mu_k = \frac{f_k}{N}$

18. Correct option: (B) Increases.

The component of the force caused by gravity operating parallel to the plane grows as the angle of inclination of an inclined plane increase. This element helps the sliding block move down the hill more quickly. As a result, the sliding block moves more quickly. This is due to a stronger effective force dragging the block down the incline, which causes a faster acceleration and speed.

19. Correct option: (A) Fluid friction.

When a solid item moves through a liquid or gas, fluid friction occurs. To lessen the fluid friction, aeroplanes are streamlined.

20. Correct option: (C) Same that of B.

The normal reaction at the common area of contact determines the amount of frictional force, which is independent of the area of contact.

Block A and the surface will experience the same amount of friction as Block B does. n is the normal force and is the frictional coefficient. The only variables that affect the normal force are the body's mass and the acceleration caused by gravity.

21. Correct option: (C) Both.

The basic need for an effective lubricant is that it has a very high viscosity.

Regardless of the environmental conditions surrounding the system, a lubricant need to have specific qualities that allow its use in all situations. It needs to be able to consistently reduce the friction between two bodies. A good lubricant needs to be both thermally and hydraulically stable. It should guard against the possibility of corrosion and protect itself

from oxidation. It is constantly accessible as a liquid. Therefore, it should have a high boiling point and a low freezing point in order to preserve its liquidity.
A good lubricant should not easily evaporate or vaporise at operating temperatures, which is referred to as non-volatility. This makes sure the lubricant won't lose its effectiveness over time and won't require constant replenishing. Viscosity, a measurement of a fluid's resistance to flow, is crucial for lubrication, but lubricants must also be non-volatile in order to be effective.

22. Correct option: (D) The frictional force reduces to zero as it is a self-adjusting force

When two surfaces move relative to one another, frictional force is created. When a body is stopped, there is no relative motion and no resulting force, hence there is no longer any friction.

23. Correct option: (B) Increase friction.

To make the railway rails less smooth, sand is sprinkled on them. Friction increases because of this.

24. Correct option: (D) Friction between rubber & concreate is less than that between iron and concreate.

Because rubber and concrete have substantially less friction than iron and concrete do, car tyres are constructed of rubber rather than iron. Driving becomes safer and more effective because of the reduced friction, which improves traction and grip on the road.
If iron tyres were utilised, the strong metal-to-concrete contact would result in significantly increased rolling resistance and excessive noise. Iron tyres would also result in greater wear and tear on the tyres and the road surface, increasing maintenance costs and shortening the lifespan of the road.
Conversely, rubber tyres are good for absorbing road vibrations and improving overall ride comfort because they offer higher shock absorption, flexibility, and the capacity to conform to road irregularities.

25. Correct option: (D) Both A & C.

A) Lubricants form a thin coating between moving surfaces, acting as ball bearings. As small ball bearings, the lubricant's molecules reduce direct contact between solid surfaces and promote smoother motion.
C) Simple relative motion between solid and liquid: Lubricants create a barrier between solid surfaces, making it easier for them to slide over one another. As a result, the forces that oppose motion are reduced through friction.

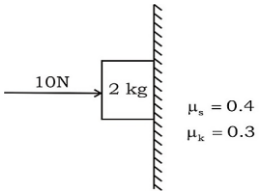
26 the correct options are (A) and (B).

(A) If the wall is frictionless and smooth, there won't be any force operating in the horizontal direction to counteract the force F, therefore it will slide down the wall. Due to the force being applied, the block will consequently slide down the wall.
B) If the wall is rough, the frictional force may counterbalance the weight. If the wall is rough and creates some friction between the block and the wall, the frictional force (Ff) between the block and the wall may be able to counterbalance the weight. In this situation, the frictional force can counteract the block's weight and keep it from sliding down the wall. The friction coefficient (μ) between the block and the wall determines the size of the frictional force.

27 The correct options are (A) and (C)

Since the body at rest on the horizontal surface, the coefficient of friction is $\mu = \tan\phi$
Maximum frictional force required to put it in motion $F = \mu mg = mg\tan\phi$

COMPREHENSION TYPE



28. Correct option: (C) 10 N

We know that the Normal reaction acting on the vertical wall is equivalent to the horizontal force acting on the wall. F=10N

Then, F = N = 10 N

29. Correct option: (B) 7 m/s²

The weight of the body acting downward =mg

When the block is sliding against the wall, the kinetic frictional force, $F_k = \mu_k mg$

Net force with which the body is moving is $ma = mg - \mu_k mg$

$a = 10 - (0.3 \times 10)$ since $\mu_k = 0.3$

a = 7 m/s²

30. Correct option: (A) 40 N

The force required to balance the block aginst the wall is $F = \mu_s N$

$F = 0.4 \times 10$

F = 40 N

31. Correct option: (A) $15\sqrt{3}\text{N}$**.**

Form the diagram,

$N = mg - \cos\theta$

$$N = 50 - \frac{30}{3} = 50 - 15 = 35\text{N}$$

Now using friction formula,

$f = \mu_k N$

$f = 30\cos 30^\circ$

$f = 15\sqrt{3}\text{N}$

So, the frictional force is $15\sqrt{3}\text{N}$.

32. Correct option: (D) $\frac{3\sqrt{3}}{7}$**.**

From the diagram,

$N = mg - \cos\theta$

$$N = 50 - \frac{30}{3} = 50 - 15 = 35\text{N}$$

Now using friction formula,

$f = \mu_k N$

$f = 30\cos 30^\circ$

$f = 15\sqrt{3}\text{N}$

Again, using formula of friction,

$f = \mu_k N$

$15\sqrt{3}\text{N} = \mu_k \times 35$

$\mu_k = \frac{15\sqrt{3}}{35}$

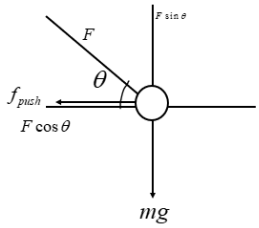
$\mu_k = \frac{3\sqrt{3}}{7}$

So, the value of μ_k is $\frac{3\sqrt{3}}{7}$.

33. Correct matching is (a)→(T),(b)→(S),(c)→(P),(d)→(R).

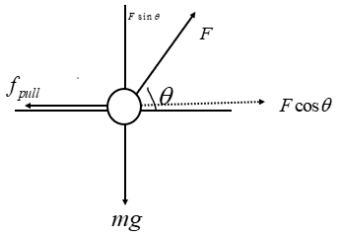
Column – I	**Column – II**
a) Pulling	T) $N = mg - F\sin\theta$
b) Pushing	S) $N = mg + F\sin\theta$
c) Angle of friction	P) $\tan\phi$
d) Coefficient of kinetic friction	R) $\frac{f_k}{N}$

(a) Pulling:



From the diagram normal reaction is given by as, $N = mg - F\sin\theta$

(b) Pushing:



Normal reaction in case of pushing is given by as, $N = mg + F\sin\theta$

(c) Angle of friction is given by as, $\mu_s = \tan\theta$

Angle of friction: This is the angle formed between the normal reaction and the resultant force of friction.

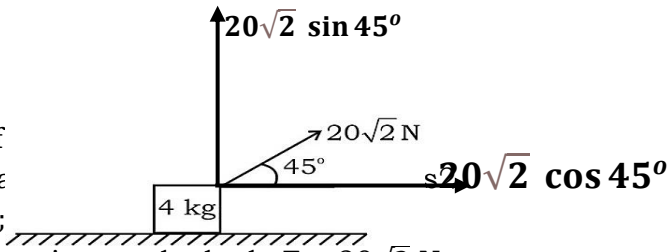
(d) The ratio of the friction force to the normal force felt by a body moving on a dry, uneven surface is known as the kinetic friction coefficient, or μ_k, $\mu_k = \frac{f_k}{N}$

34. Correct matching is (a)→(Q); (b)→(S); (c) →(P); (d)→(R), (T)

Column – I	**Column – II**
a) Static friction	Q) Self-adjusting force
b) Limiting friction	S) Maximum static frictional force
c) Co-efficient of friction	P) Dimensionless quantity
d) Frictional force	R) Tangential to surface of contact
	T) Non-conservative force

(a) Static friction is a self-adjusting force. It will increase up to certain limit by following the amount force applied on it.
(b) Limiting friction, is the maximum frictional force that can withhold by a surface. It is also defined as the threshold force that is required to put a body in motion.
(c) Co-efficient of friction – it is defined as the ratio of frictional force that opposes the motion to the normal force that is acting on the surfaces that are in contact. Since, it is the ration of two forces, co-efficient of friction is a dimensionless quantity.
(d) Frictional force – a non-conservative force is the one for which the work-done by the force depends on the path. So, frictional force is also a non-conservative force because, the work done by the frictional force depends on the path. And the frictional force between two surfaces acts along the surfaces that are in contact and opposes its motion. So, its direction is tangential direction.
So, the correct answer is (a)→(Q); (b)→(S); (c) →(P); (d)→(R), (T)

35 Correct matching is: (a)→(Q); (b)→(P); (c)→(S); (d)→(R)



Mass of
Accelera
$\mu_s = 0.8$;

External force acting on the body F = $20\sqrt{2}$ N
Angle at which the force acting is $\theta = 45^o$
Horizontal component of the force $F_h = 20\sqrt{2}\ \cos 45^o$
Vertical component of the force $F_v = 20\sqrt{2}\ \sin 45^o$

(a) Normal Reaction N = mg - F_v

$N = 4 \times 10 - 20\sqrt{2}\ \sin 45^o$

$N = 40 - 20\sqrt{2} \times \frac{1}{\sqrt{2}}$

$Normal\ Reaction = 40 - 20 = 20\ N$

(b) Force of friction under the motion = $\mu_k N = 0.6 \times 20 = 12\ N$

(c) Acceleration of the block, $a = \frac{F_h - Frictional\ force\ under\ motion}{Mass}$

$a = \frac{20\sqrt{2}\ \cos 45^o - \mu_k N}{m}$

$a = \frac{20\sqrt{2} \times \frac{1}{\sqrt{2}} - 0.6 \times 20}{4}$

$a = \frac{20-12}{4} = 2\ m/s^2$

(d) Limiting Friction $F_{lim} = \mu_s N$

$F_{lim} = 0.8 \times 20 = 16\ N$

So, (a)→(Q); (b)→(P); (c)→(S); (d)→(R)

6. Sound
Solutions

LEVEL-I

1. Correct option: (C) Sound will be louder, but the pitch will not be different.

The frequency of the wave determines the pitch of every sound wave. The sound will be louder but the pitch will remain the same when the mechanical piano is struck more forcefully the second time because the frequency has not altered. There won't be a change in sound pitch because the pitch is dependent on the frequency of the specific key being struck. If the key is struck more forcefully, the amplitude will increase, increasing the volume.

2. Correct option: (D) 30 Hz.

The number of complete cycles per second determines a wave's frequency.

Here, the wave's frequency will be: $\frac{24}{0.8}=30\text{Hz}$

3. Correct option: (B) It is directly proportional to wavelength and frequency.

Wavelength is inversely proportional to the frequency.

$$\text{Wave velocity}=\text{Wavelength}\times\text{Frequency}$$

$$\text{Wavelength}=\frac{\text{Wave velocity}}{\text{Frequency}}$$

According to this equation, wavelength and frequency have a direct correlation with the speed of sound waves. A sound wave's velocity grows as its frequency increases, and in a similar way, a light wave's velocity increases as its wavelength does.

4. Correct option: (A) Aluminium.

Since solids have a higher density and stronger intermolecular interactions than liquids and gases, the speed of sound is in fact greatest in solids, allowing for quicker sound propagation.

The wave of sound is longitudinal. It spreads by pushing the particles in the material medium. The density of the medium that sound is passing through affects its speed. The speed of sound propagation increases with the density of the medium.

Since aluminium has the highest density of the available materials, its sound speed is also the highest.

5. Correct option: (A) Frequency.

A sound wave's frequency is independent of the medium in which it travels. A specific sound wave maintains its invariance across all media. As a result, when a sound wave travels from one medium to another, its frequency doesn't change.

6. Correct option: (C) Frequency, wavelength, and speed.

When sound travels through one medium, it collides with the surface of another and rebounds in a different direction. The reflection of sound is the name given to this occurrence. The wave's frequency, speed, and wavelength are unaffected by reflection.

7. Correct option: (C) Longitudinal waves.

Compression occurs when the spring's component parts go near to one another. Particles get more apart from one another when the spring is expanded. The direction of wave propagation coincides with the direction of particle motion, leading to longitudinal waves.

8. Correct option: (A) Both transverse and longitudinal waves.

Whether the waves are transverse or longitudinal, crests and troughs are features of waves that entail amplitude fluctuations. The crest of a transverse wave is its highest point, while the trough is its lowest. Regions of compression correlate to crests and regions of rarefaction to troughs in a longitudinal wave, like a sound wave. As a result, both types of waves have crest and trough structures.

9. Correct option: (A) 20 Hz - 20,000 Hz.

The spectrum of sounds that people can hear is (A) 20 Hz - 20,000 Hz.

10. Correct option: (B) The speed of the wave doesn't depend on the properties of the medium.

The distance between the crest and trough is half of the wavelength. When a sound is 120 dB loud, it has a $1\frac{W}{m^2}$ intensity. Electronic oscillators are used to create ultrasonic waves. The characteristics of the medium influence the wave's speed.

It is true that the characteristics of the medium in which a wave is propagating affect its speed. The densities, elasticities, and other characteristics of various media influence how quickly a wave may move through them.

11. Correct option: (B) The roaring of a lion is shriller than the buzz of an insect.

The statement is untrue. Higher frequencies and pitches are related to shrillness. The high-frequency buzz of an insect is typically regarded as being louder than the roar of a lion. In fact, when compared to an insect's buzzing sound, a lion's roar typically has a lower frequency and is regarded as having a lower pitch.

12. Correct option: (A) Reverberation of sound.

In order to lessen reverberation, the walls and roof of the theatre or cinema are typically coated with sound-absorbing materials like compressed fibreboard or drapes. These materials absorb sound waves, which lessens the occurrence of echoes.

13. Correct option: (C) The density of the medium decreases.

Particles move closer to one another in an area of the medium known as compression, where their average distance from one another decreases. As a result, the volume temporarily decreases and the medium's density rises as a result.

Similar to rarefaction, which results in a drop in density as particles become more distant from one another.

14. Correct option: (D) Amplitude.

The amplitude of a sound determines how loud it is. A loud sound therefore has a higher amplitude, whereas a weak sound has a lower amplitude. Therefore, to transform a quiet sound into a loud one, we must raise the sound's amplitude.

15. Correct option: (D) A longitudinal wave as well as a transverse wave.

When one end of a slinky is stationary while the other is stretched and given a jolt at a right angle to its length, transverse waves are created.

When compressions are applied to its free end at regular intervals of time, it generates longitudinal waves.

16. Correct option: (A) Waveform.

The sound waveform is the primary factor affecting quality. A tuning fork's sound waveform is a straightforward sine curve. But the majority of human bodies vibrate in a very intricate way. They also include frequencies like 2f, 3f, 4f, etc. in addition to the fundamental frequency f. Overtones or harmonics are the names for these additional tones. According to Helmholtz, the quantity, relative frequency, and intensities of the individual simple tones that make up a complex vibration are what determine the quality.

17. The correct option (B) Decibel.

The reaction of sound to human ears is measured by sound level, often known as loudness of sound. Loudness of sound not only indicates how much energy is reaching the human ear, but also how sensitive the human ear is to detecting this energy. A sound's volume is expressed in decibels (dB).
Since the amount of energy that reaches the ear relies on the square of amplitude, the loudness of a sound depends on a number of variables, including the amplitude of sound waves, the sensitivity of the ear, the distance from the sound source, etc.

18. Correct option: (B) Stationary waves.

A point of zero velocity and hence zero energy is known as a displacement node. Think about the following exercise. With standing waves on a lengthy vibrating string, you could clamp two nodes to the base and cut off the remaining portion, making it genuinely impossible for energy to travel beyond the nodes while leaving the section you kept unchanged. As a result, there is no energy transfer through permanent zero energy sites.

19. Correct option: (B) 2 cm.

Given this, the distance between the fifth and fifteenth crests is 20 cm. We must ascertain the sound's wavelength.
We are aware that a wave's wavelength is the separation between its two crests and troughs.
The distance between the fifth and fifteenth crests is 10λ. So,

$10\lambda = 20$

$\lambda = 2\,\text{cm}$

So, the wavelength of the sound is 2 cm.

20. Correct option: (B) They are easy to produce.

Piezoelectric materials can be used to generate ultrasonic waves with ease.
These waves have a frequency of over 20,000 Hz, which is above the human ear's range of hearing.
In order to vibrate and stir the chemical solutions, these won't be audible.

21. Correct option: (B) Longitudinal waves.

Density-distance graph is used to depict longitudinal waves. Compression and rarefaction of the medium cause longitudinal waves to form. While the air density in the rarefaction zone is lower than that of the surrounding normal air, it is higher in the compression region. Periodically, the density changes as the wave moves through space.

22. Correct option: (B) Lesser.

The Loudness of the sound is given by,

$$L = 10\log_{10}\frac{I}{I_0}$$

Where, I is the intensity of sound.
Greater the surface area of the vibrating body, the lower the sound's amplitude and, consequently, its intensity.
As a result, the volume of the sound will be lower.

23. Correct option: (B) Square.

Greater the surface area of the vibrating body, the lower the sound's amplitude and, consequently, its intensity.
As a result, the volume of the sound will be lower.

24. Correct option: (C) 17.2 m.

Our brains retain the sound-related feeling for roughly 0.1 seconds. This means that there must be a 0.1-second gap between the original sound and the reflected sound in order to clearly hear an echo.

Air's speed of sound at a specific temperature, let's say 20 degrees, $C = 344\frac{\text{m}}{\text{s}}$.

Here, the sound must travel to the obstruction and, after 0.1 s of reflection, return to the listener's ear.

Consequently, the total distance travelled by the sound from its source to its reflecting surface and back $= 344 \times 0.1 = 34.4\text{m}$

In order to hear the echo, the obstruction must be at least half of this distance from the source of the sound.

So, the distance of a person from a sound-reflecting surface to hear an echo is 17.2 m.

25. Correct option: (C) Multiple reflections.

In a stethoscope, sound is picked up by the chest piece and transferred through a lengthy tube by several reflections to the earpieces. By listening to the sound with a stethoscope, doctors can determine the health of an organ.

LEVEL-II

1. Correct option: (D) 0.2 Hz.

The frequency is given by as,

$$\text{Frequncy}(\nu) = \frac{\text{Wave velocity, V}}{\text{Wave length,}\lambda}$$

The wavelength of the wave is equal to the distance between two successive crests, therefore, $\lambda = 100\text{m}$.

$$\nu = \frac{V}{\lambda}$$

$$\nu = \frac{20}{100} = 0.2\text{Hz}$$

So, the frequency is 0.2 Hz

2. Correct option: (C)

The sound travels a distance in time t if t is the time interval for hearing an echo.

$$S = 2 \times 85 = 170\text{m}$$

Now we know that,

$$S = vt$$

$$t = \frac{S}{v}$$

$$t = \frac{170}{240} = 0.5\text{s}$$

So, the time interval is 0.5s.

3. Correct option: (C) Be the same as that of the source.

The frequency of a sound does not vary when it travels through different media. Therefore, whether a source of 20 kHz is produced in water or air, its frequency doesn't change.

4. Correct option: (A) Statement 1 is true but statement 2 is false.

Echoes are produced when sound waves bounce off of a big obstruction. The correct circumstances are necessary for the echo to be audible separately. The size of the object that the sound wave is reflecting from must be greater than the size of the sound wave. It needs to have a polished, firm surface. The original sound must be very brief and the echo's must be loud enough or powerful enough to be heard.

5. Correct option: (B) 3000 m.

As the echo took 4 seconds to return to the surface of the echo receiver after travelling to the bottom. It takes time to go to the ocean floor of $\frac{4\text{Sec}}{2}=2\text{sec}$.

$\text{Distance=speed}\times\text{time}$

$\text{Distance=}1500\times 2$

$\text{Distance=}3000\text{m}$

So, the depth of the sea is 30000m.

6. Correct option:

Distance of person from nearer cliff,

$d_1=\frac{v\times t}{2}$

$d_1=\frac{330\times 2}{2}$

$d_1=330\text{m}$

Distance of person from farther cliff,

$d_2=\frac{v\times t}{2}$

$d_2=\frac{330\times 4.5}{2}$

$d_2=742.5m$

Total distance between two cliffs,

$d=d_1+d_2$

$d=330+743.5$

$d=1320\text{m}$

$\text{d=}1072.5\text{m}$

So, the distance between the two cliffs is 1072.5m.

The option is not available.

7. Correct option: (A)

Here, No. of waves produced in 3 seconds

So, No. of waves produced in 1 second= $\frac{1500}{3}=500$

Consequently, this sound wave's frequency is 500 Hz.

The following equation can be used to determine a sound wave's velocity:

$v=f\times\lambda$

Here, frequency, $f=500\text{Hz}$

And wavelength, $\lambda=68\text{cm}=\frac{68}{100}=0.68\text{m}$.

Putting, these values of f and λ in the above formula, we get,

$v=500\times 0.68$

$v=340\frac{\text{m}}{\text{s}}$

So, the velocity (or speed) of the sound waves is $340\frac{\text{m}}{\text{s}}$.

8. Correct option: (B) 300m.

The distance is given by,

$$d = \frac{1500 \times 0.4}{2}$$

$$d = 300\,\text{m}$$

So, the depth of the shoal is 300m.

9. Correct option: (A) (ii. Frequency of the sitar string with the frequency of other musical instruments.

Before playing the symphony, the sitar's string's frequency must be changed to match that of the other musical instruments. An electronic signal's balance between its many frequency components is adjusted during equalisation in order to bring it into harmony with the frequencies of the other instruments. The length of the string, the tension in the string, and the thickness of the string all affect the frequency of the sound the string makes. By adjusting the sitar's strings, the sitarist balances frequency with other musical instruments. The same physical event is described by the terms loudness, intensity, and amplitude, and they are all influenced by how hard you strike the string.

10. Correct option: (C) Frequency of vibration.

The intensity or loudness of sound depends upon,

Distance (r) of listener from the source ,

$I \propto r$

Amplitude (A) of vibration of the source ,

$I \propto A^2$

Surface area (a) of vibrating body,

$I \propto a$

The amplitude of the wave determines how loud it is.

The biggest deviation a particle makes from its typical position is known as amplitude.

The loudness increases with amplitude. Decibels are units used to measure loudness.

In actuality, frequency affects pitch.

So, the loudness does not depend on Frequency of vibration.

11. Correct option: (D) Density and elasticity of gas.

A mechanical wave is sound. It needs a medium to spread.

A medium is continuously stretched and compressed as sound travels through it. As a result, the medium is disrupted, and the disruption grows. This is how sound propagates throughout the medium.

As a result, the pace will depend on certain material properties of the medium, which will either support or hinder this ongoing stretching and compression action.

For gases, these values are the gas density and the bulk modulus of elasticity, respectively. The bulk modulus of density affects a gas's elasticity and how easily it can be compressed and stretched.

The density prevents this constant stretching and compression. Consequently, based on the Newton-Laplace equation:

$$v = \frac{K}{\rho}$$

Where, v is the speed of the sound in the bus.

K is the gas bulk modulus of elasticity

ρ is the gas density

As a result, the density and elasticity of a gas determine its sound speed.

12. Correct option: (A).

The velocity of sound in gas,

$$V = \sqrt{\frac{\gamma RT}{M}}$$

As long as the temperature is constant, regardless of the gas's pressure, the speed of sound in a gas is independent of that pressure.

Only temperature affects velocity. Therefore, Velocity only depends on only temperature.

13. Correct option: (A) The temperature of the air.

The velocity of sound in gas,

$$V = \sqrt{\frac{\gamma RT}{M}}$$

As long as the temperature is constant, regardless of the gas's pressure, the speed of sound in a gas is independent of that pressure.

In a perfect gas approximation, air pressure has no impact at all on sound speed. This is due to the fact that temperature has no effect on pressure or density in an ideal gas, which means that only temperature can affect the velocity of sound.

Only temperature affects velocity. Therefore, Velocity only depends on only temperature

14. Correct option: (A) The longitudinal movement of air molecules produces pressure fluctuations.

Compressions and rarefactions make up longitudinal waves. When molecules are squeezed or forced together, compression occurs. The converse, known as rarefaction, happens when molecules are given more room and let to expand. Therefore, compression and rarefaction are caused by pressure changes caused by the longitudinal movement of air molecules.

15. Correct option: (A) 1-D,2-A,3-C,4-B.

A disturbance that moves from one place to another across a medium is referred to as a wave. Mechanical waves are waves that move through a solid, liquid, or gas with a wave speed that is influenced by the elastic and inertial characteristics of that medium. A transverse wave is a moving wave made up of oscillations that happen right angles (perpendicular) to the direction of energy transfer or wave propagation. Transverse waves oscillate on the y-z plane in the up and down directions while they are travelling in the positive x direction. Oscillatory motion is defined as repeated motion, such as to-and-fro or back-and-forth motion, where an item repeatedly performs the same motion.

So, the correct matching is 1-D, 2-A, 3-C, 4-B.

7. LIGHT
SOLUTIONS

Refraction of Light
LEVEL-I

1. Correct option: (B)

Given: $_a\mu_g = \frac{3}{2}$

We know that

$$_a\mu_g = \frac{1}{_g\mu_a}$$

So, $_g\mu_a = \frac{1}{\frac{3}{2}} = \frac{2}{3}$

So, refractive index of air w. r. t glass is $2/3$

2. Correct option: (C)

Given: $_a\mu_g = 1.5$

$$_a\mu_w = \frac{4}{3}$$

$$_w\mu_g = \frac{1.5}{\frac{4}{3}} = 1.125$$

Hence, refractive index of glass w. r. t water = 1.125

3. Correct option: (D)

The refractive index depends on the nature of the material of the medium and on the wavelength (or colour) of the light used. The value of refractive index of a substance is a characteristic property of that substance which can be used to identify it.

A substance having higher refractive index is optically denser than another substance having lower refractive index.

4. Correct option(B)

Given: $_a\mu_w = \frac{4}{3}$

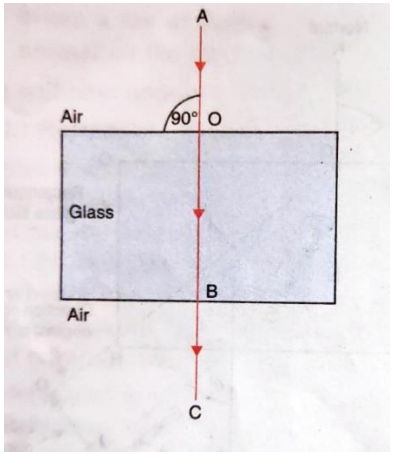
We know that

$$_a\mu_w = \frac{1}{_w\mu_a}$$

So, $_w\mu_a = \frac{1}{\frac{4}{3}} = \frac{3}{4}$

So, refractive index of air w. r. t water is ¾

5. Correct option: (D)



If the incident ray falls normally (or perpendicularly) to the surface of a glass slab, then there is no bending of the ray of light, and it goes straight. For example, in above figure, a ray of light AO travelling in air falls on a glass slab normally (or perpendicularly) at point O, so it does not bend on entering the glass slab or on coming out of the glass slab. It goes straight in the direction AOBC. Since the incident ray goes along the normal to the surface, the angle of incidence in this case is zero (0) and the angle of refraction is also zero (0).

6. Correct option: (D)

Given :

Refractive index of water, μ= 4/3, h=12 cm

Calculation of radius

The ray after refraction grazes along the surface when incident at critical angle (C),

$\mu=\frac{1}{\sin C}$

Sin C $=\frac{1}{\mu}$

Sin C $=\frac{3}{4}$

Applying Pythagoras theorem in the right angled triangle

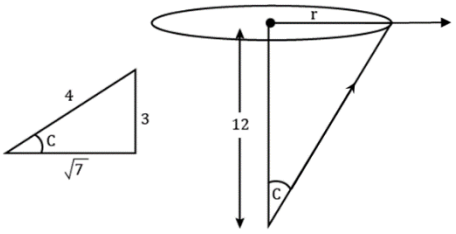
$H^2 = P^2 + B^2$

$B^2 = H^2 - P^2$

$B^2 = 4^2 - 3^2$

$B^2 = 16 - 9$

$B^2 = 7$



Or

B=√7

Tan C = $\frac{3}{\sqrt{7}} = \frac{r}{12}$

Or r = $\frac{12 \text{ x } 3}{\sqrt{7}}$

Radius of circle is r = $\frac{12 \text{ x } 3}{\sqrt{7}}$ cm

7. Correct option: (A)

When light is going from vacuum to another medium, then the value of refractive index is called the absolute refractive index.

The ratio of speed of light in vacuum to the speed of light in a medium, is called the refractive index of that medium. That is:

Refractive index of medium $\mu = \frac{c}{v}$----------(1)

where c = speed of light in vacuum

v= speed of light in medium

given: refractive index of transparent liquid(μ) $= \frac{4}{3}$

speed of light in vacuum(c) = $3x10^8$m/s

putting the values in equation 1

$$\frac{4}{3} = \frac{3x10^8 m/s}{v}$$

$$v = \frac{3x10^8 m/s}{\frac{4}{3}}$$

$v = 2.25x\ 10^8 m/s$

hence, the speed of light in liquid = $2.25x10^8 m/s$

8. Correct option(A)

Given: apparent depth = 2 m

Refractive index = 1.33

Apparent depth (y) = $\frac{\text{actual depth(x)}}{\text{refractive index}(\mu)}$

actual depth(x)= *apparent depth* X refractive index

actual depth(x) = 2 x 1.33 = 2.66

Hence, actual depth of swimming pool = 2.66 m

9. Correct option: (B)

The change in direction of light when it passes from one medium to another obliquely, is called refraction of light. In other words, the bending of light when it goes from one medium to another obliquely is called refraction of light.

10. Correct option: (A)

Both microwaves as well as UV radiations of any wavelength travels in vacuum with the speed of light in vacuum.

Hence speed/velocity of the microwaves and UV radiations is the same.

11. Correct option: (B)

When light travels from a rarer to denser medium, its frequency remains unchanged. It is because, frequency is an inherent property of light.

12. Correct option: (A)

When a ray of light goes from an optically rarer medium to a denser medium, it bends towards the normal (at the point of incidence).

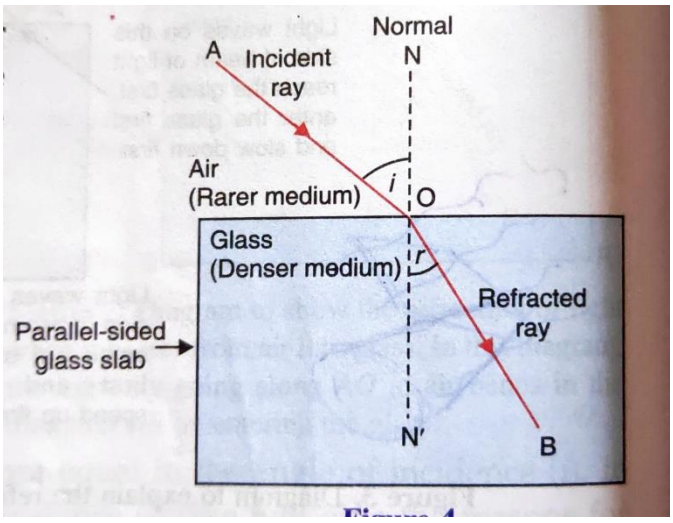


Figure above shows a ray of light AO going from air (a rarer medium) into glass (which is denser medium). In this case, the incident ray AO gets refracted at point O and bends towards the normal ON' in the direction OB. Thus When a ray of light goes from an optically rarer medium to a denser medium, it bends towards the normal (at the point of incidence).

13. Correct option(B)

When a ray of light goes from an optically denser medium to a rarer medium, it bends away from the normal (at the point of incidence).

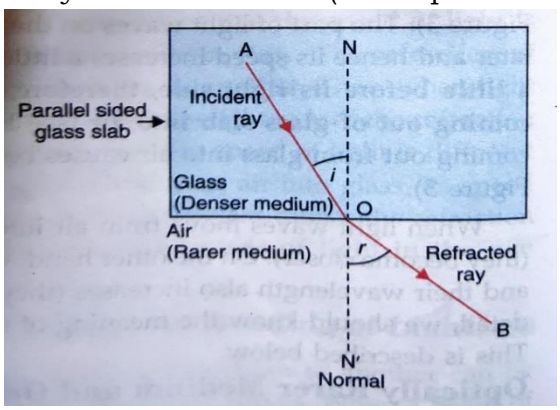


Figure above shows a ray of light AO going from glass (a denser medium) into air (which is rarer medium). In this case, the incident ray AO gets refracted at point O and bends away from the normal ON' in the direction OB. Thus When a ray of light goes from an optically denser medium to a rarer medium, it bends away from the normal (at the point of incidence).

14. Correct option: (B)

Speed of light in vacuum is constant.

Hence

$V_r = V_v$

15. Correct option: (A)

According to Snell's law

$\mu = \frac{\sin i}{\sin r} - - - - - - (1)$

where $\mu = Refractive\ index\ of\ the\ medium$

i= angle of incidence

r= angle of refraction

Given: angle of incidence i = 90˚ - 30˚ =60˚

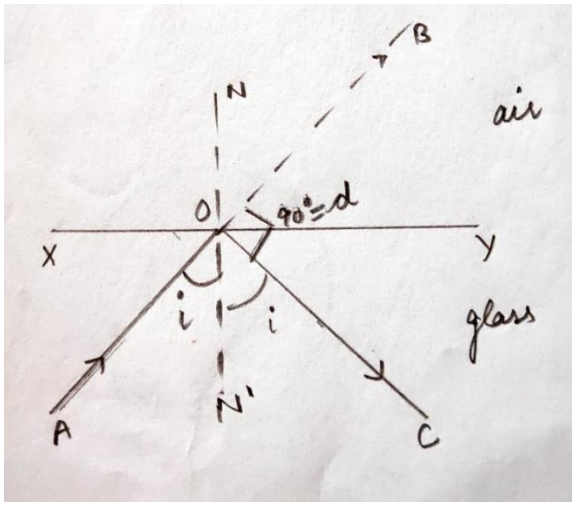
Angle of refraction r = 30˚

Putting in the equation 1 $\mu = \frac{sin60°}{sin30°}$

$\mu = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$

Hence, refractive index of the medium = $\sqrt{3}$

16. Correct option: (B)



This is case of total internal reflection
The ray OC is the reflected ray
Given: angle of deviation (d) = 90°
From the figure above

$$\angle AON' + \angle CON' + \angle BOC = 180° \quad (linear\ pair)$$

$$\angle CON' = \angle AON' = i\ (by\ law\ of\ reflection\ \ i = r)$$

$$2i + d = 180°$$
$$2i + 90° = 180°$$
$$i = \frac{90°}{\mathbf{2}} = 45°$$

17. Correct option: (B)
According to Snell's law

$$\mu = \frac{sin\ i}{sin\ r} - - - - - - (1)$$

where $\mu = Refractive\ index\ of\ the\ medium$
i= angle of incidence
r= angle of refraction
Given: angle of incidence i = 45˚
To find :Angle of refraction
Putting in the equation 1

$$1.526 = \frac{sin\ 45°}{\sin\ r}$$
$$sinr = \frac{1/\sqrt{2}}{1.526} = 0.463$$
$$sinr = sin27.61°$$

Or

$$r = 27.61°$$

Angle of deviation = angle of incidence – angle of refraction
d = 45˚-27.61° = 17.39°

18. Correct option: (A)
Using snell's law:

$$\mu = \frac{\sin i_1}{\sin r_1}$$
$$\sqrt{3} = \frac{sin\ 60°}{\sin r_1}$$
$$\sin r_1 = \frac{sin\ 60°}{\sqrt{3}}$$
$$\sin r_1 = \frac{\sqrt{3}}{2\ \mathrm{x}\ \sqrt{3}}$$
$$\sin r_1 = \frac{1}{2} = \sin 30°$$
$$r_1 = 30°$$

Also from the given figure

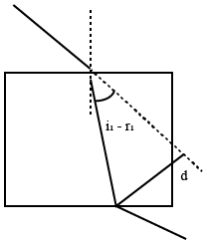
$$\sin(i_1 - r_1) = \frac{\mathrm{d}}{5}$$
$$\sin(60° - 30°) = \frac{\mathrm{d}}{5}$$

$$\sin(30°) = \frac{d}{5}$$

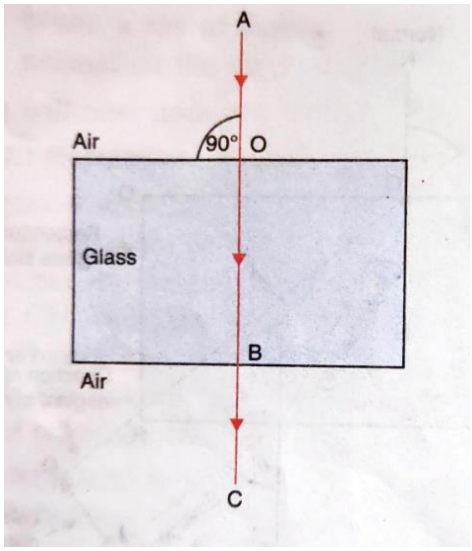
$$5 \times \frac{1}{2} = d$$

Or d = 2.5 cm

So, perpendicular distance = 2.5 cm



19. Correct option: (B)



If the incident ray falls normally (or perpendicularly) to the surface of a glass slab, then there is no bending of the ray of light, and it goes straight. For example, in above figure, a ray of light AO travelling in air falls on a glass slab normally (or perpendicularly) at point O, so it does not bend on entering the glass slab or on coming out of the glass slab. It goes straight in the direction AOBC. Since the incident ray goes along the normal to the surface, the angle of incidence in this case is zero (0) and the angle of refraction is also zero (0).

20. Correct option: (B)

According to Snell's law

$$\mu = \frac{\sin i}{\sin r} - - - - - - (1)$$

where $\mu = Refractive\ index\ of\ the\ medium$

i= angle of incidence

r= angle of refraction

Given: angle of incidence i = 90˚ - θ_1

Angle of refraction r = 90˚ - θ_2

Putting in the equation 1

$$\mu = \frac{\sin 90° - \theta_1}{\sin 90° - \theta_2}$$

$$\mu = \frac{\cos\theta_1}{\cos\theta_2}$$

So, refractive index of glass = $\frac{\cos\theta_1}{\cos\theta_2}$

21. Correct option: (B)

According to Snell's law

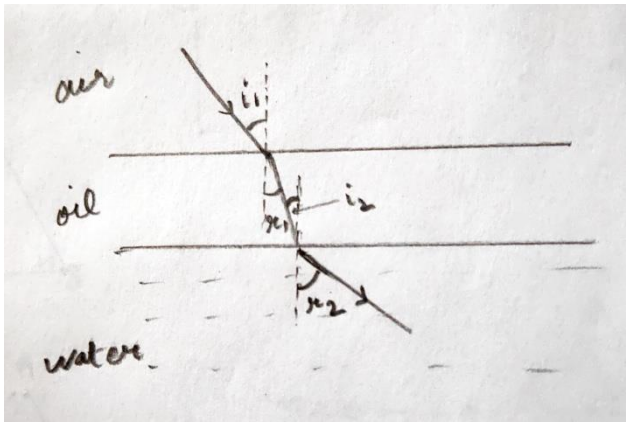
$$\mu = \frac{sin\, i}{sin\, r} - - - - - - (1)$$

where $\mu = Refractive\ index\ of\ the\ medium$
i= angle of incidence
r= angle of refraction
Given: angle of incidence = i
i +r = 90˚
Angle of refraction r = 90˚ - i
Putting in the equation 1

$$\mu = \frac{sin\, i}{sin 90° - i}$$

$$1.6 = \frac{\sin i}{cosi} = \tan i$$
$$\tan 58° = \tan i$$
$$i = 58°$$
$$r = 90° - 58° = 32°$$

22. Correct option: (C)



Applying snell's law at the air – oil interface

$$\mu_o = \frac{sin\, i_1}{sin\, r_1} - - - - - - (1)$$

Applying snell's law at the oil – water interface

$$_o\mu_w = \frac{\mu_w}{\mu_o} = \frac{1.33}{1.45} = 0.92 = \frac{sin\, i_2}{sin\, r_2} - - - - - - (2)$$
$$r_1 = i_2\ (\ alternate\ interior\ angles\ )$$

So, equation 1 becomes

$$\mu_o = \frac{sin\, i_1}{sin\, i_2} - - - - - - (1)$$

Given : $\mu_o = 1.45$
$i_1 = 40°$
Putting in eq 1

$$1.45 = \frac{sin\, 40°}{sin\, i_2} - - - - - - 3$$
$$\sin i_2 = \frac{\sin 40°}{1.45}$$

$\sin i_2 = \frac{0.643}{1.45}$

$\sin i_2 = 0.443$

Putting in eq 2

$0.92 = \frac{0.443}{sin\, r_2}$

$\sin r_2 = \frac{0.443}{0.92} = 0.481$

$r_2 = 29\ (approx)$

23. Correct option: (B)

Applying Snell's law for glass and water,

$\mu_g \sin i = \mu_w \sin r$ ..... (1)

Applying Snell's law for water and air,

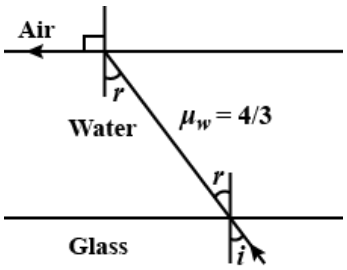
$\mu_w \sin r = \mu_a \sin 90°$ ..... (2)

Using (1) and (2), we can apply Snell's Law for glass and air

$$\mu_g \sin i = \mu_a \sin 90°$$

$$\mu_g \sin i = 1\ x\ 1$$

$$\mu_g = \frac{1}{\sin i}$$



24. Correct option: (B)

The ratio of speed of light in vacuum to the speed of light in a medium, is called the refractive index of that medium. That is:

Refractive index of medium $\mu = \frac{c}{v}$----------(1)

where c = speed of light in vacuum

v = speed of light in medium

given: speed of light in vacuum(c) = $3x10^8$m/s

speed of light in glass(v) = $2x10^8$m/s

putting the values in equation 1

Refractive index of glass w. r.t air $_a\mu_g = \frac{3x10^8}{2\ x\ 10^8} = \frac{3}{2}$

25. Correct option: (C)

Given: $_a\mu_g = \frac{3}{2}$

$_a\mu_w = \frac{4}{3}$

$$_w\mu_g = \frac{\frac{3}{2}}{\frac{4}{3}} = \frac{9}{8}$$

26. Correct option: (B)

The refractive index of medium 2 with respect to medium 1 is equal to the ratio of speed of light in medium 1 to the speed of light in medium 2.

This can be written as:

$$_1\mu_2 = \frac{v_1}{v_2} - - - (1)$$

Where v_1 =Speed of light in medium 1

And v_2 =Speed of light in medium 2

$refractive\ index\ of\ glass\ w.r.t\ water \quad {}_w\mu_g = \frac{\mu_g}{\mu_w} = \frac{\frac{3}{2}}{\frac{4}{3}} = \frac{9}{8}$

Also from equation 1 $_w\mu_g = \frac{v_w}{vg}$

$$\frac{9}{8} = \frac{v_w}{2x10^8}$$

$$\frac{9}{8} x\ 2x10^8 = v_w$$

$$\frac{9}{4} x\ 10^8 m/s = v_w$$

Hence, speed of light in water = $\frac{9}{4} x\ 10^8 m/s$

27. Correct option: (A)

Refractive index of medium $\mu = \frac{v_o}{v}$-

Or $v = \frac{v_o}{\mu}$---------(1)

where v_o= speed of light in vacuum

v= speed of light in medium

we know that

speed of light in a medium is given by v=$f\ \lambda$

where f= frequency

$\lambda = wavelength$

for vacuum

$v_o = f_o\lambda_o$----------(2)

For medium

$v = f\ \lambda$----------(3)

frequency doesn't change when light rays travel from one medium to other

$$so,\ \ f = f_o$$

Dividing equation 3 by 2

$$\frac{v}{v_o} = \frac{f}{f_o}\ x\frac{\lambda}{\lambda_o}$$

From equation 1 v = $\frac{v_o}{\mu}$

$$\frac{\frac{v_o}{\mu}}{v_o} = \frac{\lambda}{\lambda_o}$$

$$\frac{1}{\mu} = \frac{\lambda}{\lambda_o}$$

$$\frac{\lambda_o}{\mu} = \lambda$$

Given : $\lambda_o = \ wavelength\ in\ air = \ 5800\ A°$

$\mu = 1.33$

$$\frac{5800}{1.33} = \lambda$$

$$4360.9\ A° = \lambda$$

Hence, wavelength in water = 4360.9 A°

28. Correct option: (D)

The refractive index of medium 1 with respect to medium 2 is equal to the ratio of speed of light in medium 2 to the speed of light in medium 1.

This can be written as:

$$_2\mu_1 = \frac{v_2}{v_1} - - - (1)$$

Where v_1 =Speed of light in medium 1

And v_2 =Speed of light in medium 2

29. Correct option (C)

The refractive index of medium 2 with respect to medium 1 is equal to the ratio of speed of light in medium 1 to the speed of light in medium 2.

This can be written as:

$$_1\mu_2 = \frac{v_1}{v_2} - - - (1)$$

Where v_1 =Speed of light in medium 1

And v_2 =Speed of light in medium 2

Also from equation 1 $_a\mu_g = \frac{v_a}{vg} \ldots (2)$

$$_a\mu_g = \frac{sin\ i\ (air)}{sin\ r\ (glass)} \ldots\ldots (3)$$

Comparing eq 2 and 3

$$\frac{\sin\ \mathrm{i}(air)}{\sin \mathrm{r}\ (glass)} = \frac{v_a}{vg}$$

Putting the values

$$\frac{\sin 60°}{\sin \mathrm{r}(glass)} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$$

$$\sin r = \frac{\frac{\sqrt{3}}{2}}{1.5} = \frac{\sqrt{3}}{3} = 0.577$$

$$\mathrm{r} = 35.26\ °$$

for second glass

Putting the values

$$\frac{\sin 45°}{\sin \mathrm{r}'(glass)} = \frac{3 \times 10^8}{2.2 \times 10^8} = 1.36$$

$$\sin r' = \frac{\frac{1}{\sqrt{2}}}{1.36} = 0.519$$

$$\mathrm{r}' = 31.23\ °$$

Ratio =

$$\frac{\mathrm{r}}{r'} = 35.26 : 31.23$$

30. Correct option: (B)

The refractive index depends on the nature of the material of the medium and on the wavelength (or colour) of the light used. The value of refractive index of a substance is a characteristic property of that substance which can be used to identify it.

LEVEL-II

1. Correct option: (C)

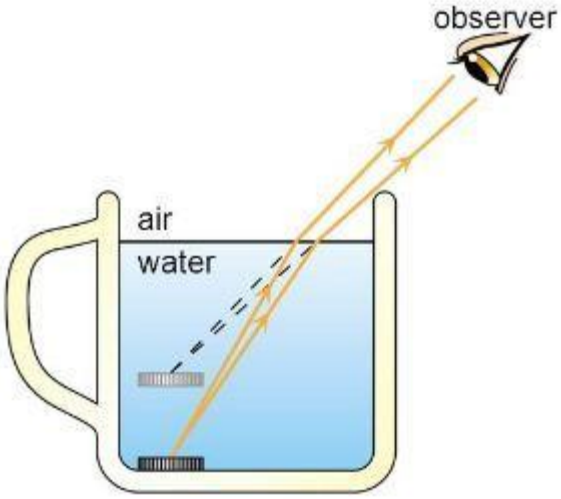
When a pencil is partly immersed in water and held obliquely to the surface, the pencil appears to bend at the water surface (when viewed from the side). This apparent bending of the pencil is due to the refraction of light when it passes from water into air.

2. Correct option:(B)

For the light to bend, walls should be very thin, as thin as 1 nanometer which is not possible hence light waves do not bend or get refracted across the brick wall

3. Correct option: (D)

This is because of refraction of light. As light passes from water to air, it bends away from the normal. The refracted rays appear to come from a point above the actual position of the penny, hence the penny appears raised.



4. Correct option: (B)

The incident angle is the angle made by the incident light ray with the normal to the surface, at the point of incidence.

The angle of emergence is formed by a light ray that exits from the second surface of the rectangular slab. It is parallel to the incident ray.

Therefore, for refraction through a glass slab, the emergent angle (e) should be equal to the incident angle (i)

So i=e

And

$r_1 = r_2 (\ alternate\ interior\ angles)$

5. Correct option: (B)

Applying Snell's law for water and air,

$\mu_w \sin i = \mu_a \sin 90°$

$$1.33 \text{ x} sini = 1 \text{ x } 1$$

$$sini = \frac{1}{1.33} = 0.75$$

$$i = \sin^{-1}(0.75)$$

6. Correct option: (C)

Applying Snell's law for vacuum and medium,

$\mu_v \sin i = \mu_m \sin r$

$$1 \sin i = \mu \sin \frac{i}{2} \qquad as\ r = \frac{i}{2}$$

$$2 \sin \frac{i}{2} \cos \frac{i}{2} = \mu \sin \frac{i}{2}$$

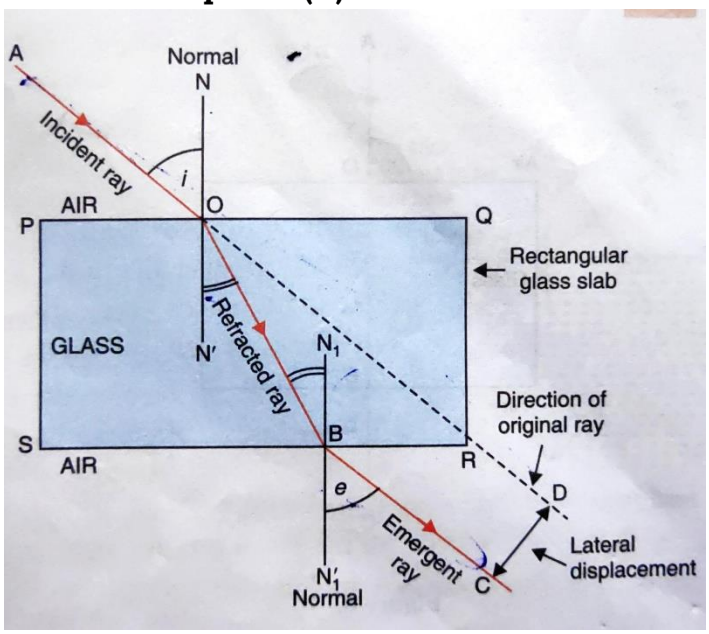
$$2\cos \frac{i}{2} = \mu$$

$$\cos \frac{i}{2} = \frac{\mu}{2}$$

$$\frac{i}{2} = cos^{-1} \frac{\mu}{2}$$

$$i = 2cos^{-1} \frac{\mu}{2}$$

7. Correct option: (B)



The incident ray AO and the emergent ray BC are parallel to each other (though the emergent ray has been displaced parallel to the incident ray). The incident ray AO and emergent ray BC are parallel to each other because the extent of bending of the ray of light at points O and B on the opposite, parallel faces (PQ and SR) of the rectangular class slab is equal and opposite. The incident ray AO bends towards the normal at point O whereas the refracted ray OB bends away from the normal at point B by an equal amount. Thus, the light emerges from a parallel-sided glass slab in a direction parallel with that in which it enters the glass slab. **Though the emergent ray BC is parallel to the incident ray AO, but the emergent ray has been sideways displaced for laterally displaced) from the original path of the incident ray** by a perpendicular distance CD the original path of incident light is AOD but the emergent light goes along BC, the lateral displacement between them being CD. The perpendicular distance between the original path of incident ray and the emergent ray coming out of the glass slab is called **lateral displacement** of the emergent ray of light.

8. Correct option: (C)

According to Snell's law

$$\mu = \frac{sin\ i}{sin\ r} - - - - - - (1)$$

where $\mu = Refractive\ index\ of\ the\ medium$

i= angle of incidence

r= angle of refraction

Given: angle of incidence i = 90° - 30° = 60°

$\mu_a = 1$, $\mu_r = 1.73$

Let Angle of refraction = r

Putting in the equation 1

$$\mu_r = \frac{sin\ 60°}{\sin r}$$

$$1.73 = \frac{\frac{\sqrt{3}}{2}}{\sin r}$$

$$sinr = \frac{\frac{\sqrt{3}}{2}}{1.73} = \frac{1}{2}$$

$$sinr = \sin 30°$$

$$r = 30°$$

So, value of x = 90 − 30 = 60°

9. Correct option: (D)

The emergent ray CD can become parallel to incident ray AB after traveling through different media, only when both of them travel in same medium of same refractive index.

Therefore $\mu_4 = \mu_1$

10. Correct option: (C)

$\mu_{eq} = \frac{t_1+t_2}{\frac{t_1}{\mu_1}+\frac{t_2}{\mu_2}} = \frac{10+15}{\frac{10}{4/3}+\frac{15}{3/2}} = \frac{25}{\frac{30}{4}+\frac{30}{3}} = \frac{25\,x12}{90+120} = \frac{300}{210} = 1.43$

11. Correct option: (B)

Apparent depth = total depth – shift

D'= (8+4.5+6)-6 cm

D' = 12.5 cm

Apparent depth = $\mu_{ob}\left(\frac{t_1}{\mu_1}+\frac{t_2}{\mu_2}+\frac{t_3}{\mu_3}\right)$

$12.5 = 1\left(\frac{6}{\frac{4}{3}}+\frac{4.5}{\mu}+\frac{8}{1.6}\right)$

$12.5 = \left(4.5+\frac{4.5}{\mu}+5\right)$

$12.5 = 9.5 + \frac{4.5}{\mu}$

$3 = \frac{4.5}{\mu}$

$$\mu = \frac{4.5}{3} = 1.5$$

12. Correct option: (B)

When the light ray goes from medium having refractive index μ_1 to medium having refractive index μ_2

It bends away from the normal which means $\mu_1 > \mu_2$

Also, When the light ray goes from medium having refractive index μ_2 to medium having refractive index μ_3

It goes away undeviated which means $\mu_2 = \mu_3$

13. Correct option: (D)

The ratio of speed of light in vacuum to the speed of light in a medium, is called the refractive index of that medium. That is:

Refractive index of medium $\mu = \frac{c}{v}$----------(1)

where c = speed of light in vacuum

v= speed of light in medium

As the refractive index is inversely proportional to the speed of light in medium

So, more is the speed of light lesser is the refractive index in that medium

It is given that speed of light is highest in water and lowest in glass

So water has lowest refractive index and glass has highest

Hence, decreasing order of refractive index is

Glass, glycerine, plastic, water

14. Correct option: (B)

Apparent depth (y) = $\frac{\text{actual depth(x)}}{\text{refractive index}(\mu)}$

As apparent height is inversely proportional to the refractive index so, lesser is the refractive index of the medium more will be the apparent height

Hence, apparent height will be maximum in water

15. Correct option: (C)

According to Snell's law

$$\mu = \frac{sin\ i}{sin\ r} - - - - - - (1)$$

where $\mu = Refractive\ index\ of\ the\ medium$

i= angle of incidence

r= angle of refraction

As refractive index is inversely proportional to the angle of refraction so, lesser is the angle of refraction, more will be the refractive index of the medium

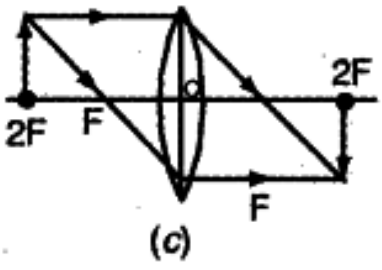
Hence decreasing order is

$\mu_3, \mu_1, \mu_4\ and\ \mu_2$

SPHERICAL LENSES
LEVEL-I

1. Correct option: (B)

when the object is placed at the centre of curvature(2F) of the convex lens, it forms a real, inverted image of same size as the object.



2. Correct option: (B)

Focal length of concave mirror and concave lens is negative.

3. Correct option: (B)

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

Given : focal length = 50 cm = 0.50 m

P = $\frac{1}{0.5}$D= 2 D

Hence, power = 2 D

4. Correct option: (D)

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

$$or\ f = \frac{1}{P}$$

Given : power = - 2. 0 D

f = $\frac{1}{-2}$m = - 0.5 m

Hence, focal length = - 0.5 m

5. Correct option: (B)

Total power of any number of lenses in contact is equal to algebraic sum of the powers of individual lenses.

$P = P_1 + P_2$----------(1)

From equation 1

P = 5 D – 5D

P = 0D

Hence, power of resulting lens = 0 D

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

$$or\ f = \frac{1}{P}$$

Given : power = 0 D

f = $\frac{1}{0}$ = $\boldsymbol{infinity}$

Hence, focal length = infinity

6. Correct option: (B)

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

$$or\ f = \frac{1}{P}$$

Given : power = - 2. 0 D

f = $\frac{1}{-2}$m = - 0.5 m

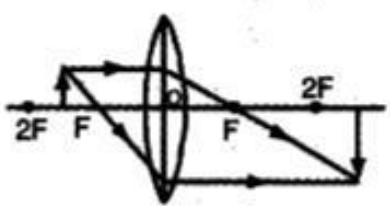
Hence, focal length = - 0.5 m = - 50 cm

7. Correct option: (B)

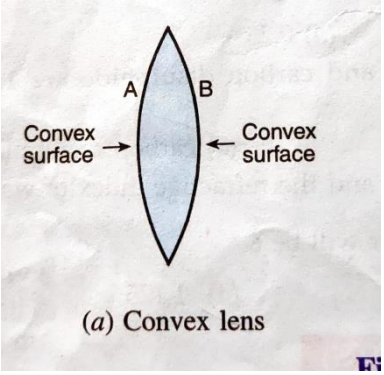
Focal length of coloured goggles is infinity as their power is 0.

8. Correct option: (C)

When the object is placed between the focus(F) and the centre of curvature(2F), it forms a real, inverted enlarged image.



9. Correct option: (A)



A convex lens is thick at the middle but thinner at the edges. Figure below shows a convex lens in which the two surfaces A and B are convex or bulging out at the centre.

10. Correct option: (C)

The rod will be invisible. Since the rod and kerosene oil have the same refractive index, the rod will be invisible inside the oil.

11. Correct option: (A)

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

Given : focal length = 25 cm = 0.25 m

P = $\frac{1}{0.25} = \frac{100}{25} = 4D$

Hence, power = 4 D

12. Correct option: (A)

To solve this question we will use the *Lens maker's formula which is given below*

$$\frac{1}{f} = \left[\frac{\mu_2}{\mu_1} - 1\right] \times \left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$here\ ,\quad f = focal\ length\ of\ the\ lens$$

$$\mu_2 = refractive\ index\ of\ the\ medium\ of\ lens$$

$$\mu_1 = refractive\ index\ of\ the\ surrounding\ \ medium$$

Given:

Refractive index of medium of lens μ_2 = 1.5

Refractive index of surrounding medium μ_1 = 1.33

As it is biconvex lens

let R_1 = +R

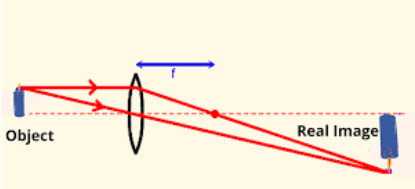
R_1 = - R

$$\frac{1}{f} = \left(\frac{1.5}{1.33} - 1\right)\left(\frac{1}{R} + \frac{1}{R}\right)$$

So, clearly the focal length will come out to be positive, therefore it will behave like a convergent lens.

13. Correct option: (A)

A real image occurs where final rays converge, whereas a virtual image occurs where rays only appear to diverge.

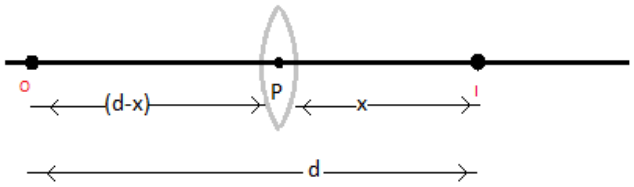


14. Correct option: (D)

The black painted region blocks some light rays entering. This makes a slight difference in the intensity. It reduces the brightness of the image more than usual. So the image formed is complete but less bright.

15. Correct option: (D)

Let I is the real image of an object which is placed at O. Let d be the distance between them. If the distance of the image from the center of the lens is x. then the distance between object and center of lens will be, (d−x).



As we know, sign convention for convex lenses. Thus, u=−(d−x) and v=+x putting the values in the lens formula

$$\frac{1}{f}=\frac{1}{x}-\frac{1}{-(d-x)}$$

$$\frac{1}{f}=\frac{1}{x}+\frac{1}{(d-x)}$$

$$\frac{1}{\mathrm{f}}=\frac{\mathrm{d}-\mathrm{x}+\mathrm{x}}{x(d-x)}=\frac{d}{dx-x^2}$$

$$\mathrm{x}^2-xd-fd=0$$

For a real image, the value of x must be a real number, for roots of equation will be real when,

$\mathbf{d^2} \geqslant 4\mathrm{fd}$

Or

$\mathrm{d} \geqslant 4\mathrm{f}$

Hence, from the above equation we get the minimum distance between an object and its real image formed by a convex lens is d=4f

16. Correct option: (D)

When two thin lenses of focal lengths f_1 and f_2 are in contact with each other , F is the focal length of the combination, then power of the combination is given by

$$P=\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

$$\frac{1}{F}=\frac{f_1+f_2}{f_1f_2}$$

Or

$$F=\frac{f_1f_2}{f_1+f_2}$$

Let $f_1=+f \quad , \quad f_2=-f$

Then focal length of combination will be $F=\frac{\mathrm{f}^2}{\mathrm{f}-\mathrm{f}}=\frac{f^2}{0}=infinity$

17. Correct option: (C)

$$power\ of\ a\ lens=\frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P=\frac{1}{f}$$

$$or\ f=\frac{1}{P}$$

Given : power of combination of lens = 1D +1D = 2D

f = $\frac{1}{2}$m = 0.5 m

Hence, focal length = 0.5 m = 50 cm

Object distance (u) = $-200\ cm$

putting the values in the lens formula

putting the values in the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f= focal length of the lens

u=object distance from the lens

v= image distance from the lens

$$\frac{1}{50} = \frac{1}{v} - \frac{1}{-200}$$

$$\frac{1}{50} - \frac{1}{200} = \frac{1}{v}$$

$$\frac{4-1}{200} = \frac{1}{v}$$

$$\frac{3}{200} = \frac{1}{v}$$

$$\frac{200}{3} = v$$

$$66.67 \text{ cm} = v$$

Hence, image is formed at a distance of 66.67 cm from the lens system.

18. Correct option: (C)

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

Given : Radius of curvature of convex lens (R) = 2m

Focal length = R/2

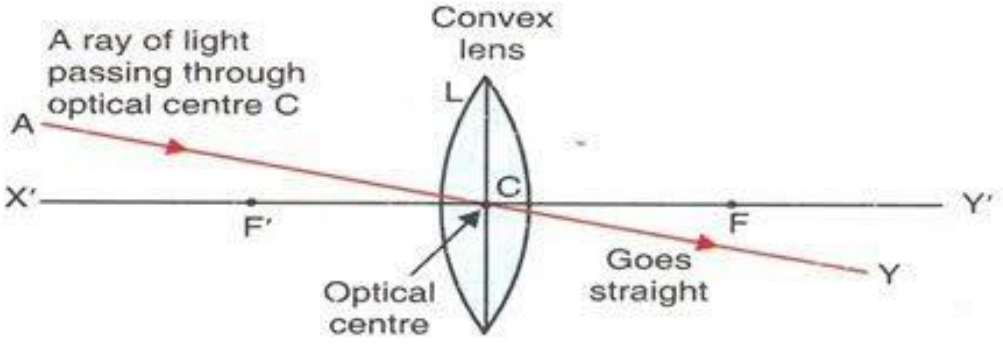
focal length = 2 m / 2 = = 1 m

P = $\frac{1}{1}$ = 1 D

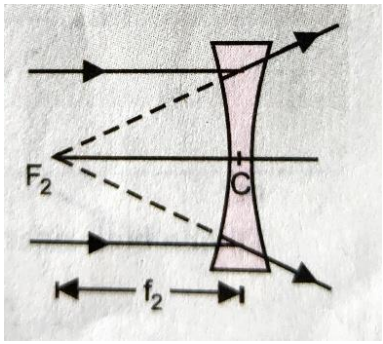
Hence, power = 1 D

19. Correct option: (C)

The rays passing through the optical centre of the lens pass undeviated. The optical centre of the lens is defined as the point on the major axis through which light rays pass without deflection, or it can be described as the central point of the lens through which light flows without deflection. A ray of light travelling through the optical centre will not change its course.



20. Correct option: (D)



When a light ray parallel to the principal axis is incident on a concave lens, the refracted ray will appear to converge at the second focus

21. Correct option: (B)

Suppose a parallel beam of light rays falls on a convex lens parallel to one another and also parallel to the axis of the lens and get refracted (or bent) according to the laws of refraction. All the rays, after passing through the convex lens, converge at the same point F on the other side (right) side) of the lens. The point F is called principal focus (or) just focus) of the convex lens. We can now say that: The principal focus of a convex lens is a point on its principal axis to which light rays parallel to the principal axis converge after passing through the lens.

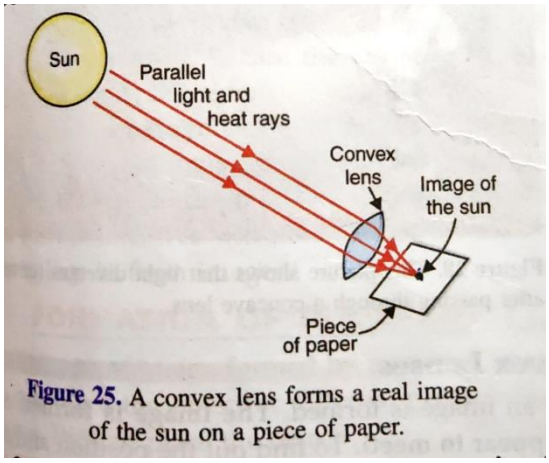
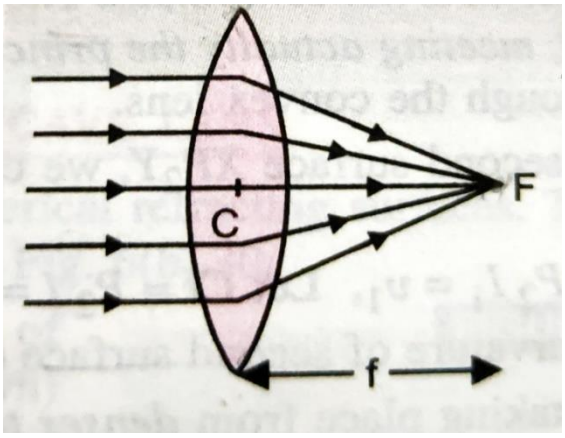


Figure 25. A convex lens forms a real image of the sun on a piece of paper.



22. Incorrect values

23. Correct option(B)

When two thin lenses of focal lengths f_1 and f_2 are in contact with each other , F is the focal length of the combination, then power of the combination is given by

$$P = \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{F} = \frac{f_1 + f_2}{f_1 f_2}$$

Or

$$F = \frac{f_1 f_2}{f_1 + f_2}$$

$$f_1 = -30cm \quad , \quad f_2 = -30cm$$

Then focal length of combination will be $F = \frac{30^2}{-30-30} = \frac{900}{-60} = -15\ cm$

24. Correct option: (B)

Formula for refraction at curved surface for light rays going from rarer to denser medium is as follows:

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$----------(1)

Where μ_2= refracting index of the denser medium
μ_1= refracting index of the rarer medium
R= Radius of curvature of the refracting surface
u=object distance from the refracting surface
v= image distance from the refracting surface

given: $\mu_1 = 1$

$$\mu_2 = 1.5$$

Radius of curvature = 20 cm

object distance from the pole of the refracting surface(u) = - 100 cm

putting the values in equation 1

$$\frac{1.5}{v} - \frac{1}{-100} = \frac{1.5-1}{20}$$

$$\frac{15}{10v} + \frac{1}{100} = \frac{0.5}{20}$$

$$\frac{3}{2v} + \frac{1}{100} = \frac{1}{40}$$

$\frac{3}{2v} = \frac{1}{40} - \frac{1}{100}$ (lcm of 100 and 40 = 200)

$$\frac{3}{2v} = \frac{5-2}{200}$$

$$\frac{3}{2v} = \frac{3}{200}$$

$$\frac{1}{v} = \frac{1x2}{200} = \frac{1}{100}\,cm$$

v = 100 cm

Hence, real image is formed at a distance of 100 cm towards the right.

25. Correct option: (D)

Given: magnification(m) = 4

Focal length (f) = 20 cm

Magnification(m) = $\frac{\text{image distance(v)}}{\text{object distance(u)}}$----------(1)

4 = $\frac{\text{image distance(v)}}{\text{object distance(u)}}$

V = 4 u…… (2)

From lens formula

$\frac{1}{\text{f}} = \frac{1}{\text{v}} - \frac{1}{\text{u}}$

Where f= focal length of the concave lens
U=object distance from the lens
V= image distance from the lens

$\frac{1}{20} = \frac{1}{4\text{u}} - \frac{1}{\text{u}}$

$\frac{1}{20} = \frac{1-4}{4\text{u}}$

$\frac{1}{20} = \frac{-3}{4\text{u}}$

$\text{u} = \frac{-3\text{x}\,20}{4} = -15\,cm$

From equation 2

V= - 4 x 15 = - 60 cm

26. Correct option: (B)

Total power of any number of lenses in contact is equal to algebraic sum of the powers of individual lenses.

$P = P_1 + P_2$----------(1)
Given:
Power of convex lens $P_1 = +6D$
Power of concave lens $P_2 = -2D$
From equation 1
P = 6D – 2 D
P = 4 D
The power of a lens is the reciprocal of its focal length in metres.

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

As
power = 4D
$focal\ length\ of\ a\ lens = \frac{1}{4}$=0.25 m

$$focal\ length\ of\ a\ lens = 0.25 \text{ x } 100\ cm = 25\ cm$$

27. Correct option: (B)

When two thin lenses of focal lengths f_1 and f_2 are in contact with each other , F is the focal length of the combination, then power of the combination is given by

$$P = \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{F} = \frac{f_1 + f_2}{f_1 f_2}$$

Or

$$F = \frac{f_1 f_2}{f_1 + f_2}$$

28. Correct option: (D)

1) The image will form at the same distance which is at the focal length of the lens ie 10 cm. The use of lens having double aperture will help in better illumination of image.
2) The image will form at a distance of 5 cm. The focal length of the lens is inversely proportional to the power. So, when another lens of same aperture but double the power will be used, the focal length will become half.

29. Correct option: (C)

Area of each square, A = $1mm^2$
Object distance (u) = $-9cm$
Focal length (f) = 10 cm
putting the values in the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f= focal length of the lens
u=object distance from the lens
v= image distance from the lens

$$\frac{1}{10} = \frac{1}{v} - \frac{1}{-9}$$

$$\frac{1}{10} - \frac{1}{9} = \frac{1}{v}$$

$$\frac{9 - 10}{90} = \frac{1}{v}$$

$$\frac{-1}{90}=\frac{1}{v}$$
$$-90=v$$
$$-90\text{cm}=v$$

Magnification, m=v/u=−90/(−9)=10

Area of each square = $10\,x\,10\,mm^2 = 100mm^2 = 1cm^2$

magnifying power of lens $\text{m}=\frac{\text{d}}{\text{u}}=\frac{25}{9}=2.8$

30. Correct option: (C), (D)

For the given concave mirror: f=−24 cm

Mirror formula is

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$
$$\text{v}=\frac{\text{uf}}{u-f}$$

Case 1 : when u = - 42 cm

$$\text{v}=\frac{-42\,\text{x}-24}{-42+24}=-56\,cm$$

Case 2: when u = - 48 cm

$$\text{v}=\frac{-48\,\text{x}-24}{-48+24}=-48cm$$

Case 3: when u = - 66 cm

$$\text{v}=\frac{-66\,\text{x}-24}{-66+24}=-37.71\,cm$$

Case 4: when u = - 78 cm

$$\text{v}=\frac{-78\,\text{x}-24}{-78+24}=-34.66cm$$

LEVEL-II

SINGLE ANSWER QUESTIONS

1. Correct option: (B)

When the object is placed at the first principal focus of the convex lens then the final image is formed at infinity.

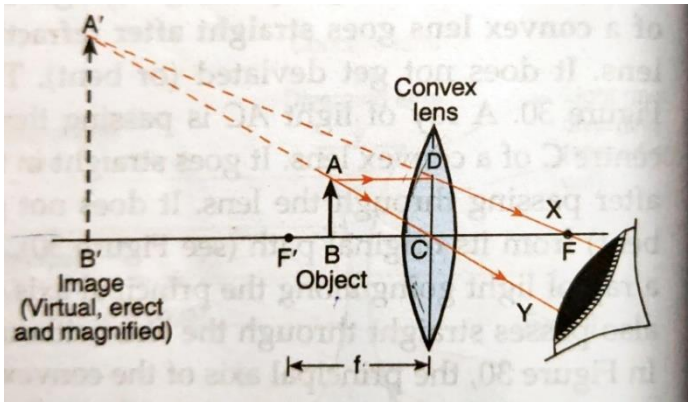
2. Correct option: (D)

As the image of object placed at infinity is formed at the focus of the lens

If the screen is moved towards the lens the image will get blur

3. Correct option: (D)

When the object is placed between its optical centre (C) and focus (F). The image formed is behind the object (on the left side of lens), virtual and erect, and larger than the object (enlarged or magnified).



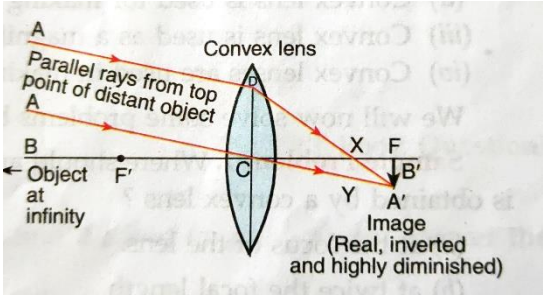
4. Correct option: (A)

When the object is placed at the infinity from the convex lens, then the image formed is

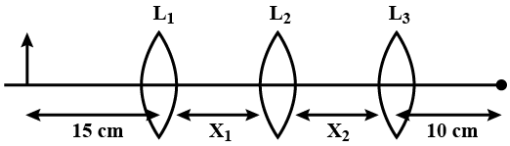
At the focus

Real and Inverted

Highly diminished



5. Correct option: (D)



Object distance (u) = $-15cm$

Focal length (f) = 10 cm

putting the values in the lens formula

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Where f= focal length of the lens

u=object distance from the lens

v= image distance from the lens

$$\frac{1}{10}=\frac{1}{v}-\frac{1}{-15}$$
$$\frac{1}{10}-\frac{1}{15}=\frac{1}{v}$$
$$\frac{3-2}{30}=\frac{1}{v}$$
$$\frac{1}{30}=\frac{1}{v}$$
$$30=v$$
$$30\text{cm}=v$$

we have seen above that the image by L_1 is formed at 30 cm right of L_1 which is at 10cm left of L_2

The separations between the lenses will be as follows:

Distance between L_2 and L_3 doesn't matter as the image formed by L_2 is at infinity so that X2 can take any value.

Image by L_3 is formed at the focus. So the object should lie at infinity for L_3

So, L_2 will form image at infinity and the object for L_2 should be at its focus.

So distance between L_1 and L_2 = 30 cm + 10 cm = 40 cm

6. Correct option: (C)

For a convex lens

An image of same size is formed when the object is placed at center of curvature 2F on the left side then image is formed at the same distance from lens on the right hand side.

From the above figure it is clear that

Distance x = 2f

Or f = $\frac{x}{2}$

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

7. Correct options: (B), (C), (D)

A convex lens is a converging lens as it converges the parallel rays of light at its principal focus

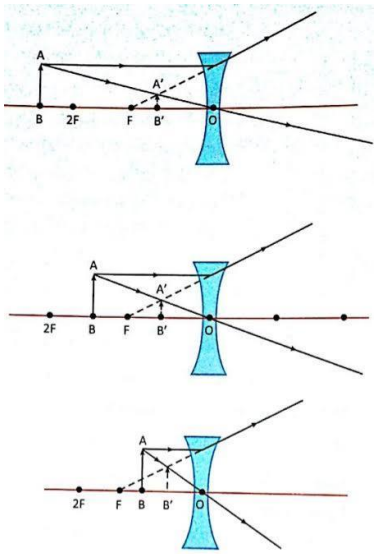
It can form

Real, inverted, diminished or magnified image

Virtual, erect and magnified image when object is placed between the optical center and the focus.

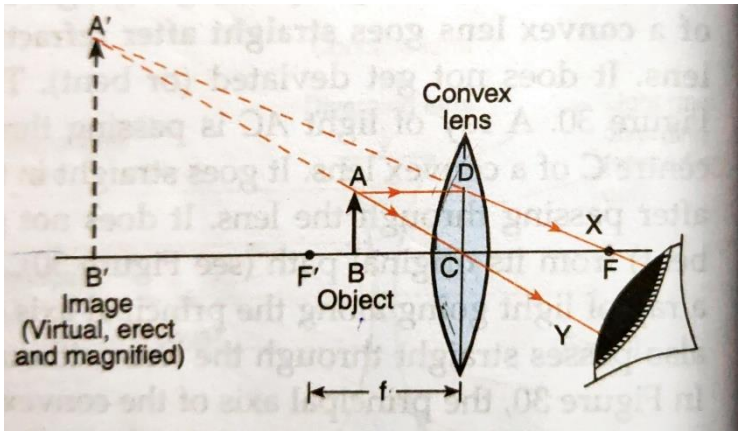
8. Correct options: (A), (C)

No matter where the object is placed in front of the concave lens, the concave lens always forms a virtual, erect and diminished image of the object.



9. Correct options: (A), (B), (C), (D)

When the object is placed between its optical centre (C) and focus (F). The image formed is behind the object (on the left side of lens), virtual and erect, and larger than the object (enlarged or magnified).



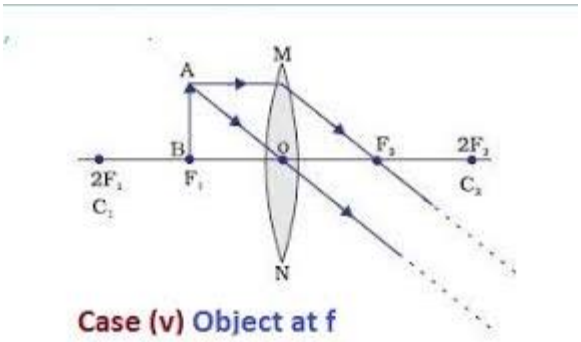
10. Correct options: (C), (D)

When the object is placed at the focus of the convex lens, the image formed is

At Infinity

Real and inverted

Highly enlarged

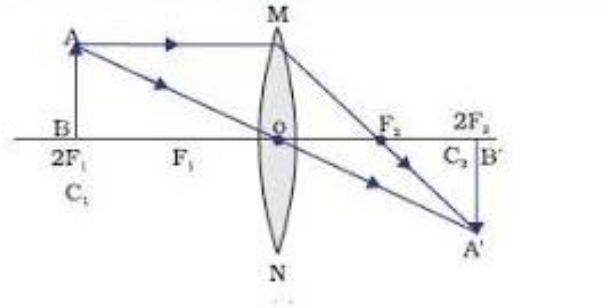


Case (v) Object at f

11. Correct options: (A), (B), (C)

when the object is placed at the centre of curvature(2F) of the convex lens, it forms a real, inverted image of same size as the object on the other side of the lens.

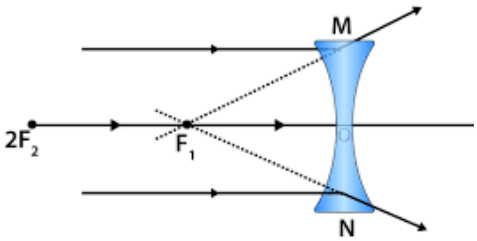
diminished



Case (iii) Object at 2f

12. Correct options: (C), (D)

No matter where the object is placed in front of the concave lens, the concave lens always forms a virtual, erect and image of the object.



13. Correct options: (A), (D)

In deriving the lens maker's formula, we use New Cartesian Sign Conventions, which are:

1. All distances are measured from the optical centre of the lens.
2. All the distances measured in the direction of incidence of light are taken as positive, whereas all the distances measured in a direction opposite to the direction of incidence of light are taken as negative.
3. For a convex lens, f is positive and for a concave lens, f is negative,

14. Correct options: (A), (D)

For a convex lens focal length is positive so radius of curvature is also taken as positive according to new cartesian sign conventions.

15. Correct options: (A), (C), (D)

(i) As the object is places to the left of the lens so it's distance is taken as negative.

(ii) As real image is formed on the right side of the convex lens, hence, its distance is taken as positive.

(iii) Virtual image is formed on the same side as the object so its distance is taken as negative.

16. Correct options: (A), (B), (C), (D)

According to New Cartesian Sign Conventions:

(i) For a convex lens, f is positive and for a concave lens, f is negative.

(ii) Virtual image is formed on the same side as the object so its distance is taken as negative.

17. Correct options: (A), (B), (C)

(i) The power of a lens is the reciprocal of its focal length in metres.

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

(ii) S. I unit of power of a lens is dioptre.

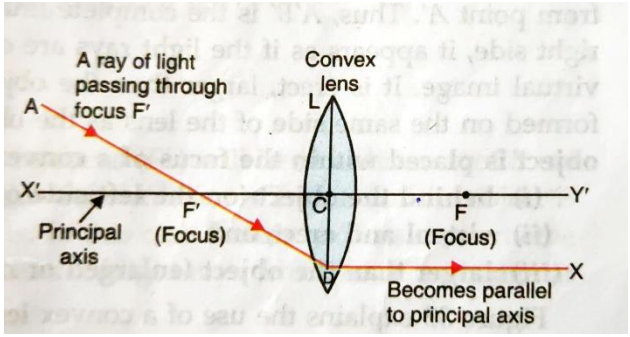
(i) The power of a lens is measured by an instrument known as a dioptremeter.

(ii) Power of concave lens is negative.

STATEMENT TYPE QUESTIONS

18. Correct option: (B)

Principal focus is a point on the principal axis through which if any ray passes, then it emerges parallel to the incident ray.



Focal plane is a plane that is perpendicular to the principal axis of a lens or mirror and passes through the focus.

19. Correct option: (A)

Because projector works by focussin an image on the screen. Light from the projector hits the screen and then that focused light forms the image which we can see.

Lamp light will throw unfocussed light and we will not be able to see the image.

The aim of the condenser lens in a projector system is to align the light rays into a straight path. This allows the maximum amount of light possible to reach the projector screen.

20. Correct option: (C)

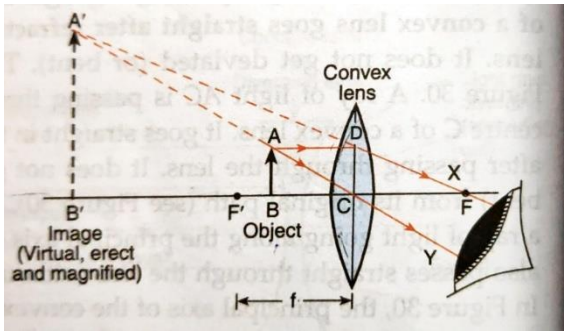
A biconvex lens behaves as a magnifying glass due to the following property

When the object is placed between its optical centre (C) and focus (F). The image formed is

(i) behind the object (on the left side of lens),

(ii) virtual and erect, and

(iii) larger than the object (enlarged or magnified).



21. Correct option: (D)

- No matter where the object is placed in front of the concave lens, the concave lens always forms a virtual, erect and image of the object.
- A convex lens can form both real and virtual image of an object depending on the position of object.

22. Correct option: (B)

- The opaque painted region blocks some light rays entering. This makes a slight difference in the intensity. It reduces the brightness of the image more than usual. So the image formed is complete but less bright.
- The principal focus of a convex lens is a point on its principal axis to which light rays parallel to the principal axis converge after passing through the lens.

The principal focus of a concave lens is a point on its principal axis from which light rays, originally parallel to the axis

Appear to diverge after passing through the concave lens.

23. Correct option: (A)

$$from\ the\ Lens\ maker's\ formula\ \ which\ is\ given\ below$$

$$P = \frac{1}{f} = \left[\frac{\mu_2}{\mu_1} - 1\right] \times \left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$here\ ,\quad f = focal\ length\ of\ the\ lens$$

$$\mu_2 = refractive\ index\ of\ the\ medium\ of\ lens(glass)$$

$$\mu_1 = refractive\ index\ of\ the\ surrounding\ \ medium$$

We can see that the power of the lens is directly proportional to the refractive index of glass.

Also refractive index is inversely proportional to the wavelength.

So as the wavelength decreases, refractive index increases hence, power increases.

24. Correct option: (A)

The focal length of a glass plate is infinite and its power is zero.

Because The power of a lens is the reciprocal of its focal length in metres.

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

25. Correct option: (C)

The power of a lens is the reciprocal of its focal length in metres.

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

HUMAN EYE
LEVEL-I

1. Correct option: (C)

A normal eye can see the distant objects as well as the nearby objects clearly. The eye is able to focus the objects lying at various distances. An eye can focus the images of the distant objects as well as the nearby objects on its retina by changing the focal length (or converging power) of its lens. The focal length of the eye-lens is changed by the action of ciliary muscles. The ciliary muscles can change the thickness of the soft and flexible eye-lens and hence its focal length which, in turn, changes the converging power of the eye-lens. Let us see how it happens.

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due

to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

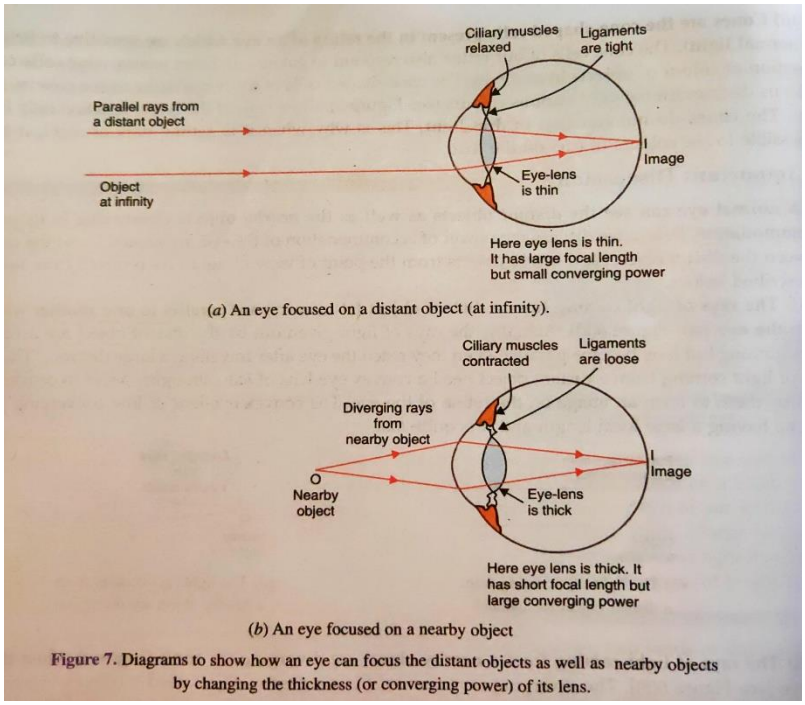


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.

2. Correct option: (D)

The Power of accommodation for a person with normal eyesight is around 4 dioptre

3. Correct option: (A)

As the person cannot see at a distance less than 40 cm, he has hypermetropia

It means the near point of the person = -40 cm

We have to use convex lens such that the image is formed at the near point of the hypermetropic person

Normal near point is 25 cm

Given:

Image distance(v) = - 40 cm
Object distance (u) = $-25\ cm$
putting the values in the lens formula
putting the values in the lens formula

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Where f= focal length of the lens
u=object distance from the lens
v= image distance from the lens

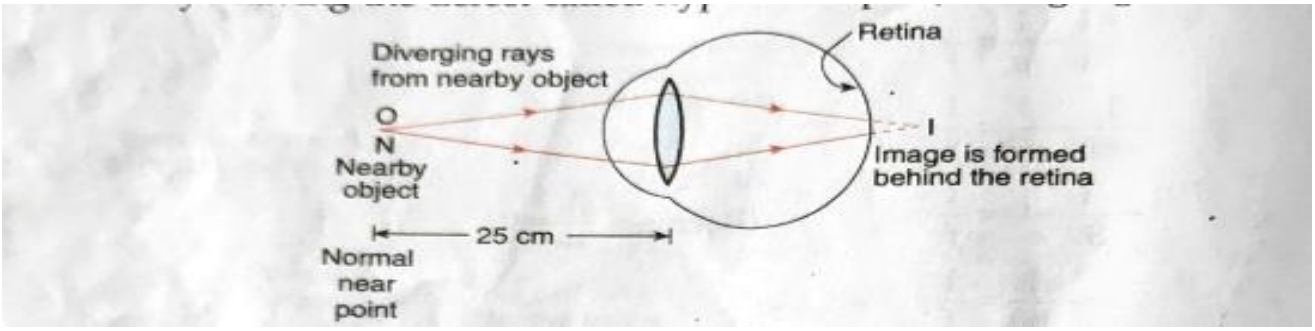
$$\frac{1}{f}=\frac{1}{-40}-\frac{1}{-25}$$
$$\frac{1}{f}=-\frac{1}{40}+\frac{1}{25}$$
$$\frac{1}{f}=\frac{-5+8}{200}$$
$$\frac{1}{f}=\frac{3}{200}cm$$

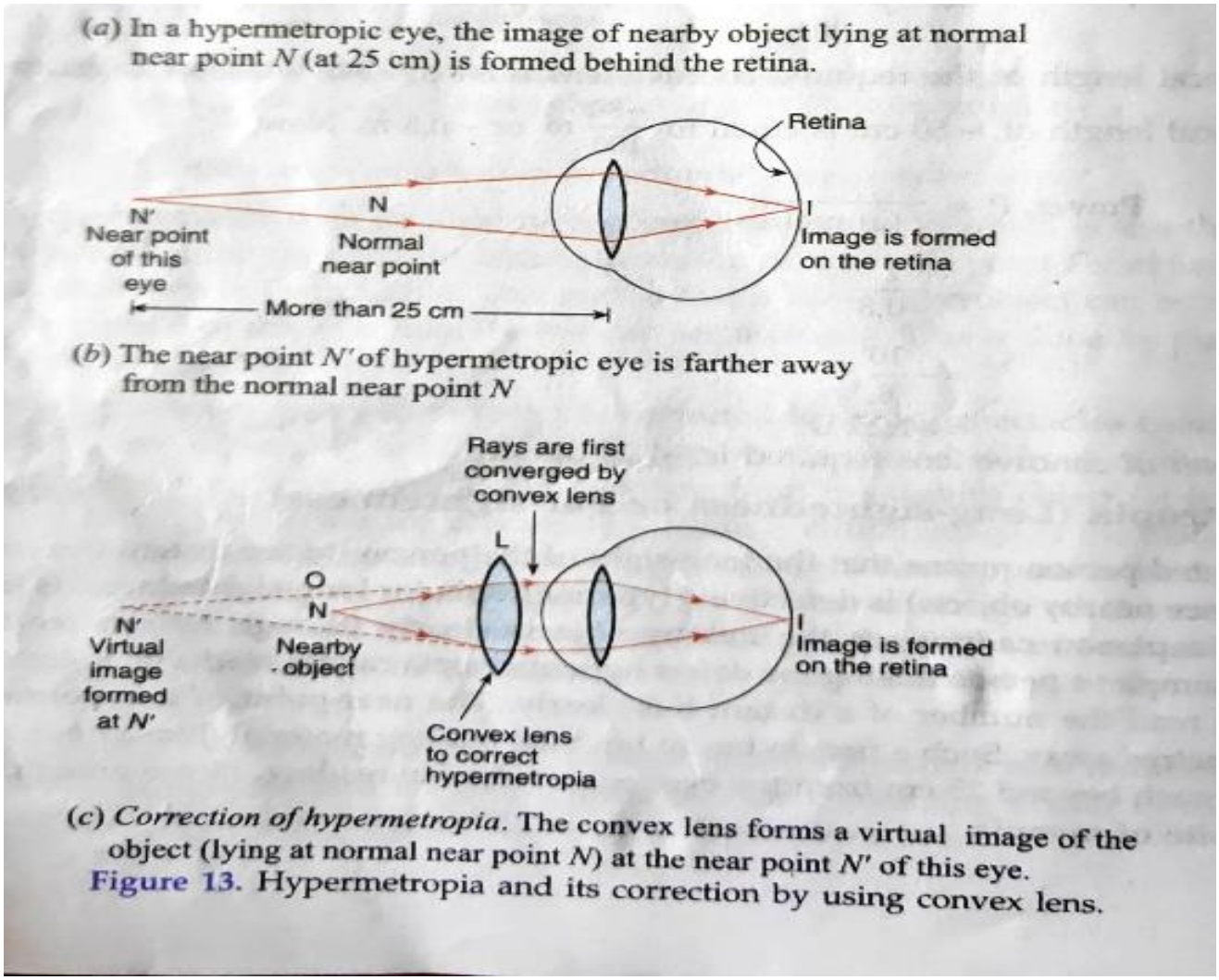
Or f= $\frac{200}{3}cm$

4. Correct option: (B)

The near-point of an eye having hypermetropia (or long-sightedness) is at point N' which is more than 25 centrimetres away [see Figure 13(b)] . The diverging rays of light coming from a hypermetropic person's near point can just be focused by his eye on the retina as shown in Figure 13(b). This means that if the object placed at the normal near point N (25 cm) can be made to appear as if it were placed at this eye's near point N, then the eye will be able to see it clearly. This can be done by putting a convex lens in front of the eye.

Hypermetropia (long-sightedness or far-sightedness) is corrected by using spectacles containing convex lenses. When a convex lens (converging lens) L of suitable power is placed in front of the hypermetropic eye as shown in Figure 13(c), then the diverging rays of light coming from the nearby object at 25 cm) are first converged by this convex lens. Due to this, the convex lens forms a virtual image of the nearby object (which is lying at the normal near point N) at the near point N' of the hypermetropic eye [see Figure 13(c)]. Since the rays of light now appear to be coming from this eye's near point N', they can be easily focused by the eye-lens to form an image on the retina.





5.Correcr option: (A)

The focal length of the human eye lens is 2.5 cm.

6.Correct option: (C)

Retina is the part of the eye that has a delicate membrane and contains a large number of light-sensitive cells.

The retina is the main part of the eye where the image of the object we see is formed. It is a delicate layer and a thin layer of tissue. “The retina contains several light-sensitive cells known as rods and cones”.

7. Correct option: (C)

A normal eye can see the distant objects as well as the nearby objects clearly. The eye is able to focus the objects lying at various distances. An eye can focus the images of the distant objects as well as the nearby objects on its retina by changing the focal length (or converging power) of its lens. The focal length of the eye-lens is changed by the action of ciliary muscles. The ciliary muscles can change the thickness of the soft and flexible eye-lens and hence its focal length which, in turn, changes the converging power of the eye-lens.

Let us see how it happens.

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small

converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

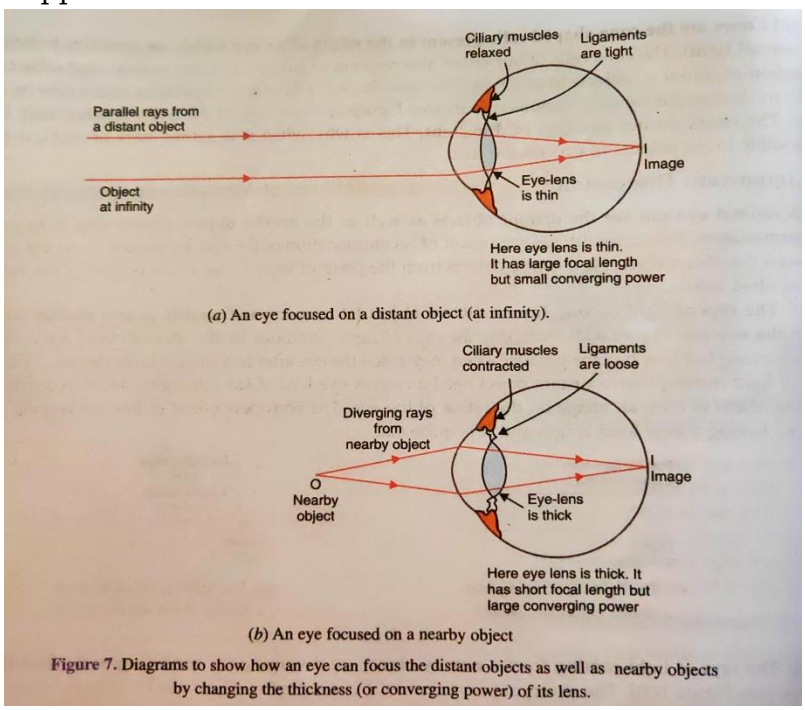


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.

8. Correct option: (B)

The light rays coming from the object kept in front of us enter through the cornea of the eye, pass through the pupil of the eye and fall on the eye-lens. The eye-lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina. (Actually, the outer surface of cornea also acts as a convex lens due to which cornea converges most of the light rays entering the eye. Only the final convergence of light rays is done by the eye-lens to focus the image of an object exactly on the retina).

The image formed on the retina is conveyed to the brain by the optic nerve and gives rise to the sensation of vision. Actually, the retina has a large number of light-sensitive cells. When the image falls on the retina then these light-sensitive cells get

activated and generate electrical signals. The retina sends these electrical signals to the brain through the optic nerve and gives rise to the sensation of vision.

9. Correct option: (B)

Crystalline eye lens Provide the focused real & inverted image of the object on the retina. It is composed of a fibrous, jelly like material. This is convex lens that converges light at retina.

10. Correct option: (B)

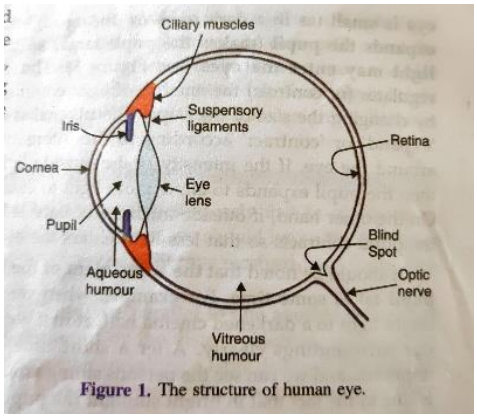


Figure 1. The structure of human eye.

11. Correct option: (A)

The iris controls the amount of light entering the eyes. The iris automatically adjusts the size of the pupil according to the intensity of light received by the eye. If the amount of light received by the eye is large (as during the day time), then the iris contracts the pupil (makes the pupil small) and reduces the amount of light entering the eye (see Figure 2). On the other hand, if the amount of light received by the eye is small (as in a dark room or during night), the iris expands the pupil (makes the pupil large) so that more light may enter the eyes (see Figure 3). Thus, the iris regulates (or controls) the amount of light entering the eye by changing the size of the pupil. The iris makes the pupil 'expand' or 'contract' according to the intensity of light around the eye. If the intensity of the outside light is low, then the pupil expands to allow more light to enter the eye. On the other hand, if outside intensity of light is high, then the pupil contracts so that less light enters the eye.

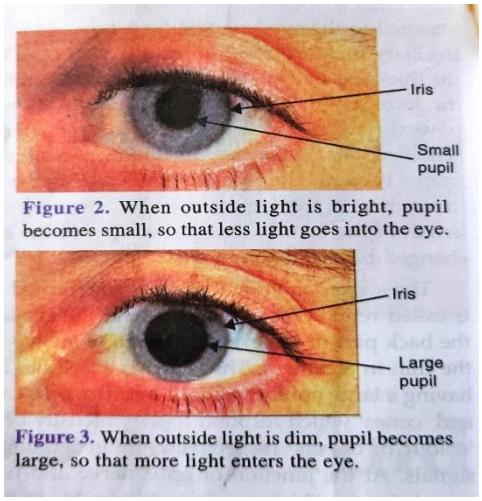


Figure 2. When outside light is bright, pupil becomes small, so that less light goes into the eye.

Figure 3. When outside light is dim, pupil becomes large, so that more light enters the eye.

12. Correct option: (A)

The iris controls the amount of light entering the eyes. The iris automatically adjusts the size of the pupil according to the intensity of light received by the eye. If the amount of light received by the eye is large (as during the day time), then the iris contracts the pupil (makes the pupil small) and reduces the amount of light entering the eye (see Figure 2). On the other hand, if the amount of light received by the eye is small (as in a dark room or

during night), the iris expands the pupil (makes the pupil large) so that more light may enter the eyes (see Figure 3). Thus, the iris regulates (or controls) the amount of light entering the eye by changing the size of the pupil. The iris makes the pupil 'expand' or 'contract' according to the intensity of light around the eye. If the intensity of the outside light is low, then the pupil expands to allow more light to enter the eye. On the other hand, if outside intensity of light is high, then the pupil contracts so that less light enters the eye

13.Correct option: (C)

Cornea is separated from the eye lens by aqueous humour.

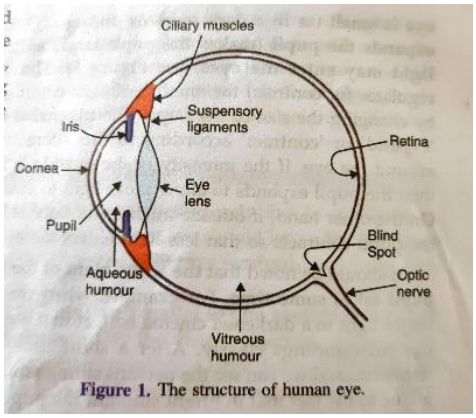


Figure 1. The structure of human eye.

14. Correct option: (D)

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: **The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.**

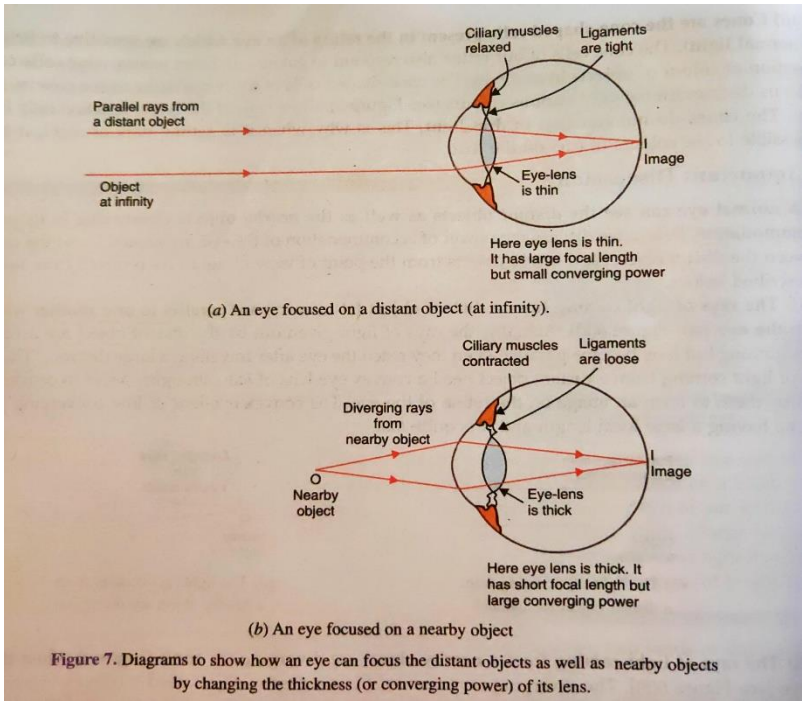


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

15. Correct option: (B)

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

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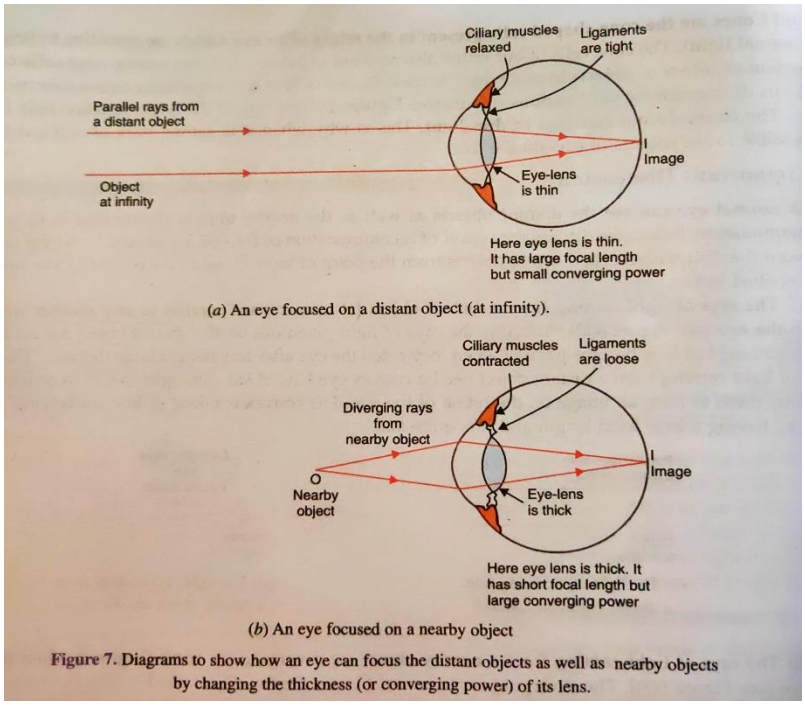


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

16. Correct option: (A)

The light rays coming from the object kept in front of us enter through the cornea of the eye, pass through the pupil of the eye and fall on the eye-lens. The eye-lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina. (Actually, the outer surface of cornea also acts as a convex lens due to which cornea converges most of the light rays entering the eye. Only the final convergence of light rays is done by the eye-lens to focus the image of an object exactly on the retina). **The image formed on the retina is conveyed to the brain by the optic nerve and gives rise to the sensation of vision.** Actually, the retina has a large number of light-sensitive cells. When the image falls on the retina then these light-sensitive cells get activated and generate electrical signals. The retina sends these electrical signals to the brain through the optic nerve and gives rise to the sensation of vision.

17.Correct option: (C)

The image formed at the retina of the human eye is real and inverted. It is due to the presence of a convex lens in the eye.

18.Correct option: (D)

The range of vision of a normal human eye is between it's near point and far point i.e, from 25cm to infinity.

19.Correct option: (B)

As the person cannot see beyond 120 cm = 1.20 m, he has myopia

It means the far point of the person = -1.20m

We have to use concave lens such that the image is formed at the far point of the myopic person

Normal far point is infinity ∞

Given:

Image distance(v) = - 1.20m

Object distance (u) = $-\infty$

putting the values in the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f= focal length of the concave lens

u=object distance from the lens
v= image distance from the lens

$$\frac{1}{f} = \frac{1}{-1.20} - \frac{1}{-\infty}$$

$$\frac{1}{f} = \frac{1}{-1.20} = -\frac{100}{120} = -0.83$$

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

P = - 0.83D
Hence, power = - 0.83 D

20.Correct option: (B)

As the person cannot see beyond 1.5 m, he has myopia
It means the far point of the person = -1.5 m
We have to use concave lens such that the image is formed at the far point of the myopic person
Normal far point is infinity ∞
Given:
Image distance(v) = - 1.5m
Object distance (u) = $-\infty$
putting the values in the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f= focal length of the concave lens
u=object distance from the lens
v= image distance from the lens

$$\frac{1}{f} = \frac{1}{-1.5} - \frac{1}{-\infty}$$

$$\frac{1}{f} = \frac{1}{-1.5} = -\frac{10}{15} = -0.67$$

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

P = - 0.67 D
Hence, power = - 0.67D

21.Correct option: (B)

As the person cannot see beyond a certain distance, he has myopia
We have to use concave lens such that the image is formed at the far point of the myopic person
Normal far point is infinity ∞
Given:
Power = - 2.5 D

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

$$f = \frac{1}{P} = \frac{1}{-2.5} = -\frac{10}{25} = -0.4m = -40cm$$

Focal length (f) = - 40 cm

Image distance = v

Object distance (u) = $-\infty$

putting the values in the lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where f= focal length of the concave lens

u=object distance from the lens

v= image distance from the lens

$$\frac{1}{-40} = \frac{1}{\mathrm{v}} - \frac{1}{-\infty}$$

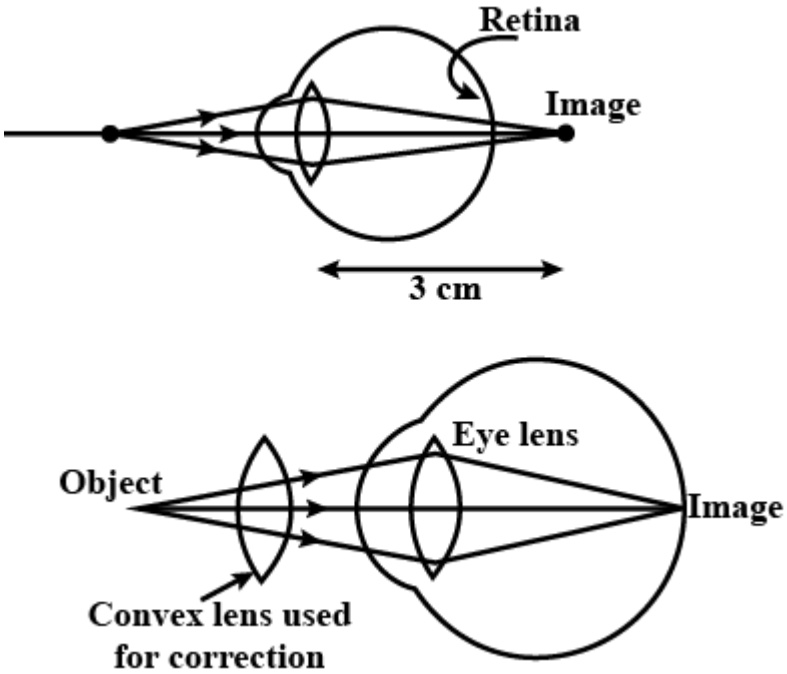
$$\frac{1}{\mathrm{v}} = \frac{1}{-40}$$

v = - 40 cm

So, far point for Billu's eyes = - 40 cm

22. Correct option: (A)

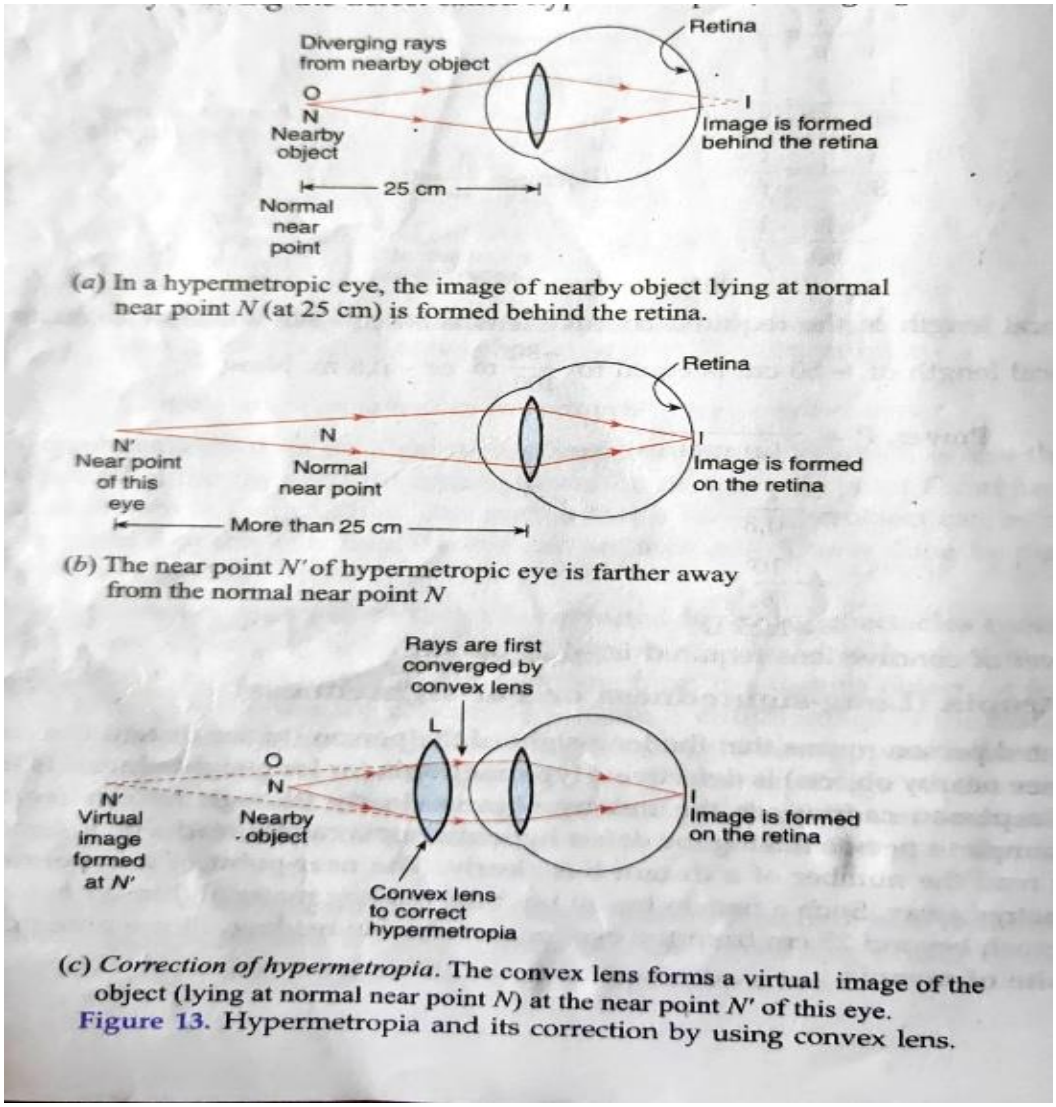
Since the distance between the pupil and the retina is 2.3 cm and the image is formed at a distance of 3 cm from the pupil, it is obvious that the image is formed behind the retina. To correct this defect in vision, Shyam needs to use a convex lens. As shown in the diagram, this would help in the formation of the image at the retina.



23. Correct option: (B)

The near-point of an eye having hypermetropia (or long-sightedness) is at point N' which is more than 25 centrimetres away [see Figure 13(b)] . The diverging rays of light coming from a hypermetropic person's near point can just be focused by his eye on the retina as shown in Figure 13(b). This means that if the object placed at the normal near point N (25 cm) can be made to appear as if it were placed at this eye's near point N, then the eye will be able to see it clearly. This can be done by putting a convex lens in front of the eye. Hypermetropia (long-sightedness or far-sightedness) is corrected by using spectacles containing convex lenses. When a convex lens (converging lens) L of suitable power is placed in front of the hypermetropic eye as shown in Figure 13(c), then the diverging rays

of light coming from the nearby object at 25 cm) are first converged by this convex lens. Due to this, the convex lens forms a virtual image of the nearby object (which is lying at the normal near point N) at the near point N' of the hypermetropic eye [see Figure 13(c)].



(*a*) In a hypermetropic eye, the image of nearby object lying at normal near point *N* (at 25 cm) is formed behind the retina.

(*b*) The near point *N'* of hypermetropic eye is farther away from the normal near point *N*

(*c*) *Correction of hypermetropia*. The convex lens forms a virtual image of the object (lying at normal near point *N*) at the near point *N'* of this eye.

Figure 13. Hypermetropia and its correction by using convex lens.

Since the rays of light now appear to be coming from this eye's near point N', they can be easily focused by the eye-lens to form an image on the retina.

24. Correct option: (D)

Hypermetropia (Long-sightedness or Far-sightedness)

A long-sighted person means that the long-sight of the person (to see distant objects) is normal but his short-sight (to see nearby objects) is defective. Hypermetropia (or long-sightedness) is that defect of vision due to which a person cannot see the nearby objects clearly (though he can see the distant objects) clearly).

As Mutum cannot see the blackboard on first bench it means he has hypermetropia.

25. Correct option: (B)

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is

looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.
The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:
When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an

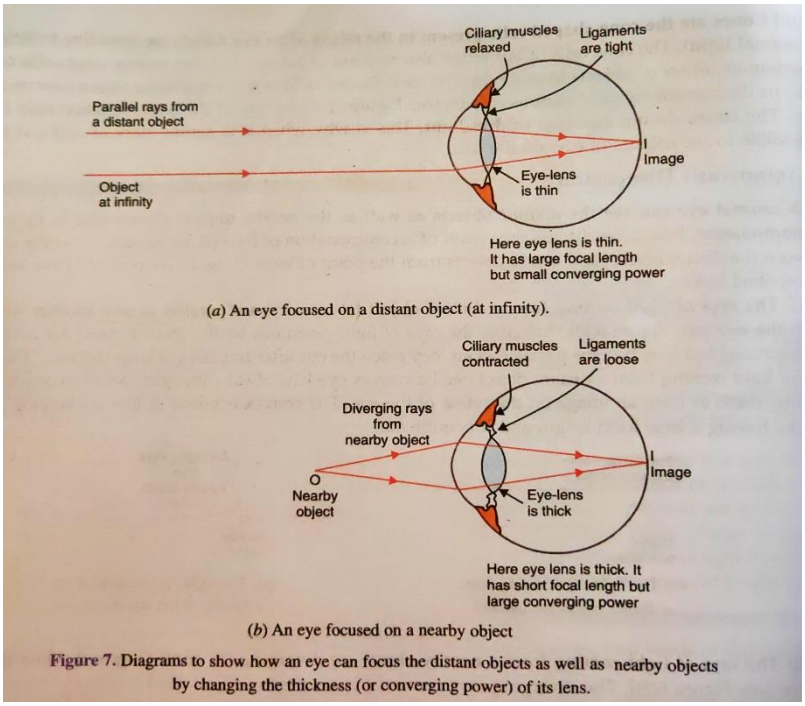


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: **The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.**

26. Correct option: (D)

The light rays coming from the object kept in front of us enter through the cornea of the eye, pass through the pupil of the eye and fall on the eye-lens. The eye-lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina. (Actually, the outer surface of cornea also acts as a convex lens due to which cornea converges most of the light rays entering the eye. Only the final convergence of light rays is done by the eye-lens to focus the image of an object exactly on the retina).
The image formed on the retina is conveyed to the brain by the optic nerve and gives rise to the sensation of vision. Actually, the retina has a large number of light-sensitive cells. When the image falls on the retina then these light-sensitive cells get activated and generate electrical signals. The retina sends these electrical signals to the brain through the optic nerve and gives rise to the sensation of vision.

27. Correct option: (C)

A normal eye can see the distant objects as well as the nearby objects clearly. The eye is able to focus the objects lying at various distances. An eye can focus the images of the distant objects as well as the nearby objects on its retina by changing the focal length (or converging power) of its lens. **The focal length of the eye-lens is changed by the action of ciliary muscles.** The ciliary muscles can change the thickness of the soft and flexible eye-lens and hence its focal length which, in turn, changes the converging power of the eye-lens. Let us see how it happens.

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an

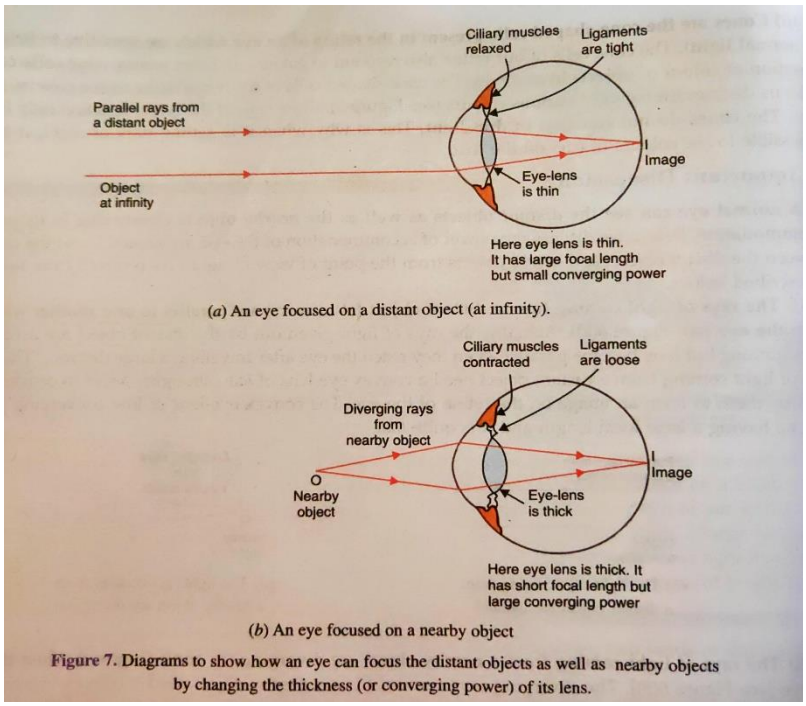


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: **The ability of an eye**

to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.

28. Correct option: (D)

the far point of the person = - 80 cm=0.8 m

We have to use concave lens such that the image is formed at the far point of the myopic person

Normal far point is infinity ∞

Given:

Image distance(v) = - 0.8m

Object distance (u) = $-\infty$

putting the values in the lens formula

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Where f= focal length of the concave lens

u=object distance from the lens

v= image distance from the lens

$$\frac{1}{f}=\frac{1}{-0.8}-\frac{1}{-\infty}$$

$$\frac{1}{f}=\frac{1}{-0.8}=-\frac{10}{8}=-1.25$$

$$power\ of\ a\ lens=\frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P=\frac{1}{f}$$

P = - 1.25 D

Hence, power = - 1.25 D

29. Correct option: (C)

The near point of the person =-1m = -100 cm

We have to use convex lens such that the image is formed at the near point of the hypermetropic person

Normal near point is 25 cm

Given:

Image distance(v) = - 100 cm

Object distance (u) = $\boldsymbol{-25\ cm}$

putting the values in the lens formula

putting the values in the lens formula

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Where f= focal length of the lens

u=object distance from the lens

v= image distance from the lens

$$\frac{1}{f}=\frac{1}{-100}-\frac{1}{-25}$$

$$\frac{1}{f}=-\frac{1}{100}+\frac{1}{25}$$

$$\frac{1}{f}=\frac{-1+4}{100}$$

$$\frac{1}{f}=\frac{3}{100}cm=\frac{3}{100}x100\ m=3m$$

$$power\ of\ a\ lens = \frac{1}{focal\ length\ of\ the\ lens\ in\ metres}$$

$$P = \frac{1}{f}$$

P = 3D

Hence, power = +3 D

30. Correct option: (D)

Hypermetropia (Long-sightedness or Far-sightedness)

A long-sighted person means that the long-sight of the person (to see distant objects) is normal but his short-sight (to see nearby objects) is defective. Hypermetropia (or long-sightedness) is that defect of vision due to which a person cannot see the nearby objects clearly (though he can see the distant objects) clearly).

As Raj cannot see the blackboard on first bench it means he has hypermetropia.

31. Correct option: (C)

The diameter of the human eyeball is 2.3cm.

The radius of the human eyeball is half of its diameter, i.e., 1.15cm.

32. Correct option: (B)

The light rays coming from the object kept in front of us enter through the cornea of the eye, pass through the pupil of the eye and fall on the eye-lens. The eye-lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina. (Actually, the outer surface of cornea also acts as a convex lens due to which cornea converges most of the light rays entering the eye. Only the final convergence of light rays is done by the eye-lens to focus the image of an object exactly on the retina). **The image formed on the retina is conveyed to the brain by the optic nerve and gives rise to the sensation of vision.**

33. Correct option: (A)

Myopia (or short-sightedness) is that defect of vision due to which a person cannot see the distant objects clearly (though he can see the nearby objects clearly).

The far point of an eye suffering from myopia is less than infinity. Such a person can see clearly only up to a distance of few metres (or even less).

The defect of eye called myopia (or short-sightedness) is caused: (i) due to high converging power of eye-lens (because of its short focal length), or (ii) due to eye-ball being too long.

In some cases, in an eye suffering from myopia, the ciliary muscles attached to the eye-lens do not relax sufficiently to make the eye-lens thinner to reduce its converging power. So, due to the greater converging power of the eye-lens in myopic eye, the image of a distant object is formed in front of the retina and hence the eye cannot see it clearly. In other cases, in the eye suffering from myopia, the eye-ball is too long due to which the retina is at a larger distance from the eye-lens. This condition also results in the formation of the image of a distant object in front of the retina (even though the eye-lens may have correct converging power).

myopia (short-sightedness) is corrected by using spectacles containing concave lens.

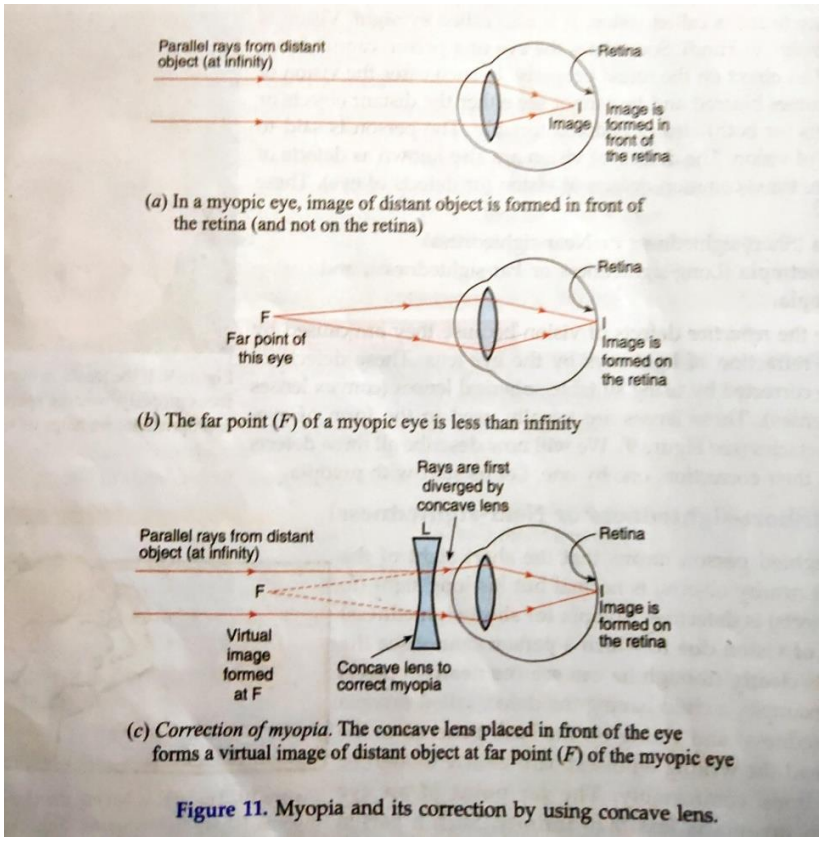


Figure 11. Myopia and its correction by using concave lens.

LEVEL-II

SINGLE ANSWER QUESTIONS

1. Correct option: (A)

This is because the red coloured light having longer wavelength is the least scattered by fog or smoke particles. Due to this the red light can be seen in the same colour even from a distance.

2. Correct option: (A)

The raindrops in the atmosphere act like many small prisms. As white sunlight enters and leaves these raindrops (or water drops), the various coloured rays present in white light are refracted by different amounts due to which an arch of seven colours called rainbow is formed in the sky. The formation of rainbow can be explained with the help of a diagram shown in Figure 25. A ray of white sunlight AB enters the raindrop at point B and

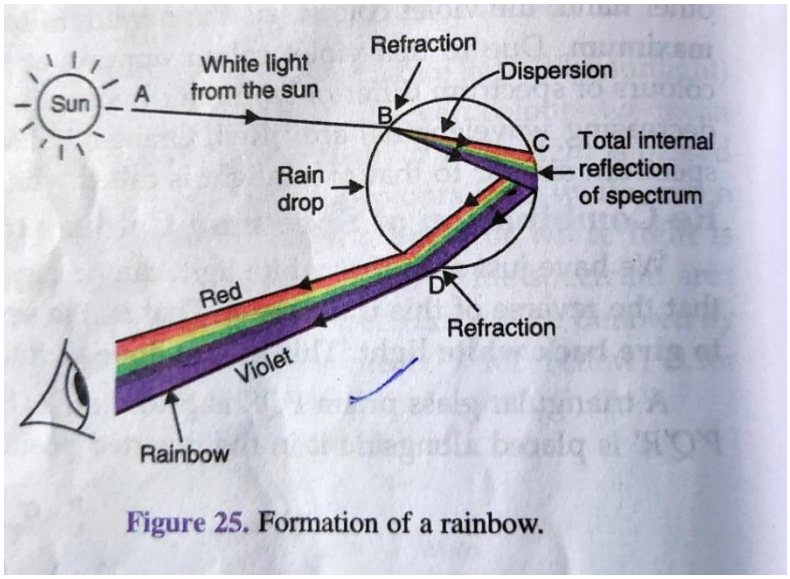
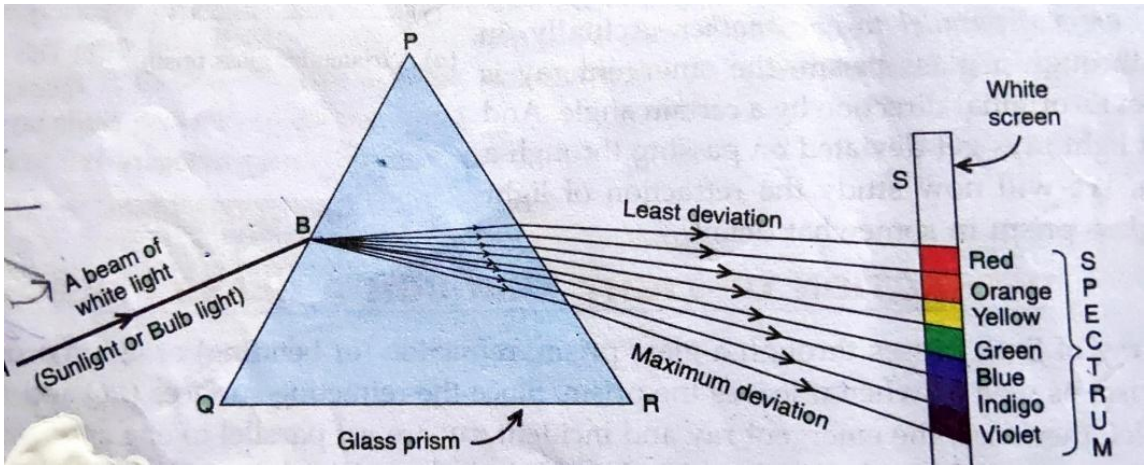


Figure 25. Formation of a rainbow.

undergoes refraction and dispersion to form a spectrum. This spectrum undergoes total internal reflection at point C within the raindrop and finally refracted out of the raindrop at point D (see Figure 25).

3. Correct option: (C)



The band of seven colours formed on a white screen, when a beam of white light is passed through a glass prism, is called spectrum of white light. The seven colours of the spectrum are Red, Orange, Yellow, Green, Blue, Indigo, and Violet. The seven colours of the spectrum can be denoted by the word VIBGYOR where V stands for Violet, I for Indigo, B for Blue, G for Green, Y for Yellow, O for Orange and R for Red.

4. Correct option: (A)

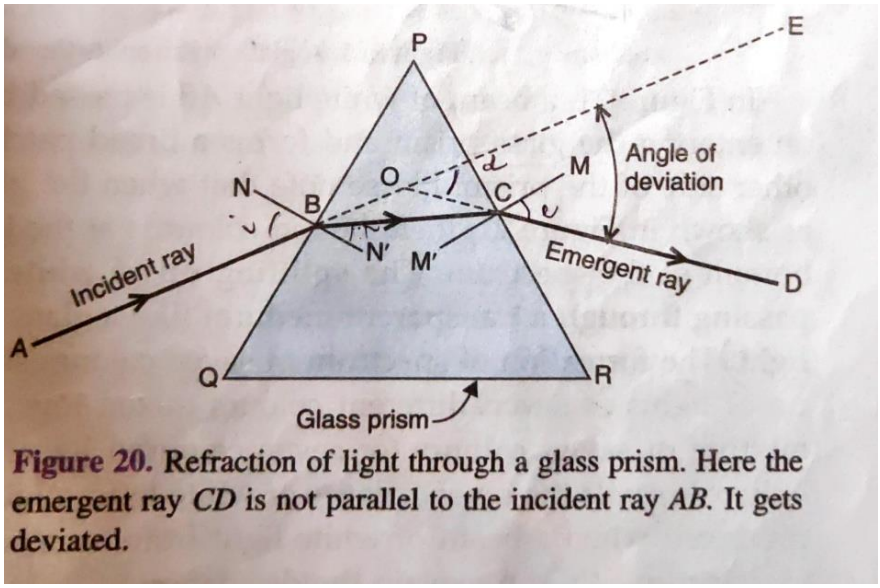


Figure 20. Refraction of light through a glass prism. Here the emergent ray *CD* is not parallel to the incident ray *AB*. It gets deviated.

In the above figure a glass prism PQR has been kept on its base QR. A ray of light AB is incident on the face PQ of the prism. The incident ray AB is going from air (rarer medium) into glass (denser medium). so it bends towards the normal BN' and goes along the direction BC inside the glass prism. Thus, BC is the refracted ray of light which bends towards the base QR of the prism) When the ray of light BC travelling in the glass prism comes out into air at point C. refraction takes place again (see Figure 20). Since the ray BC is going from glass (denser medium) into air (rarer medium),so it bends away from the normal MC and goes along the direction CD in the form of emergent ray. Here also, the emergent ray of light CD bends towards the base QR of the prism. From this discussion we conclude that when a ray of light passes through a prism, it bends towards the base of prism. In other words, when a ray of light passes through a prism, it bends towards the thicker part of the prism

5. Correct option: (A)

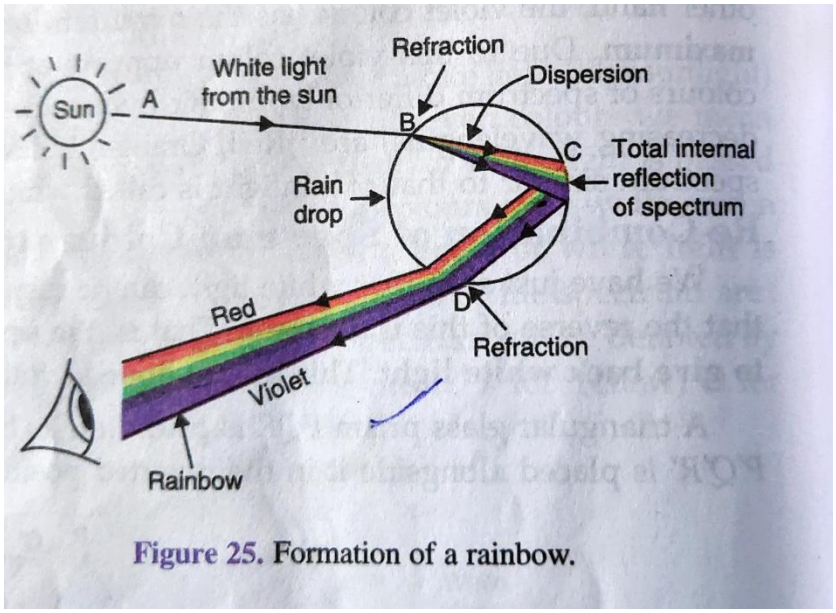


Figure 25. Formation of a rainbow.

The raindrops in the atmosphere act like many small prisms. As white sunlight enters and leaves these raindrops (or water drops), the various coloured rays present in white light are refracted by different amounts due to which an arch of seven colours called rainbow is formed in the sky. The formation of rainbow can be explained with the help of a diagram shown in Figure 25. A ray of white sunlight AB enters the raindrop at point B and undergoes refraction and dispersion to form a spectrum. This spectrum undergoes total internal reflection at point C within the raindrop and finally refracted out of the raindrop at point D (see Figure 25).

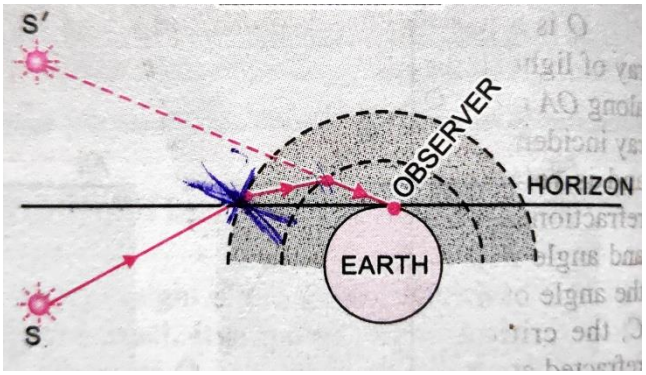
6. Correct option: (D)

If the refractive index of the earth's atmosphere is decreased then there will be less bending of light coming from distant star, hence the new position of star would be between 1 and 2.

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

7. Correct options: (B), (C)

- Atmospheric refraction: We know that when light goes from one medium to another medium having different optical densities, then refraction of light rays (or bending of light rays) takes place. Now, the cooler air layers of the atmosphere behave as optically denser medium for light rays whereas the warmer air layers (or hotter air layers) of the atmosphere behave as optically rarer medium for the light rays. And when light rays pass through the atmosphere having air layers of different optical densities, then refraction of light takes place. The refraction of light caused by the earth's atmosphere (having air layers of varying optical densities) is called atmospheric refraction



the air in the upper atmosphere is optically rarer(less refractive index) and as we come down, the air in the lower atmosphere is optically denser(more refractive index).

The actual sunrise takes place when the sun is just above the horizon. But due to refraction of sunlight caused by the atmosphere, we can see the rising sun about 2 minutes before it is actually above the horizon. This happens as follows: When the sun is slightly below the horizon, then the sun's light coming from less dense air to more dense air is refracted downwards as it passes through the atmosphere. Because of this atmospheric refraction, the sun appears to be raised above the horizon when actually it is slightly below the horizon. For example, in Figure above, the actual position of the sun is at S just below the horizon but it appears to be at position S' above the horizon, due to atmospheric refraction of light rays coming from it (This is because our eye will see the sun at that position from where light enters it in the straight line direction).

It is also due to atmospheric refraction that we can still see the sun for about 2 minutes even after the sun has set below horizon. From the above discussion we conclude that we can see the sun 2 minutes before the actual sunrise time and for 2 minutes after the actual sunset time. So, the time from sunrise to sunset is lengthened by about 2 + 2 = 4 minutes because of atmospheric refraction.

- We can see the effect of atmospheric refraction by a simple observation as follows: If we look at objects through the hot air over a fire, the objects appear to be moving slightly. This can be explained as follows: The air just above the fire becomes hotter (than the air further up). This hotter air is optically rarer but the colder air further up is optically denser. So, when we see the objects by the light coming from them through hot and cold air layers having different optical densities, then refraction of light takes place randomly due to which the objects appear to be moving slightly.
- The raindrops in the atmosphere act like many small prisms. As white sunlight enters and leaves these raindrops (or water drops), the various coloured rays present in white light are refracted by different amounts due to which an arch of seven colours called rainbow is formed in the sky. The formation of rainbow can be explained with the help of a diagram shown in Figure 25. A ray of white sunlight AB enters the raindrop at point B and undergoes refraction and dispersion to form a spectrum. This spectrum undergoes total internal reflection at point C within the raindrop and finally refracted out of the raindrop at point D (see Figure 25).

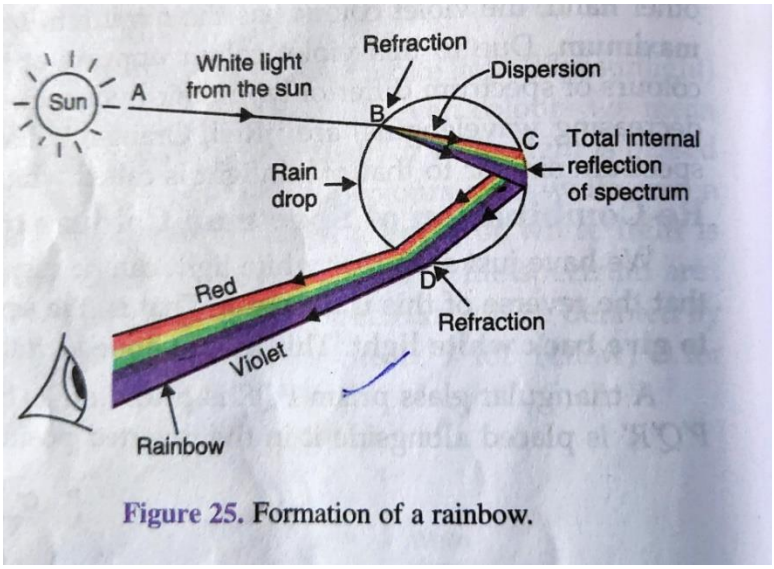


Figure 25. Formation of a rainbow.

8. Correct options: (A), (B), (D)

- The scattering of light by particles in its path is called Tyndall effect When a beam of sunlight enters a dusty room through a window, then its path becomes visible to us. This is because the tiny dust particles present in the air of room scatter the beam of light all around the room. And when this scattered light enters our eyes, we can see the beam of light. Thus, an example of Tyndall effect is the way a beam of sunlight becomes visible as it passes through dust particles in the air of a room.
- The heterogeneous mixture shows the Tyndall effect.
- A beam of light can only be seen when it hits matter particles and is scattered or reflected by them. In a vacuum, however, it is invisible.
- The colour of the scattered light depends on the size of the scattering particles in the atmosphere:

(a) The larger particles of dust and water droplets present in the atmosphere scatter the light as such due to which the scattered light also appears white.
(b) The extremely minute particles such as the air molecules present in the atmosphere scatter mainly the blue light present in the white sunlight.

STATEMENT TYPE QUESTIONS

9. Correct option: (D)

- The defect of eye called myopia (or short-sightedness) is caused: (i) due to high converging power of eye-lens (because of its short focal length), or (ii) due to eye-ball being too long.
- The near-point of an eye having hypermetropia (or long-sightedness) is more than 25 centrimetres away. This means that if the object placed at the normal near point N (25 cm) can be made to appear as if it were placed at this eye's near point N, then the eye will be able to see it clearly. This can be done by putting a convex lens in front of the eye.
- Presbyopia (old sight)
  With increasing age, the ciliary muscles holding the eye lens weaken and the lens loses some of its elasticity. Therefore, power of accommodation of the eye decreases with age. This defect is called presbyopia. To remove this defect, converging spectacle lenses are employed (as in the case of hypermetropia). The person needs to wear these specs for reading or similar close work. A person suffering from presbyopia will normally be prescribed bifocal specs. The upper half of each lens is diverging and corrects the myopia, when the wearer is looking ahead at distant objects. The lower half of each lens is convex and corrects the defect when the wearer looks through this part while reading.

10. Correct option: (A)

A normal eye can see the distant objects as well as the nearby objects clearly. The eye is able to focus the objects lying at various distances. An eye can focus the images of the distant objects as well as the nearby objects on its retina by changing the focal length (or converging power) of its lens. **The focal length of the eye-lens is changed by the action of ciliary muscles.** The ciliary muscles can change the thickness of the soft and flexible eye-lens and hence its focal length which, in turn, changes the converging power of the eye-lens. Let us see how it happens.

When the eye is looking at a distant object (at infinity), then the ciliary muscles of the eye are fully relaxed. The relaxed ciliary muscles of the eye pull the suspensory ligaments attached to the eye-lens tightly. The suspensory ligaments, in turn, pull the eye-lens due to which the eye-lens gets stretched and becomes thin (or less convex) [Figure (a)]. The

thin eye-lens has large focal length but its converging power is small. The small converging power of thin eye-lens is sufficient to converge the parallel rays of light coming from a distant object to form an image on the retina of the eye (Figure (a)). When the eye is looking at a distant object, the eye is said to be unaccommodated because it is the relaxed state of the eye.

The thin eye-lens is not powerful enough to converge the diverging light rays coming from the nearby objects onto the retina. So, to look at the nearby objects, the eye-lens has to change its shape and become thick (or more convex) to increase its converging power. This happens as follows:

When the same eye has to look at a nearby object, the ciliary muscles of the eyes contract. The contracted ciliary muscles make the suspensory ligaments loose. When the suspensory ligaments become loose, they stop pulling the eye-lens. The eye-lens bulges under its own elasticity and becomes thick (or more convex) [Figure (b). The thick eye-lens has small focal length but its converging power is large. Since the converging power of eye-lens increases, the thick eye-lens can converge the diverging light rays coming from the nearby object to form an image on the retina of the eye. This is shown in Figure (b) in which an object O is near to the eye. It has been focused by the thick eye-lens to form an

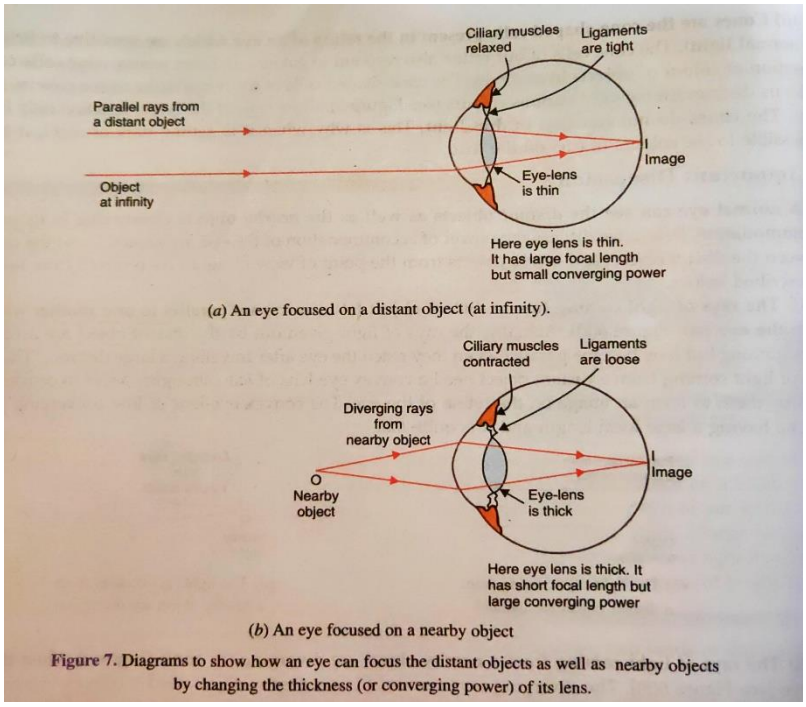


Figure 7. Diagrams to show how an eye can focus the distant objects as well as nearby objects by changing the thickness (or converging power) of its lens.

image I on the retina. When the eye-lens becomes more convex to focus the nearby objects, the eye is said to be 'accommodated". We can now say that: **The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length (or converging power) of its lens is called accommodation.**

11. Correct option: (A)

The maximum "accommodation" of a normal eye is reached when the object is at a distance of about 25 cm from the eye. After this the ciliary muscles cannot make the eye-lens more thick (or more convex). So, an object placed at a distance of less than 25 cm cannot be seen clearly by a normal eye because all the power of accommodation of the eye has already been exhausted. Thus, a normal eye has a power of accommodation which enables objects as far as infinity and as close as 25 cm to be focused on the retina.

12. Correct option: (A)

When the light coming from a star enters the earth's atmosphere, it undergoes refraction due to the varying optical densities of air at various altitudes. The atmosphere is

continuously changing (due to which the optical densities of air at different levels in the atmosphere keep on changing). The continuously changing atmosphere refracts the light from the stars by different amounts from one moment to the next. When the atmosphere refracts more star-light towards us, the star appears to be bright and when the atmosphere refracts less star-light, then the star appears to be dim. In this way, the star-light reaching our eyes increases and decreases continuously due to atmospheric refraction. And the star appears to twinkle at night.

13. Correct option: (A)

Planets do not twinkle at all. This can be explained as follows: The stars appear very, very small to us (because they are very, very far off). So, stars can be considered to be point sources of light. The continuously changing atmosphere is able to cause variations in the light coming from a point-sized star (due to refraction) because of which the star appears to be twinkling. On the other hand, the planets appear to be quite big to us (because they are much nearer to the earth). So, a planet can be considered to be a collection of a very large number of point sources of light. The dimming effect produced by some of the point sources of light in one part of the planet is nullified by the brighter effect produced by the point sources of light in its other part. Thus, on the whole, the brightness of a planet always remains the same and hence it does not appear to twinkle. We can now say that: The continuously changing atmosphere is unable to cause variations in the light coming from a big-sized planet (due to refraction) because of which the planet does not twinkle at all.

14. Correct option: (A)

The sun and the surrounding sky appear red at sunrise and at sunset because at that time most of the blue colour present in sunlight has been scattered out and away from our line of sight, leaving behind mainly red colour in the direct sunlight beam that reaches our eyes This can be explained as follows.

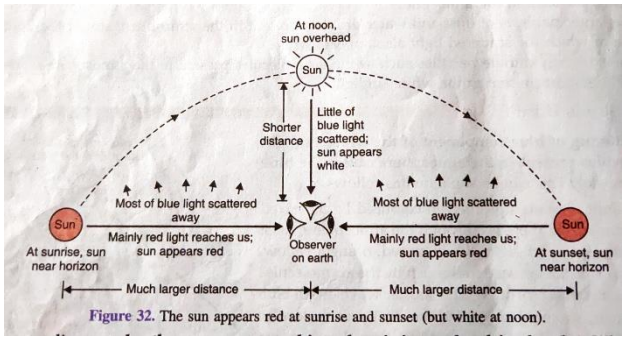


Figure 32. The sun appears red at sunrise and sunset (but white at noon).

At the time of sunrise and sunset when the sun is near the horizon, the sunlight has to travel the greatest distance through the atmosphere to reach us. During this long journey of sunlight, most of the shorter wavelength blue-colour present in it is scattered out and away from our line of sight. So, the light reaching us directly from the rising sun or setting sun consists mainly of longer wavelength red colour due to which the sun appears red (see Figure 32). Due to the same reason, the sky surrounding the rising sun and setting sun also appears red. Thus, at sunrise and sunset, the sun itself as well as the surrounding sky appear red.

15. Correct option: (A)

In outer space, the sky looks dark and black instead of blue. This is because there is no atmosphere containing air in the outer space to scatter sunlight. Since there is no scattered light to reach our eyes in outer space, therefore, the sky looks dark and black there. This is why the astronauts who go to outer space find the sky to be dark and black instead of blue.

MATCHING TYPE QUESTIONS

16.

a-------S

b-------R

c-------P

d-------Q

- The iris automatically adjusts the size of the pupil according to the intensity of light received by the eye. If the amount of light received by the eye is large (as during the day time), then the iris contracts the pupil (makes the pupil small) and reduces the amount of light entering the eye (see Figure 2). On the other hand, if the amount of light received by the eye is small (as in a dark room or during night), the iris expands the pupil (makes the pupil large) so that more light may enter the eyes (see Figure 3). Thus, the iris regulates (or controls) the amount of light entering the eye by changing the size of the pupil. The iris makes the pupil 'expand' or 'contract' according to the intensity of light around the eye. If the intensity of the outside light is low, then the pupil expands to allow more light to enter the eye. On the other hand, if outside intensity of light is high, then the pupil contracts so that less light enters the eye.

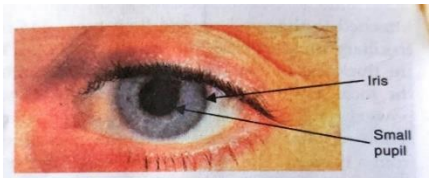


Figure 2. When outside light is bright, pupil becomes small, so that less light goes into the eye.



Figure 3. When outside light is dim, pupil becomes large, so that more light enters the eye.

- **The focal length(power) of the eye-lens is changed by the action of ciliary muscles.** The ciliary muscles can change the thickness of the soft and flexible eye-lens and hence its focal length which, in turn, changes the converging power of the eye-lens.
- **The light rays coming from the object kept in front of us enter through the cornea of the eye**, pass through the pupil of the eye and fall on the eye-lens. The eye-lens is a convex lens, so it converges the light rays and produces a real and inverted image of the object on the retina.

8. Electricity

Solutions

LEVEL – I

1. Correct option: (A)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub a glass rod with silk, electrons are transferred from glass rod to silk. The glass rod becomes positively charged and silk acquires an equal negative charge.

2. Correct option: (A)

electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

3. Correct option: (C)

In charging by induction, a charged body imparts some charge of opposite sign to another body without any actual contact between them.

4. Correct option: (C)

A gold leaf electroscope is an instrument which is used for detecting the presence of electric charge and its polarity (that is whether charge is negative or positive).

5. Correct option: (A)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell or a battery. Due to the chemical reactions going on inside the cell or battery, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell or battery is connected.

6. Correct option: (B)

A **primary cell** or battery is one that cannot easily be *recharged* after one use, and are discarded following discharge.

A **secondary cell** or battery is one that can be electrically recharged after use to their original pre-discharge condition, by passing current through the circuit in the opposite direction to the current during discharge.

Therefore, a primary cell require replacement after usage.

7. Correct option: (C)

Electromotive force is defined as the maximum potential difference between the two electrodes of the cell when no current is drawn from the cell.

SI unit of electromotive force is volts(V).

8. Correct option: (D)

The full form of MCB is **Miniature Circuit Breaker**. It is a kind of electronic switch that functions automatically. It can detect a problem occurring inside a circuit by identifying the extreme amount of current flow in the circuit. If there is a large amount of power available in the loop, it can break the loop at any moment.

9. Correct option: (C)

The heating effect of electric current is utilized in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

10. Correct option: (B)

- Similar charges (or Like charges) repel each other. For example, a positive charge repels a positive charge; and a negative charge repels a negative charge.
- Opposite charges (or Unlike charges) attract each other. For example, a positive charge attracts a negative charge.

11. Correct option: (B)

Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical to electrical.

12. Correct option: (D)

Electrolysis is used extensively in metallurgical processes such as in **extraction** (electrowinning) or **purification** (electro refining) of metals from ores or compounds and in deposition of metals from solution (**electroplating**).

13. Correct option: (B)

During electrolysis of water, electricity is passed through acidified or salt water. Water undergoes decomposition and the process liberates hydrogen gas at the cathode and oxygen gas at the anode.

14. Correct option: (D)

Properties of electrolyte are:

- It conducts electric current as a result of its dissociation into ions.
- It generally ionizes into positive and negative ions when it is dissolved in a solvent.

15. Correct option: (D)

The electrodes decide which element or compound is deposited as a result of the chemical reaction triggered by the electric current passing through them.

An electrode is a nonmetallic component of a circuit that makes contact with an electrical conductor.

chemical reaction depends on Electrodes, Nature of electrolyte, magnitude of current.

16. Correct option: (A)

Drinking water is a good conductor of electricity because it contains small amounts of various salts dissolved in it.

17. Correct option: (B)

Electroplating is used to give objects made of a cheap metal coating of a more expensive metals to make them look more attractive. Less expensive metal is electroplated with more expensive metals like silver and gold to make Jewellery or ornaments. These ornaments have the appearance of silver or gold but they are much less expensive.

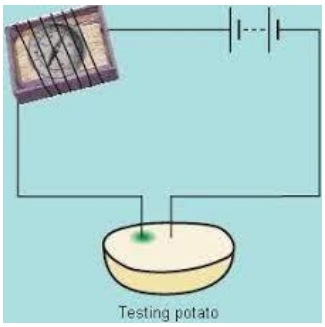
18. Correct option: (C)

Electrical conductivity of distilled water can be increased by adding some type of an acid or a salt to it. Vegetable oil, kerosene and sugar are bad conductors of electricity.

19. Correct option: (C)

Suppose we want to pass current through a potato. So, we connect ends of a battery by copper wire with potato by inserting in it. The electrons from negative potential of battery through copper wire reach in potato where they undergo in electrolysis (chemical reaction) inside the potato in the presence of potato juice (starch).

This process releases electrons in potato and increases the negative ions. When these negative ions reach the other end of copper wire (positive terminal) it reacts with part of the copper wire inside the potato and makes a compound of copper which is bluish-green in color.



This is the reason for greenish spots in potato on passing a current through it. It always appears around the copper wire connected to the positive terminal of the battery. This is an example that electricity causes chemical reactions. This chemical reaction causes color change.

20. Correct option: s(B)

The passage of an electric current through a solution causes chemical effects. The solution decomposes into positive and negative ions when electricity is passed through it. The process of decomposition of the solution is a chemical effect.

21. Correct option: s(C)

The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

22. Correct option: (C)

The metallic rods or plates used in the electrolysis are called conductors or electrodes.

23. Correct option: (B)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub a glass rod with silk, electrons are transferred from glass rod to silk. The glass rod becomes positively charged and silk acquires an equal negative charge.

LEVEL-II

1. Correct option: (A)

The heating effect of electric current is utilized in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the

circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

2. Correct option: (B)

The same charge is transferred to the gold leaves through the metal rod. The leaves repel each other and diverge as shown in. By measuring the divergence of leaves, the amount of charge on the body can be estimated.

Now, since the leaves are getting closer to each other it means the negative charge on the leaves is reducing which can happen if the rod is positively charged.

3. Correct option: (C)

When two charged bodies at different electric potentials are connected by a metal wire, then electric charges will flow from the body at higher potential to the one at lower potential (till they both acquire the same potential).

This flow of charges in the metal wire constitutes an electric current. It is the potential difference between the ends of the wire which makes the electric charges (or current) to flow in the wire.

We now know that the electric charges whose flow in a metal wire constitutes electric current are the **negative charges called electrons.**

4. Correct option: (A), (B)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell or a battery. Due to the chemical reactions going on inside the cell or battery, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell or battery is connected.

5. Correct option: (A), (C)

- Large Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured.
- A galvanometer can measure small electric currents.

6. Correct option: (B), (C)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

If we put charge Q = 1 coulomb and time = 1 second in the above formula,

then current I becomes 1 ampere. This gives us the following definition of ampere: When 1 coulomb of charge flows through any cross-section of a conductor in 1 second, the electric current flowing through it is said to be 1 ampere. That is,

1 ampere = $\frac{1\ \text{coulomb}}{1\text{sec}}$

Hence, units of electric current are Ampere and coulomb/sec

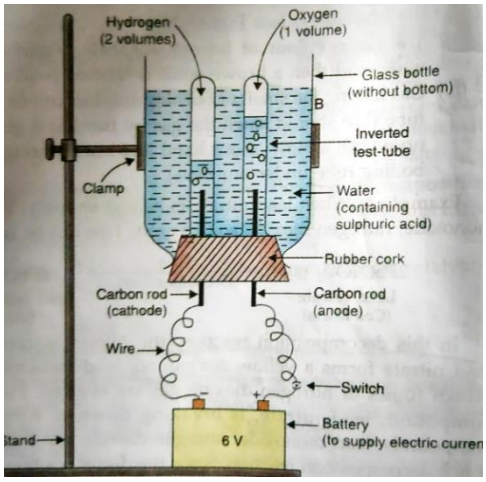
7. Correct options: (A), (B), (C)

The most familiar uses of electroplating are:

- Used in ornamentation and decoration purposes
- Purification of metals
- Repairing worn out parts of machinery.
- Preventing corrosion/ rusting of metals

8. Correct options: (B), (C)

Electrolysis of water is using electricity to split water into oxygen gas and hydrogen gas by electrolysis.



9. Correct options: (B), (C), (D)

- When salt is dissolved in distilled water, we get salt solution. Thus, distilled water becomes a good conductor of electricity on dissolving a little of salt in it.
- $CuSO_4$ dissociate completely into ions so it is strong electrolyte which conducts electricity.
- Lemon juice has charged particles in it. Hence, it acts as an electrolyte.

10. Correct options: (A), (B)

Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, Bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them.

11. Correct options: (A), (B), (C)

Liquids that are solutions of acids, bases and salts are good conductors. Distilled water is a poor conductor.

12. Correct options: (A), (B)

Rubbed area of non-conducting body/ insulator gets charged, and this charge does not move to other parts of the body. The charge is static on rubbed portion only that is outer surface.

13. Correct option: (B)

Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, Bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them**.**

14. Correct option: (C)

Those substances through which electric charges can flow, are called conductors. But the flow of electric charges is called electricity, so we can also say that: Those substances through which electricity can flow are called conductors. All the metals like silver, copper and aluminum, etc., are conductors

15. Correct option: (A)

All the conductors (like metals) have some electrons which are loosely held by the nuclei of their atoms. These electrons are called "free electrons" and can move from one atom to

another atom throughout the conductor. The presence of "free electrons" in a substance makes it a conductor (of electricity).

16. Correct option: (D)

Electroplating a thin layer of chromium onto a metal item is known as chrome plating. It is done

- To make objects heavy
- To make the objects strongest
- To protect the objects from rusting

17. Correct option: (A)

The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

18. Correct option: (A)

Electroplating is based on chemical effect of electric current. In electroplating, electric current is used to reduce metal cations which are dissolved in solvent to form thin metal coating on another electrode.

19. Correct Matching is: (a)----(R) (b)----(T) (c)----(P) (d)---(Q)

(a) The SI unit of electric charge is coulomb which is denoted by the letter C.

An electron possesses a charge of -1.6 x 10^{-19}C.

(b) The SI unit of electric potential is volt which is denoted by the letter V.

(c) The SI unit of electric current is ampere which is denoted by the letter A.

(d) The SI unit of electric charge is coulomb which is denoted by the letter C.

20. Correct Matching is: (a)----(r) (b)----(t) (c)----(q), (s) (d)---(p)

(a)Anode is the positive part of electrolyte where oxidation takes place

(b)cathode is the negative part of the cell where reduction takes place.

(c) The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

Electroplating is based on chemical effect of electric current. In electroplating, electric current is used to reduce metal cations which are dissolved in solvent to form thin metal coating on another electrode.

(d) An electrolyte is a substance that produces an electrically conducting solution when dissolved in a polar solvent such as water.